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Silicon to Software™

PV (Physical Verification)

Synopsys University Program

WEEK 15 PV (Physical Verification)

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WEEK 15 : Introduction to PV (Physical Verification)

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Introduction to PV(Physical Verification) : Overview to DRC

- **What is DRC (Design Rule Check)**
  - DRC verifies as to whether a specific design meets the constraints imposed by the process technology to be used for its ma nufacturing
  - DRC checking is an essential part of the physical design flow and ensures the design meets manufacturing requirements an d will not result in a chip failure
  - Process specific design rules must be followed when drawing layouts to avoid any manufacturing defects during the fabricati on of an IC
  - The process technology rules are provided by process engineers and/or fabrication facility
  - Process design rules are the minimum allowable drawing dimensions which affects the X and Y dimensions of layout and no t the depth/vertical dimensions

  - Technology shrinks
    - Number of Design Rules are increasing
    - Complexity of Routing Rules is increasing
    - Increasing the number of objects involved
    - More Design Rules depending on width, Halo, Parallel Length

  - Violating a design rule might result in a non-functional circuit or low yield

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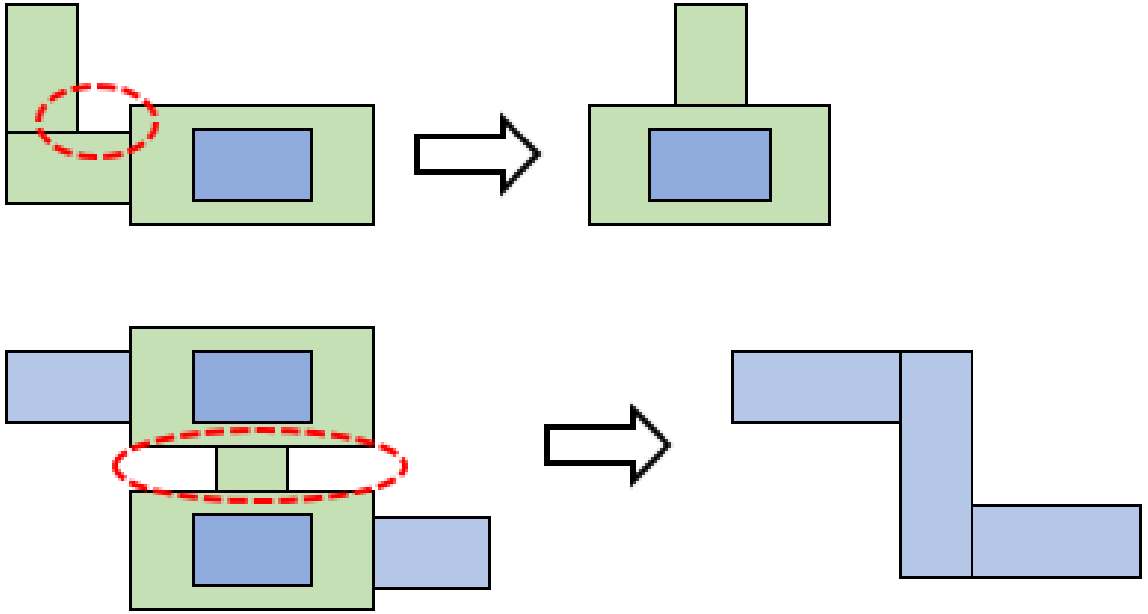
Introduction to PV(Physical Verification) : Overview to DRC

- **Example of Design Rules**
  - **Maximum Rules**

    Manufacturing of large continuous regions can lead to stress cracks (wide Metal must be slotted)
  - **Angles**

    Usually only multiples of 45 degree are allowed
  - **Grid**

    All corner points must lie on a minimal grid, otherwise an “off-grid” Error is produced
  - **Notch spacing violation**



- **Notch**

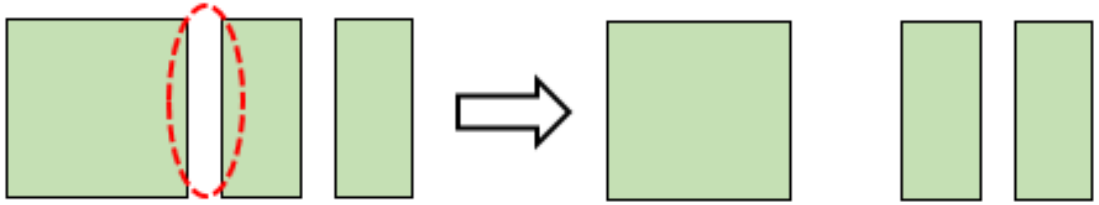
  The rule specifies the minimum spacing rule for objects on the same net, including defining the minimum notch on a single-layer, merged object

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Introduction to PV(Physical Verification) : Overview to DRC

- **Example of Design Rules**

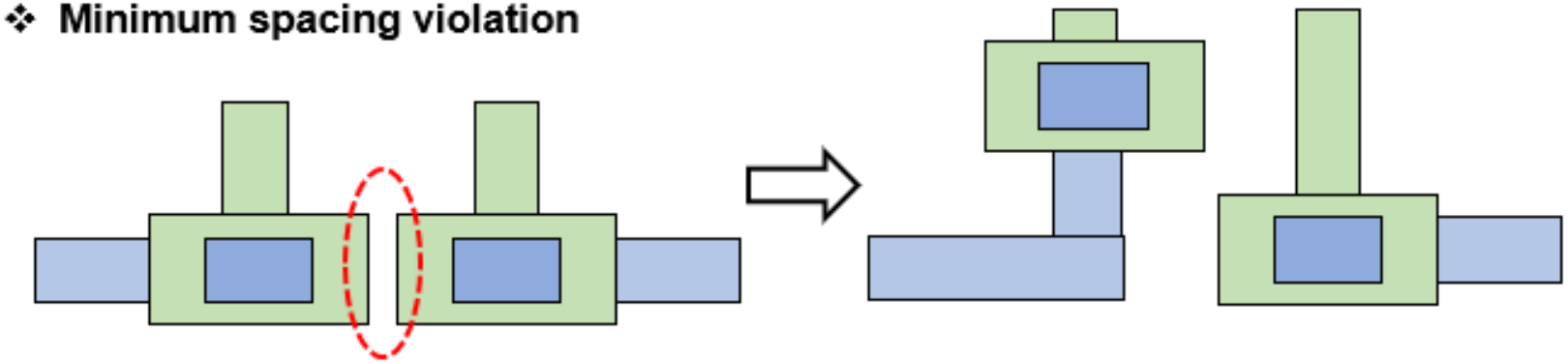
❖ **Thin & Fat spacing violation**



- **Thin & Fat spacing**

  The minimum spacing between objects on Thin & Fat single-layers

❖ **Minimum spacing violation**



- **Minimum spacing**

  The minimum spacing between objects on a single-layers

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Introduction to PV(Physical Verification) : Overview to DRC

- **Example of Design Rules**

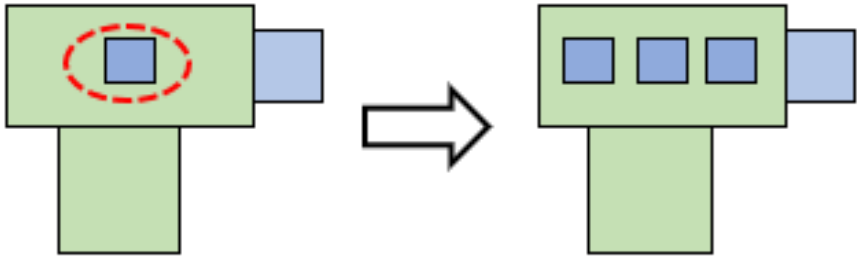
❖ **Minimum width violation**



- **Minimum Width**

  The minimum width rule specifies the minimum width of individual shapes on a single layer

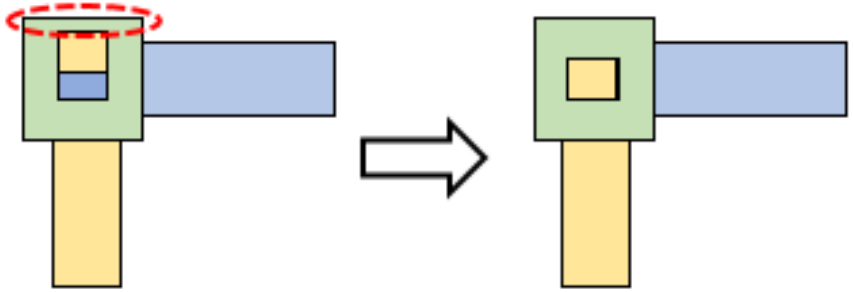
❖ **Minimum cut violation**



- **Minimum Cut**

  The minimum number of cuts a via must have when it is on a wide wire

❖ **Minimum Enclosure violation**



- **Minimum Enclosure/Overlap**

  Implies that the second layer is fully enclosed by the first one

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Introduction to PV(Physical Verification) : Overview to DRC

- **Categories of DRC (Base Layer DRC & Metal Layer DRC)**
  - **Base Layer DRC**
    - Base Layer is all layers up to contact/metal1 (In Base Layer, DRC flow rules will be checked)
    - Base Layer DRCs are spacing rules for geometries inside transistor (WELL spacing, POLY spacing, POLY width)
    - TAP cell requirement Error
    - Well continuity (After routing fill empty space using Filler cells)

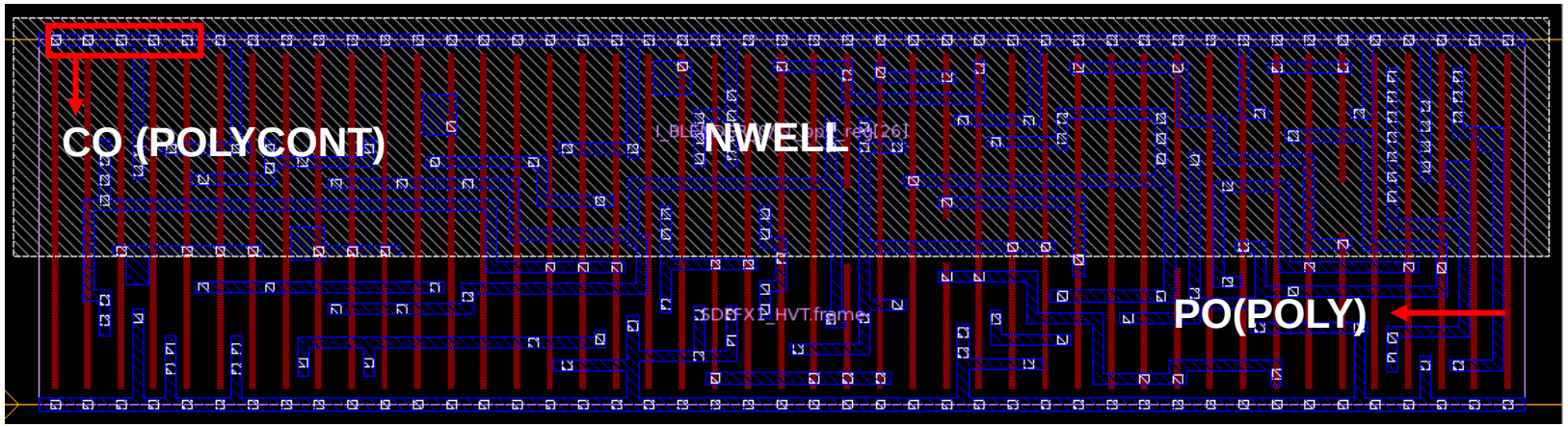
    - **How to fix Base Layer DRC**
      - Make sure design is placed legally
      - No cell overlaps & no gaps

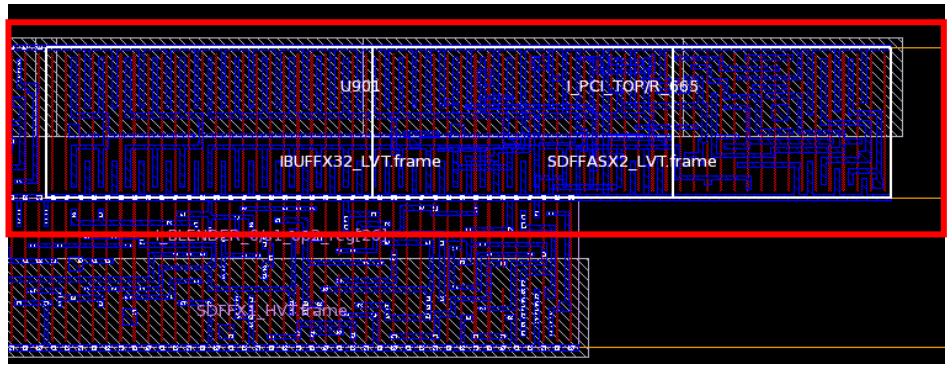
  - **Metal Layer DRC**
    - Basic Layer means from contact to all the routing layers
    - **Basic Metal Layer DRC**
      - Width (min & max)
      - Spacing (min)
      - Via enclosure (size of minimum cut, minimum via to via spacing in multi cut Vias)

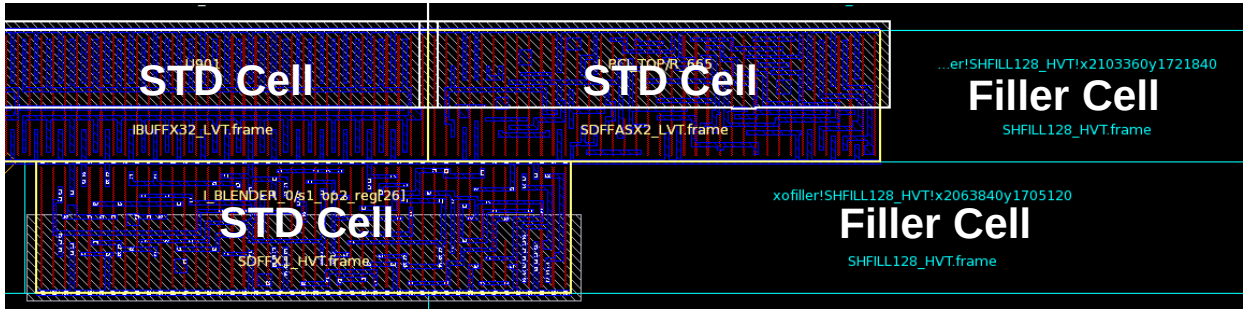
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Introduction to PV(Physical Verification) : Overview to DRC

- **Categories of DRC (Base Layer DRC & Metal Layer DRC)**
  - **Base Layer DRC**







- **No cell overlaps**
  Overlapping of cells can cause internal layers to short each other

- **No gaps**
  For continuity of the WELL layer, there should be no gap between all cells

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Introduction to PV(Physical Verification) : Overview to DRC

- **Categories of DRC (Base Layer DRC & Metal Layer DRC)**
  - **Antenna Rule DRC**
    - **What is Antenna Effect in VLSI**
      - The Antenna effect in VLSI, also known as plasma-induced gate-oxide damage or plasma-induced damage, is one of them
      - In VLSI “Antenna effect” refers to the charge collecting effect, not to the antenna device
      - A Large quantity of charge is often during plasma etching and other procedures
    - **Antenna Rule**
      - The Antenna Rule specifies the Max tolerance of the ratio of a Metal line area to the area of connected gates
    - **Antenna violation**
      - Antenna Violation occurs when the Antenna ratio exceeds the value specified in the PDK (Process Design Kit)
      - Long Metal lines and Vias are occurred Antenna violations
      - VLSI process starts from the (substrate, device Layer, Metal Layers)
      - These charges cause a high voltage spike, which may damage the gates connected to the Metals

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Introduction to PV(Physical Verification) : Overview to DRC

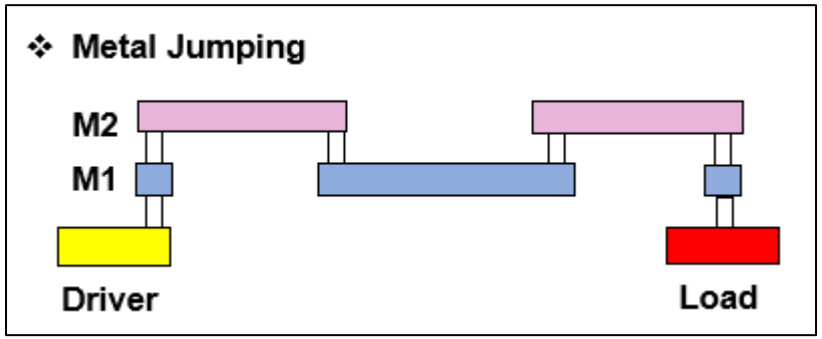
- **Categories of DRC (Base Layer DRC & Metal Layer DRC)**
  - **Antenna Rule DRC**
    - **3 types of Antenna violation**
      - **Metal area antenna rule**

        The max limit to the ratio of the metal line area to the connected gates area
      - **Perimeter antenna rule**

        The max limit to the ratio of the metal line perimeter connected gates area
      - **Via or contact area**

        The max limit to the ratio of the via or contact area to the connected gates area

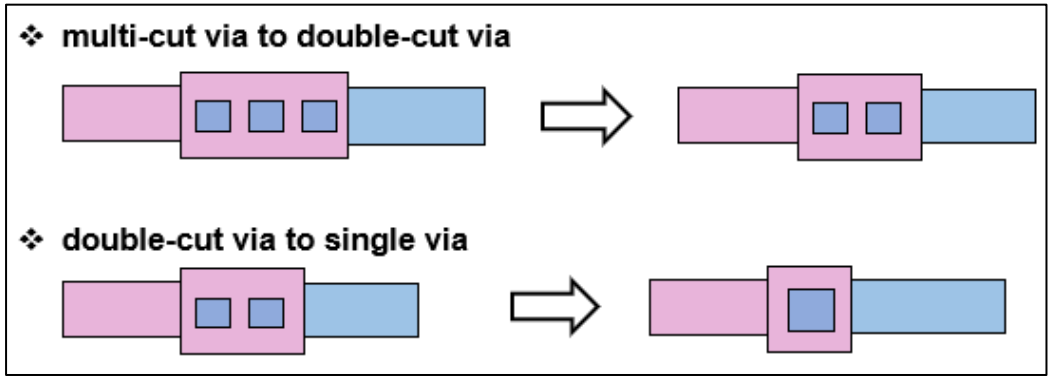
    - **Fix Antenna violation**
      - **Routing on higher Metal Layer**
      - ❖ Long Metal can be taken to higher Metal Layer (Metal jumping)
      - ❖ Metal jumping will break the long interconnect and hence the charge collected on the long interconnect will not discharge through gate oxide



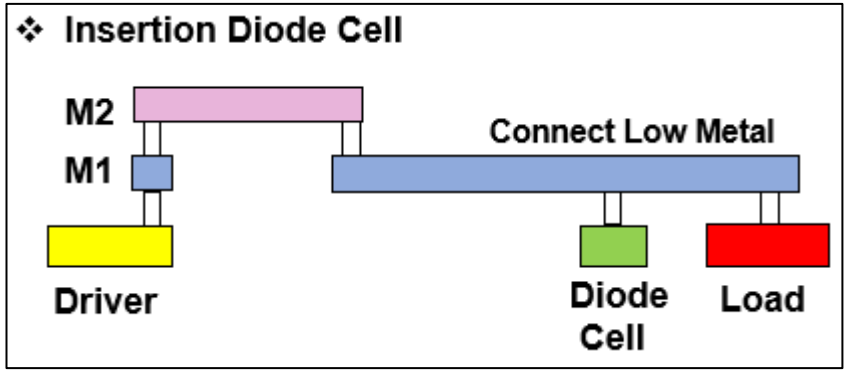
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Introduction to PV(Physical Verification) : Overview to DRC

- **Categories of DRC (Base Layer DRC & Metal Layer DRC)**
  - **Antenna Rule DRC**
    - **Fix Antenna violation**
      - **Reduce the area of Vias**
        - ❖ Large via area occurred Antenna violation
        - ❖ Converting Multi-cut vias to double-cut or single-cut via reducing the area



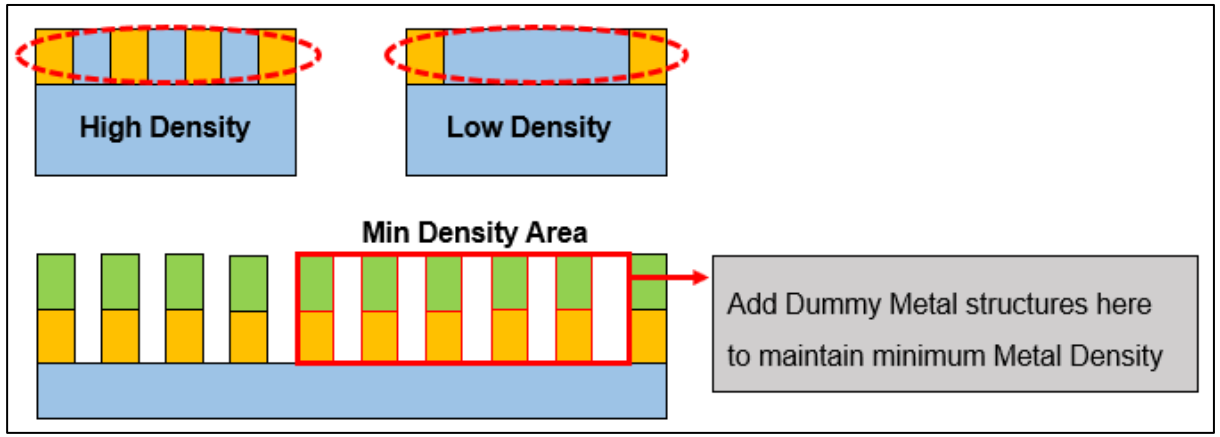
      - **Insertion Diode Cell**
        - ❖ Diode helps dissipate charges accumulated on Metal Layer
        - ❖ Diode should be placed as near as possible at the gate of device on Low Metal
        - ❖ Connect diode to the gate electrode which provides a discharging path for the static charge present on the Metal Layer



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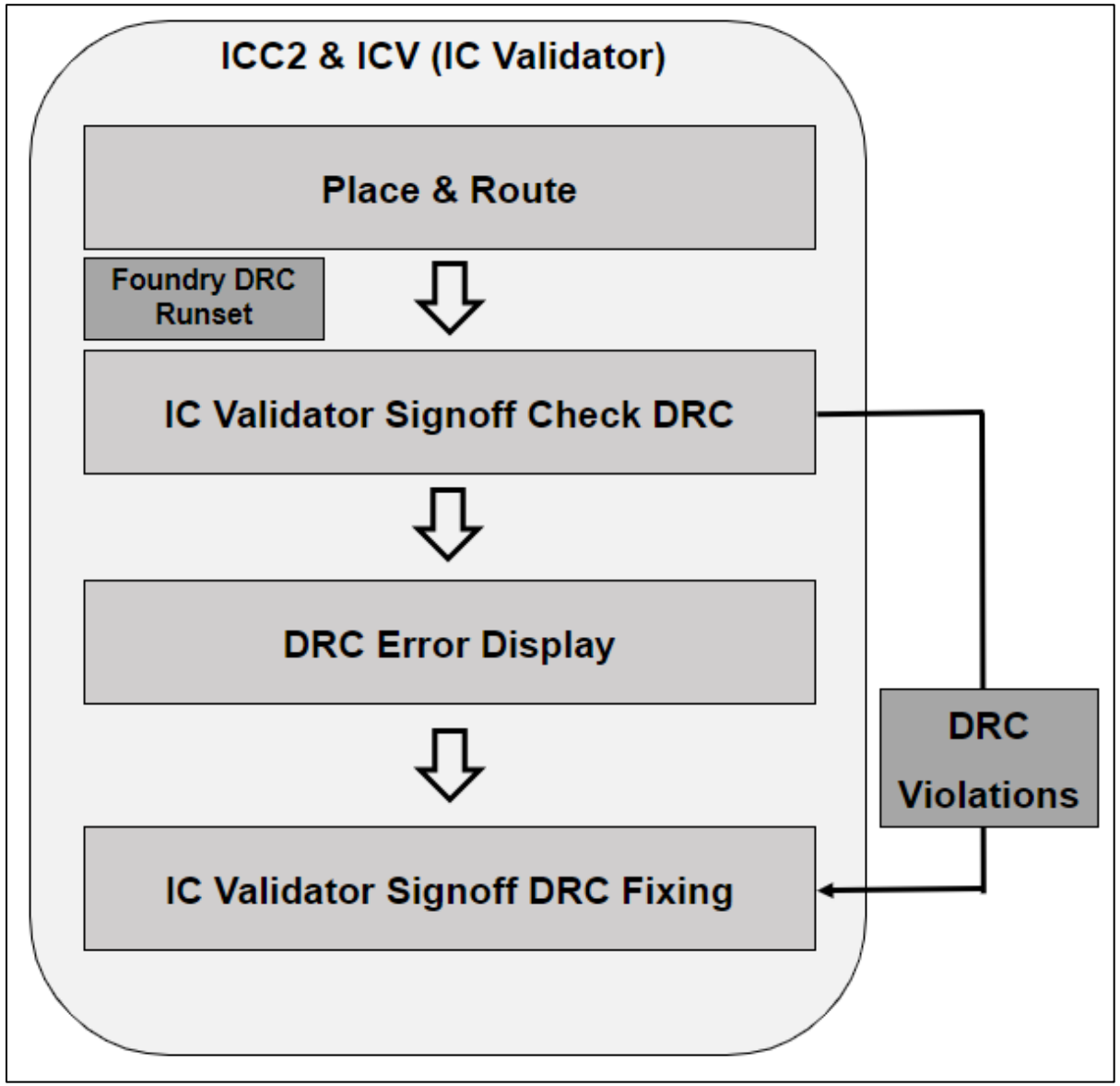
Introduction to PV(Physical Verification) : Overview to DRC

- **Categories of DRC (Base Layer DRC & Metal Layer DRC)**
  - **Metal Density DRC**
    - **Metal Density**
      - Density of Metal usage has a Min and Max constraint
      - Limits are typically 30% < Density < 80% for Metal
      - Poly also has density requirements (typically 15% < Density < 80%)
    - **Metal Density (Min & MAX)DRC**
      - Density check is performed to check the even Metal density through out the chip which required for manufacturing process to ensure th e mechanical sturdiness of the chip to achieve planarity during CMP (Chemical Mechanical Polishing)
      - In the case of Min Density violation, the violation can be fixed by filling the empty area with the Dummy Metal Layer after the Metal Fill
      - In the case of Max Density violation, after the Metal Fill, the violation can be fixed by deleting the Metal Layer in the area where the violation occurred



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Introduction to PV(Physical Verification) : Setup for DRC



- **Eliminates the need to stram out the design to run DRC**
- **Signoff quality results using foundry DRC signoff runset**
- **ICV guidance : Directs Zroute to perform targeted DRC fixing (Negligible impact on timing due to limited scope of expected changes)**

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Introduction to PV(Physical Verification) : Setup for DRC

- **DRC checking**
  - Signoff DRC checking on all routing Metal & Via layers, by default
- **DRC fixing**
  - Auto-fix performs sign-off DRC repair
  - The design should be near timing closure with only a small number of DRC violations left
- **ICV (IC Validator) opens the block on disk**
  - Save the block in iCC2, before invoking ICV

```
save_block -all

set_app_options -list {signoff.check_drc.runset "my_runset(runset file name)"}
signoff_check_drc -select_layers {M1 VIA1 M2 ....}

set_app_options -list {
        signoff.fix_drc.init_drc_error_db "signoff_check_drc_run" }

signoff_fix_drc
```



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Introduction to PV(Physical Verification) : Setup for DRC

- **Full-chip DRC checking (Metal & Via Layers) or specific rules**

```
signoff_check_drc
signoff_check_drc -select_rules {"M1*" "M3.s.*"}
```

- **Use design view or merge GDSII/OASIS, for double-pattern DRCs**

```
set_app_options -list {signoff.check_drc.read_design_views {*}}

                        - or -

set_app_options -list {signoff.physical.merge_stream_files \
                       {ORCA_TEST.gds ORCA_TEST.oas} }

signoff_check_drc [-check_all_runset_layers true]
```

- **For full-layer signoff DRC checking, merge full-layer GDSII or OASIS stream files with existing NDM libs**

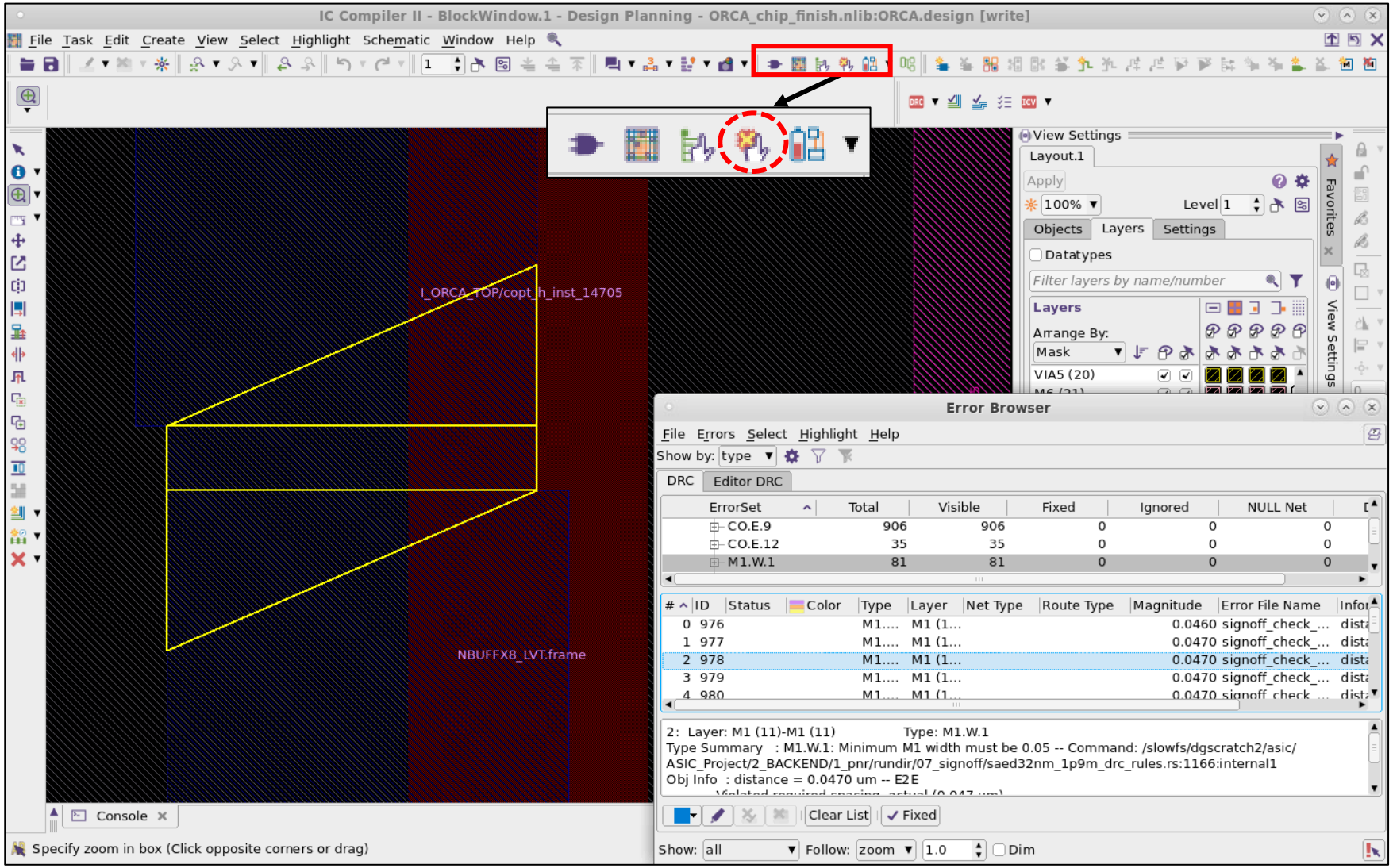
- **If .tech file and runset (GDSII) layer names differ**

```
set_app_options -list {signoff.physical.layer_map_file "map_filename"}
```


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Introduction to PV(Physical Verification) : Setup for DRC

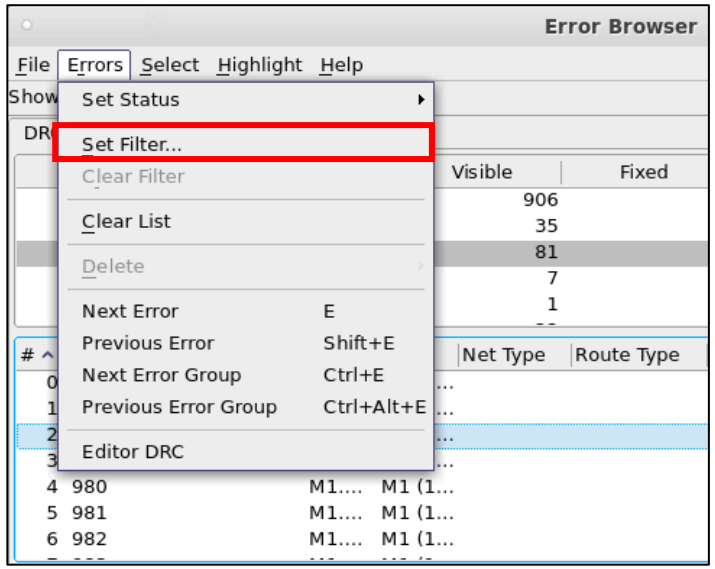
- **Viewing Error Data in the Error Browser in ICC2**

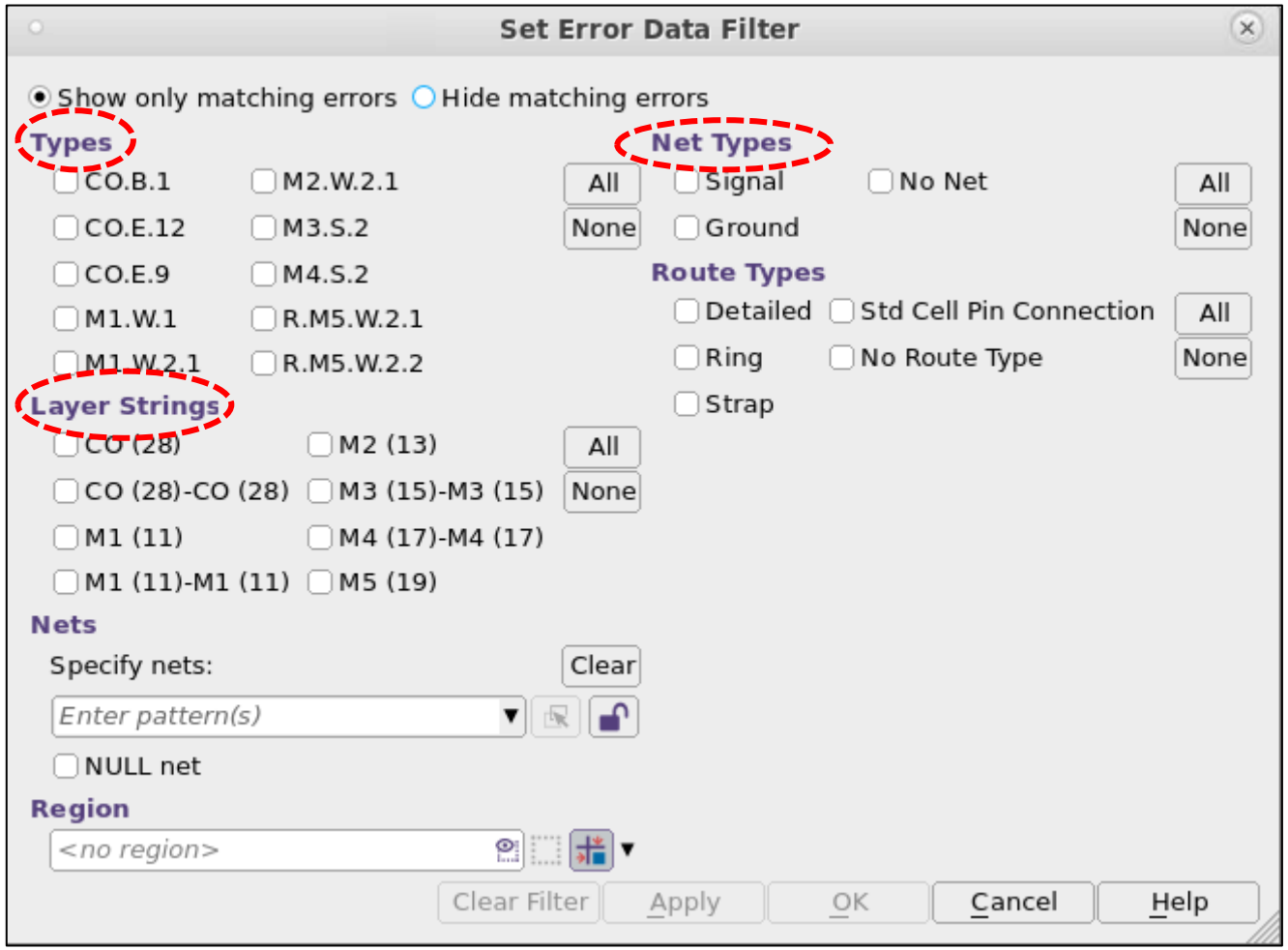


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Introduction to PV(Physical Verification) : Setup for DRC

- **Viewing Error Data in the Error Browser in ICC2**

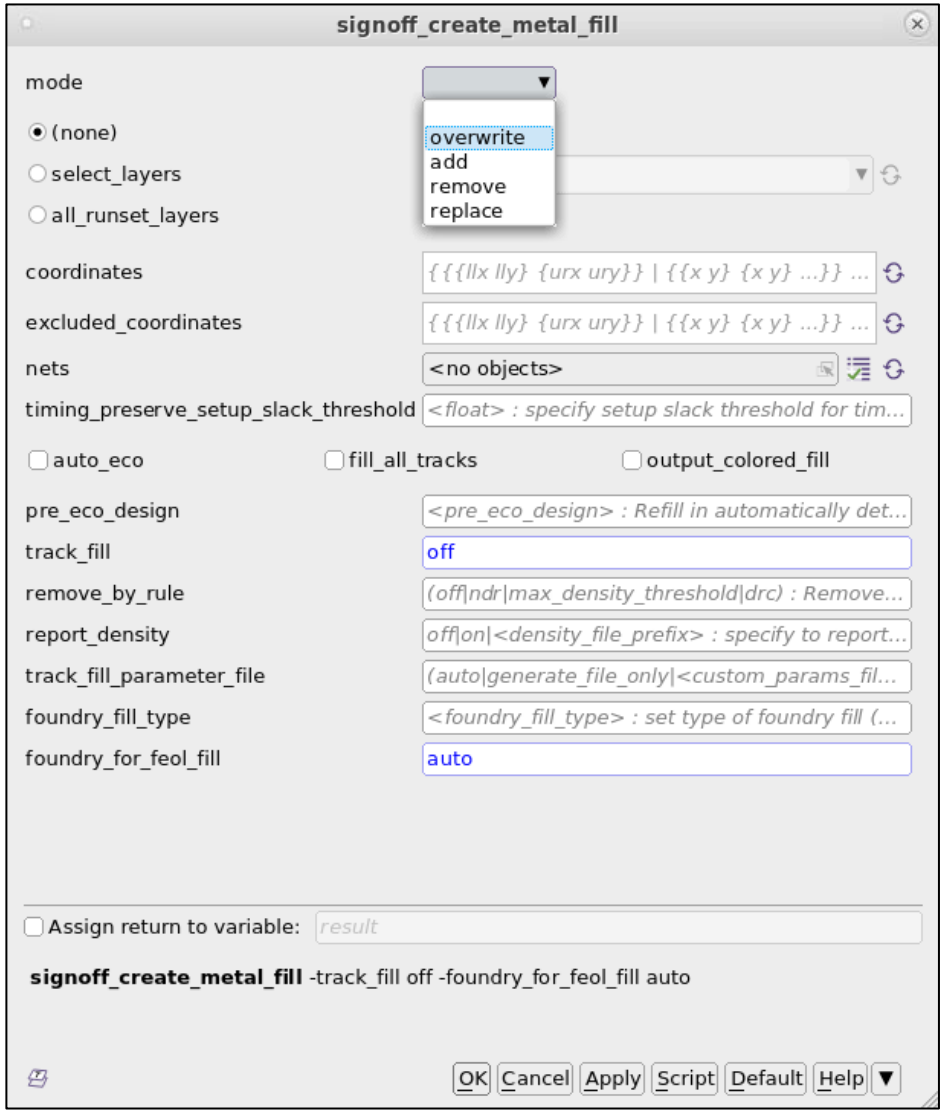




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Introduction to PV(Physical Verification) : Setup for Metal Fill

- **Full-chip Metal Fill insertion & removal using IC Validator**
  - Pattern-based flow (user-specified runset)
  - Track-based fill flow (automatically-generated runset)
- **Timing-driven Metal Fill**
- **Auto-eco : Automatically refill changed areas**
  - If route changes are > 20% (default), full Metal Fill is run



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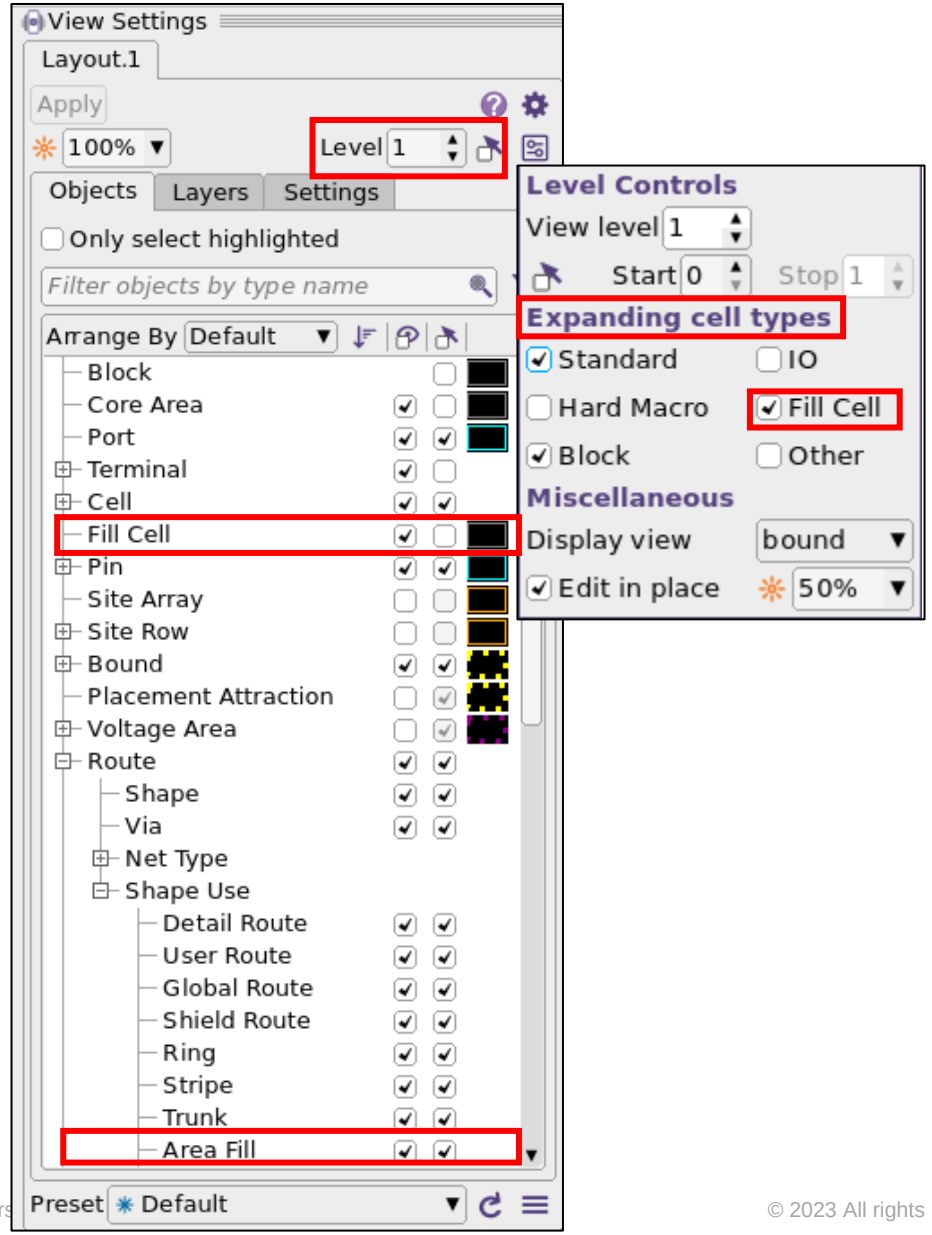
Introduction to PV(Physical Verification) : Setup for Metal Fill

- **Viewing & Editing Metal Fill**
  - **In the View Settings panel :**

1. Set the view level to 1
2. Enable "Fill Cell" under Expanding cell types
3. Set "Fill Cell" visible (default setting)
4. Set "Area Fill" visible (default)

  - **To select & edit shapes:**

1. Check "Multiple levels active" checkbox
2. Delete, resize, etc...



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Introduction to PV(Physical Verification) : Setup for Metal Fill

- **Example script of Metal Fill**

```
#set app_option for Metal Fill
set_app_options -list {
        signoff.create_metal_fill.runset {./runset/saed32nm_1p9m_mfill_rules.rs} \
        signoff.create_metal_fill.read_design_views {*} }

                        -or-

        signoff.physical.merge_stream_files {stream_file1.gds stream_file2.oas}

#create Metal fill
signoff_create_metal_fill \
        -timing_preserve_setup_slack_threshold 0.05

#Timing analysis shows new small violations
#remvoe Filler Cells, re-run roue_opt, then re-insert Filler Cells

#Run auto-eco Metal Fill
signoff_create_metal_fill \
        -auto_eco true \
        -timing_preserve_setup_slack_threshold 0.05
```

- **For full-layer signoff DRC checking, merge full-layer GDSII or OASIS stream files with existing NDM libs**

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Introduction to PV(Physical Verification) : Overview to LVS

- **What is LVS (Layout Versus Schematic)**
  - LVS(Layout Versus Schematic) verifies the connectivity of a Verilog Netlist and Layout Netlist (Extracted Netlist from GDS after routing)
  - Tool extracts circuit devices and interconnects from the Layout and Saved as SPICE format Layout Netlist
  - LVS performs comparison between 2 Netlist (Verilog, SPICE), it does not compare the functionalities of both the Netlist

- **LVS input Requirements**
  - LVS Rule deck
  - Verilog Netlist
  - SPICE Netlist
  - Physical Layout Database (GDS)

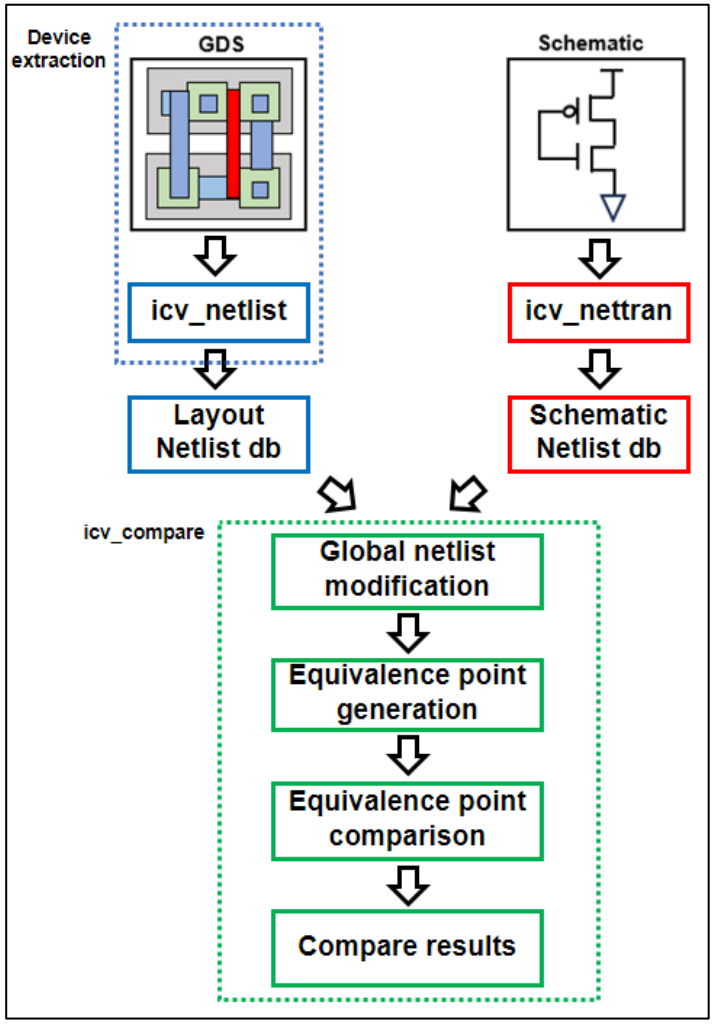
  - **LVS rule deck**
    - LVS rule deck is a set of code written in SVRF (Standard Verification Rule Format) or TVF (TCL Verification Format)
    - This file guides the tool to extract the devices and the connectivity of IC
    - Rule deck contains the layer definition to identify the layers used in layout file and to match it with the location of layer in GDS file (It also contains device structure definitions)

- **LVS check list**
  - Short Error (Net), Open Error (Net), Extract Error, Compare Error

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Introduction to PV(Physical Verification) : Overview to LVS

- **Flow of LVS (Layout Versus Schematic)**
  - **A Layout Netlist (hierarchical) is extracted from the physical layout of the**

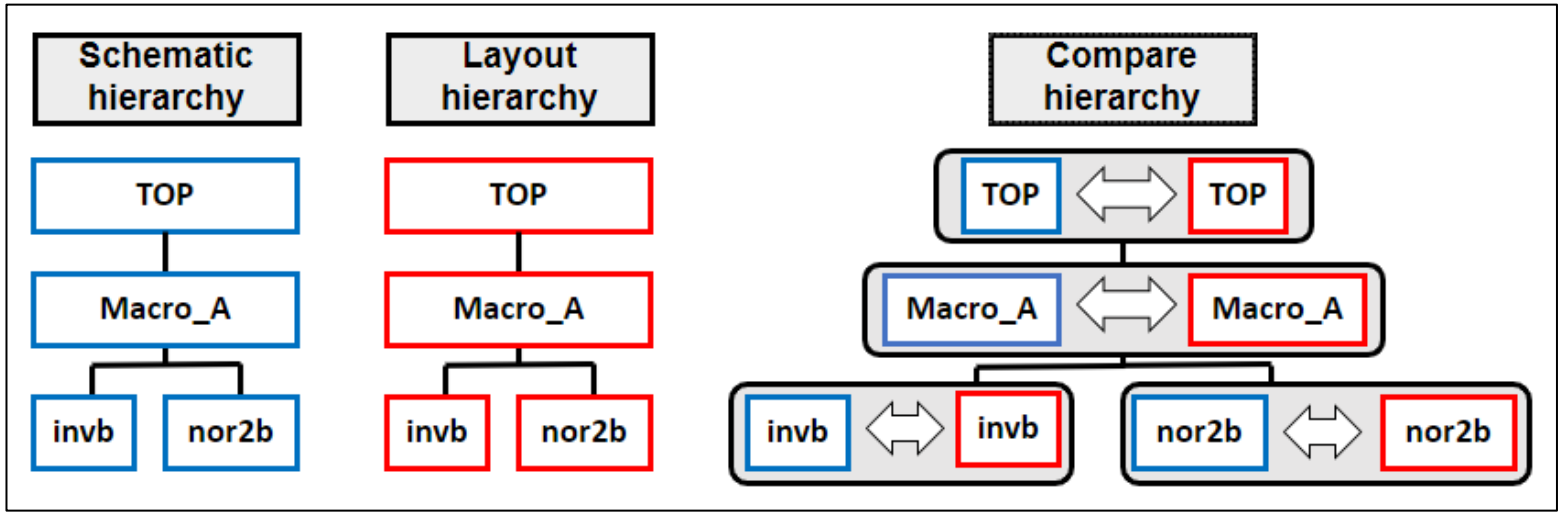


- **A Layout Netlist (hierarchical) is extracted from the physical layout of the physical design during device extraction**
  - ❖ ***Devices to be extracted are defined in the runset file***
- **A schematic netlist is converted into an ICV format using the ICV Nettran utility**
- **Once both netlists are generated, ICV will perform global netlist modifications**
- **ICV groups cell pairings from each netlist to be compared (Equivalent Cell Pairings)**

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Introduction to PV(Physical Verification) : Overview to LVS

- **Flow of LVS (Layout Versus Schematic)**
  - **LVS comparison engine verifies the logical equivalence of each Equivalence Cell Pairing**



  - **The compare engine performs bottom-up validation of each pairing**
  - **Optional device filtering and merging operations can also take place during the compare stage**
  - **Once all compare settings are configured, ICV compares the logic of the schematic and layout(gds) cell**

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Introduction to PV(Physical Verification) : Overview to LVS

- **Flow of LVS**
  - **Extraction**
    - ICV Tool takes GDSII file containing all the layers and uses polygon based approach to determine the components like (transistors, diodes, capacitors, registers) and connectivity information between devices presented in the layout by their layers of construction
    - Tool Extraction about All device layers, terminals, size of device, nets, vias, pin location and given an unique identification
  - **Reduction**
    - All defined informations are extracted in the form of netlist (spice Netlist)
  - **Comparison**
    - The extracted Layout Netlist (After routing) is then compared to the netlist of the same stage using LVS rule deck (runset file)
    - In Comparison stage, the number of instances, nets and ports are compared
    - All the mismatches such as shorts, opens, and pin mismatches are reported (The ICV also checks topology and size mismatch)

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Introduction to PV(Physical Verification) : Overview to LVS

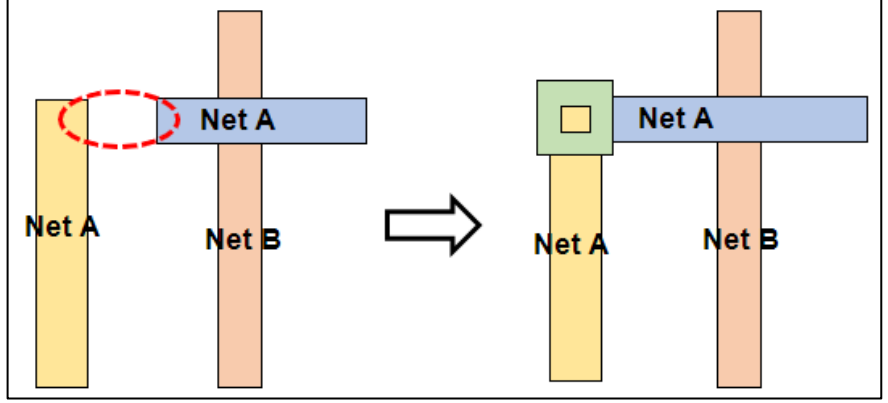
- **Types of LVS Errors**
  - **shorts**
    - two or more wires which should be connected together are connected
  - **Opens**
    - Wires or components which should be connected together are left floating or partially connected
  - **Component mismatch**
    - Components of different types are used in design (RVT cells instead of HVT cells)
  - **Component missing**
    - If an expected component is left out from the layout
  - **Parameter mismatch**
    - All components has it’s own properties in design
    - ICV is configured to compare these properties with some tolerance
    - If tolerance is not MET, then it will occur parameter mismatch

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Introduction to PV(Physical Verification) : Overview to LVS

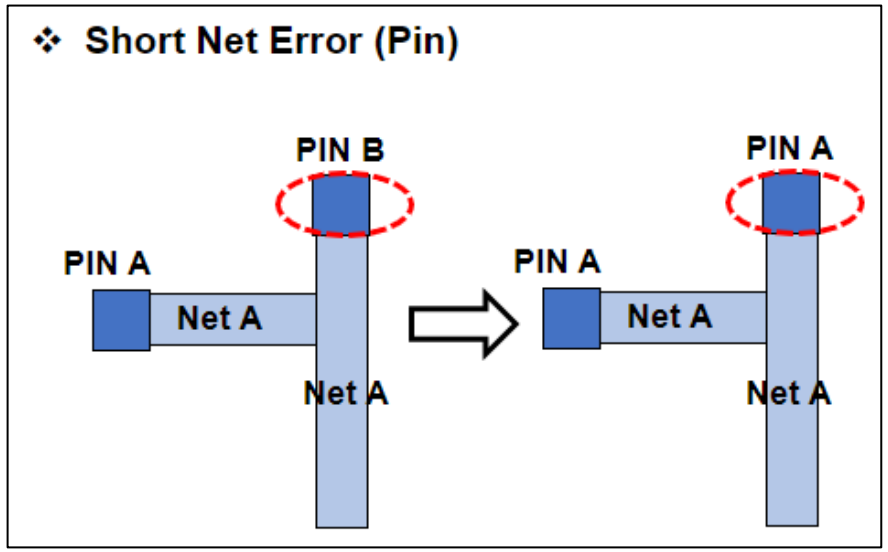
- **Type of LVS Errors**

❖ Open Net Error



Same net is routed in two different Metal Layers but not connected each other

❖ Short Net Error (Pin)



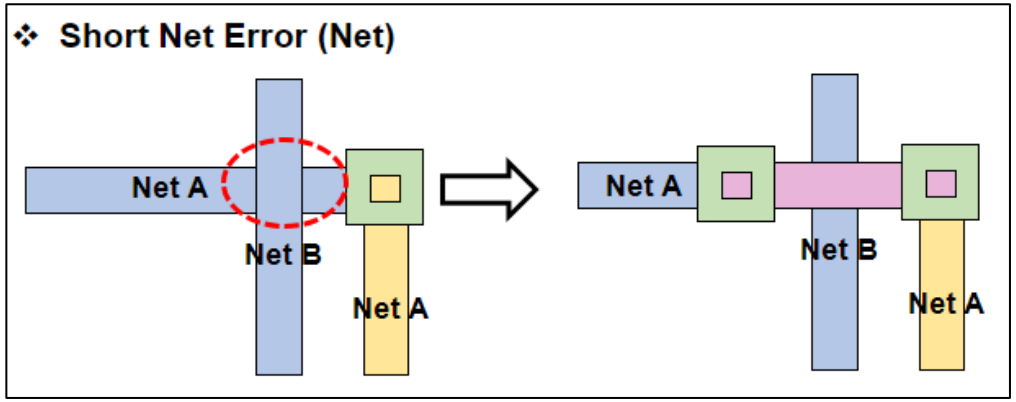
Same net with different pin names

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Introduction to PV(Physical Verification) : Overview to LVS

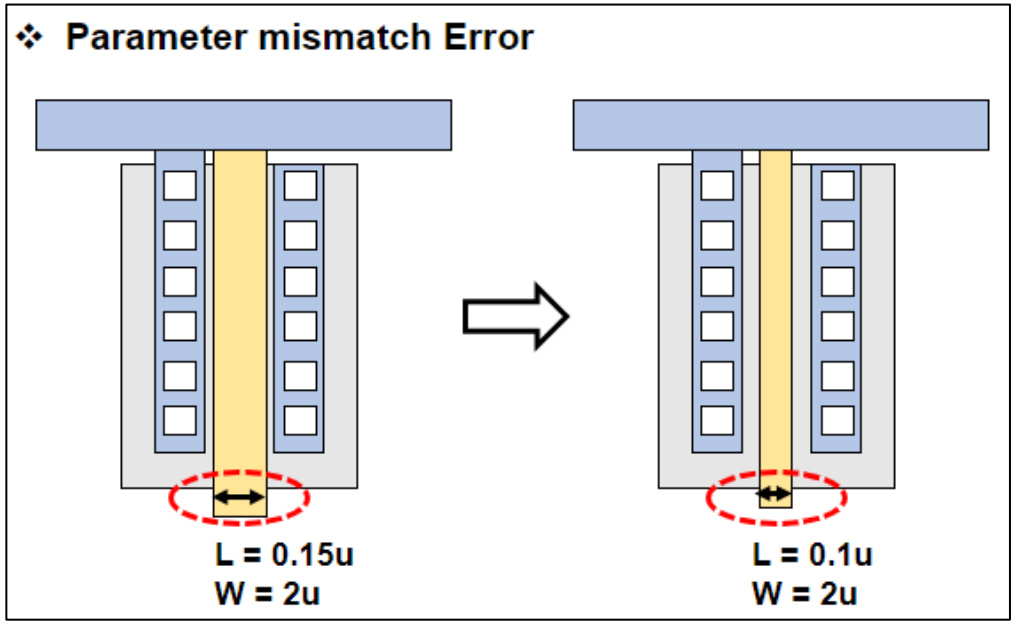
- **Type of LVS Errors**

❖ Short Net Error (Net)



Two different nets short together

❖ Parameter mismatch Error



Extract Error

- Parameter Mismatch Error
- Device parameters on schematic & layout are compared
- LVS checks are necessary parameters like width, length, multiplication factor etc ...

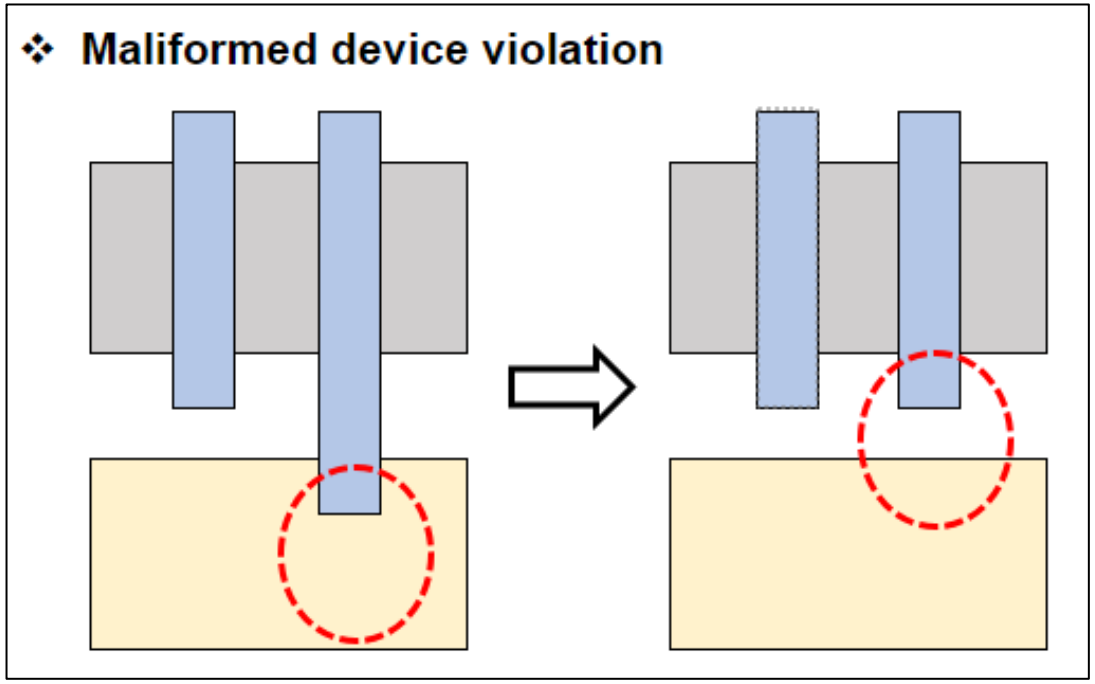
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Introduction to PV(Physical Verification) : Overview to LVS

- **Type of LVS Errors**

Compare Errors

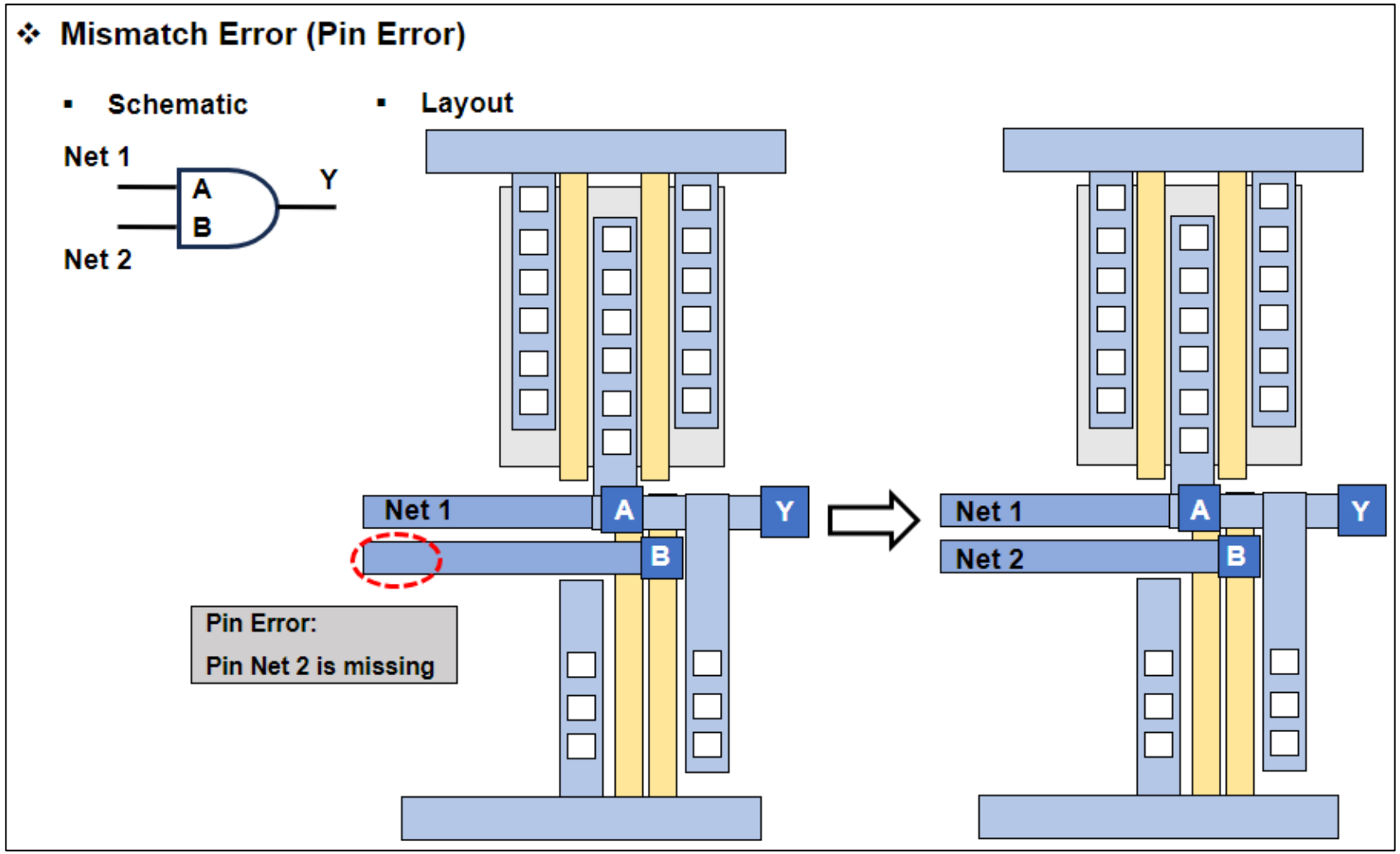
- Malformed Devices
- Pin Errors
- Device Mismatch
- Net Mismatch



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Introduction to PV(Physical Verification) : Overview to LVS

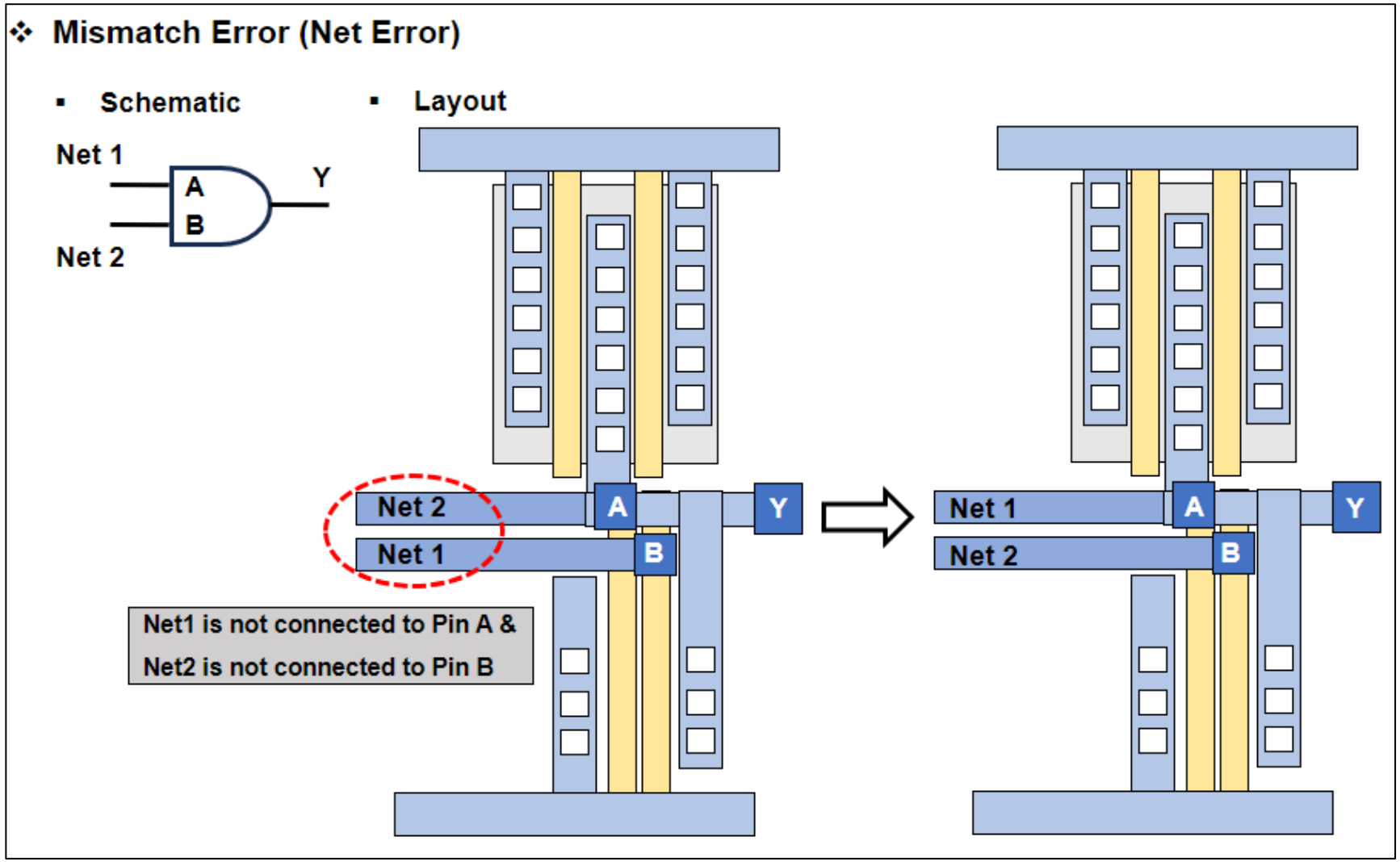
- **Type of LVS Errors**



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Introduction to PV(Physical Verification) : Overview to LVS

- **Type of LVS Errors**



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Introduction to PV(Physical Verification) : Run LVS using ICV

- **LVS Input files**
  - **Input Layout**

    GDSII, OASIS, NDM, OpenAccess, or Milkyway format
  - **Schematic Netlist**

    SPICE, VERILOG & ICV format
  - **LVS Runset**

Runset Section	Description
Run Configuration	Control how input data is processed
Layer Assignment	Maps layers in the layout to a polygon layer variable
Layer Derivation	Defines device layers
Device configuration	Configures the devices to be extracted
Device extraction	Extracts the physical layout to a layout netlist
Merge, Filter & Compare	Configures the compare settings and performs the layout vs schematic check

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Introduction to PV(Physical Verification) : Run LVS using ICV

- **Example of Run LVS using ICV (IC Validator)**
  - **lvs_rules.rs (runset file)**

```
/*Defined all needed variables*/



DECK_TYPE : string = "PEX_DECK";
CROSS_REFERENCE : string = "yes";
ICV_RUN_FILE_DIR: string = ".";


openaccess_options(
    view="layout",
    layer_mapping_file= "./Include/saed32nm_1p9m_gdsout.map"
    );



library(
library_name= "./Include/ORCA_TOP.gds",
format=GDSII,
cell= "ORCA"
);

layout_netl_fh:handle;
device_db:device_database;
xref_db_h:xref_database_handle;

schem_netl_fh = schematic(

schematic_file    = {{filename = "./Include/ORCA.lvs",format = SPICE}}

);
```

- **Runset file**

  Runset file contains a full set of conditions for a program to be run (extraction, simulation), plus the locations of all necessary (input & output) files

- **Layer mapping file**

  Layer map file maps the "GDS II" stream layer numbers and datatypes to "df II" layer names and purpose pairs

- **GDS II**

  GDS II is a database file format which is the industry standard for data exchange of integrated circuit or IC layout (extract after P&R flow)

Top Cell Name

- **SPICE Netlist**

  A SPICE Netlist is a text-based representation of a circuit

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Introduction to PV(Physical Verification) : Run LVS using ICV

- **Example of Run LVS using ICV (IC Validator)**
  - **lvs_rules.rs (runset file)**

```
error_options(db_path="./TOPCELLNAME_err",
create_vue_output=true );

run_options(
instance_prefix="X",
uppercase=false
);

hierarchy_options(flatten={"*"});

resolution_options(snap_resolution=0.001);
text_options(
        net_prefix="N",
        layout_power={"VDD", "AVDD", "DVDD" ,"VDDIO"},
        layout_ground={"VSS","AVSS", "VSSIO"}
);
net_options(
schematic_power={"VDD", "AVDD", "DVDD" ,"VDDIO"},
schematic_ground={"VSS","AVSS", "VSSIO"},
schematic_global={"VDD", "AVDD", "DVDD" ,"VDDIO","VSS","AVSS", "VSSIO"}
);
prototype_options(false);
```

Layout Power & Ground list

Schematic Power & Ground list

```
if( DECK_TYPE=="LVS_DECK" || DECK_TYPE=="PEX_DECK" ) {

  xref_db_h =  compare(
        state = compare_settings,
        schematic=schem_netl_fh,
        layout=layout_netl_fh,
        user_functions_file ="./Include/func.rs",
        memory_array_compare =(DECK_TYPE=="PEX_DECK")?(false):(true),
        //remove_equiv_sum_files = NONE,
        write_equiv_netlists   = ALL,
        action_on_error    = EXPLODE,
        remove_dangling_net = LAYOUT_UNTEXTED,
        multiply_width = false,
        push_down_pins = false
        //generate_equiv= {FULL_NAME, CASE_INSENSITIVE_NAME}
    );
};
```

- **User function file**
- In digital logic, this was too restrictive since exactly the same function can be implemented in many different ways
- LVS has been augmented by foraml equivalence checking, which checks whether two circuits (GDS II, SPICE Netlist) perform exactly the same function

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Introduction to PV(Physical Verification) : Run LVS using ICV

- **Example of Run LVS using ICV (IC Validator)**

```
#Run LVS

icv -host_init 2 -vue -vueshort -create_lvs_short_output -c <cell name> -i <library_name (ex .gds)> \
-s <schematic file> -sf <schematic_format (SPICE | VERILOG | ICV)> <runset path>
```

❖ **Command line options**

-i library_name	• ICV uses cell as the name of the block to be checked • It also overrides the cell name specified in the library function
-f format	• Input library/design that contains the cell to be compared • It overrides the library_name file specified the library function in runset file
-s schematic file	• Input library format (Overrides format specified in library function in runset file)
-sf schematic_format	• Specifies input schematic netlist format (SPICE | VERILOG | ICV)
-stc cell_name	• Top cell name of schematic file • If this is not specified then the same cell name from "-c" command line will be considered as top cell of schematic file
-host_init <no. of cups>	• Specifies the number of cores required to perform DRC flow
-vue	• Creates VUE output for debugging

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Introduction to PV(Physical Verification) : Run LVS using ICV

- **Example of Run LVS using ICV (IC Validator)**

  ❖ **Command line options**

-create_lvs_short_output [output_limit | all]	• Creates VUE output and generates LVS Short Finder output • You can specify to report only a set number of shorts or all shorts (default : 200)
-vueshort [integer | all]	• Extracts either a specified number of shorts or all shorts (default : 200)
-ex	• ICV will perform only device extraction • Compare function is not invoked
-e file	• Specifies a file which contains *equiv_options* and *lvs_black_box_options*

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Introduction to PV(Physical Verification) : Run LVS using ICV

- **LVS output files (IC Validator)**

  ❖ **Output files**

<cellname>.LVS_ERRORS	• Indicates whether or not the LVS comparison was successful • This file is the starting point for analyzing LVS issues and is generated by the compare functio n
<cellname>.net.gz	• The IC Validator tool creates an extracted layout netlist file • This file is a hierarchical description of the structures found in the layout database
equiv.run	• If you do not supply an equivalence file, then an LVS compare run automatically produces an e quivalence file that is used to compare the layout and schematic netlists
compare	• Detailed output files from an LVS comparison are placed in this directory ❖ Subdirectories are created for each equivalence point in the equivalence file ❖ Subdirectories are named based on the layout structure name

  ❖ **Compare directory structure**

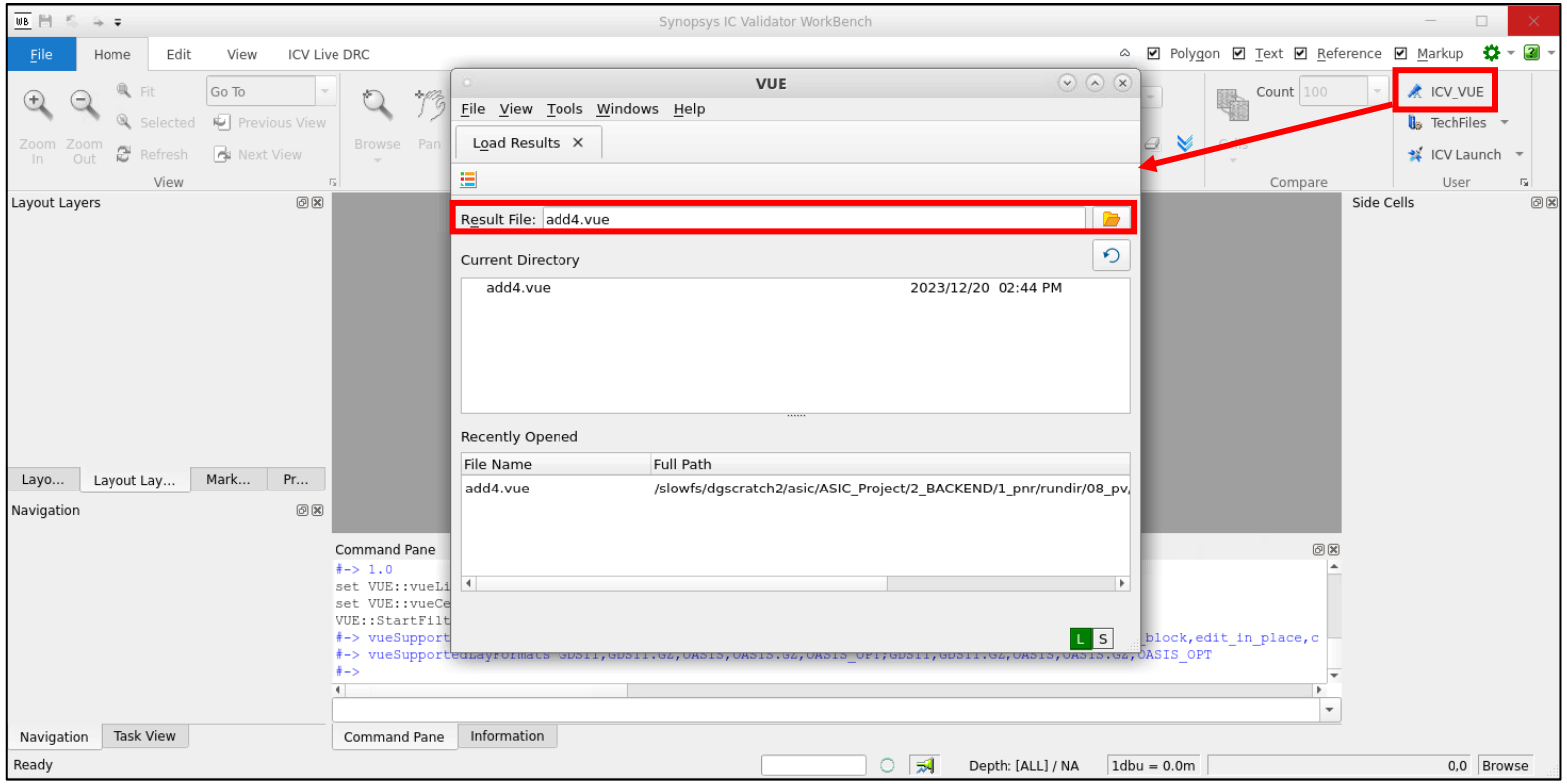
```
  < user@linux compare >> pwd
/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/08_pv/LVS/run_details/compare
  < user@linux compare >> ls
add4  cs_add  invb  nand2b  nand3b  nor2b  nor3b
```

```
  < user@linux add4 >> pwd
/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/08_pv/LVS/run_details/compare/add4
  < user@linux add4 >> ls
lay.add4  lay.add4.gz  sch.add4  sch.add4.gz  sum.add4.add4
```


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Introduction to PV(Physical Verification) : Run LVS using ICV

- **Load LVS Results in VUE**

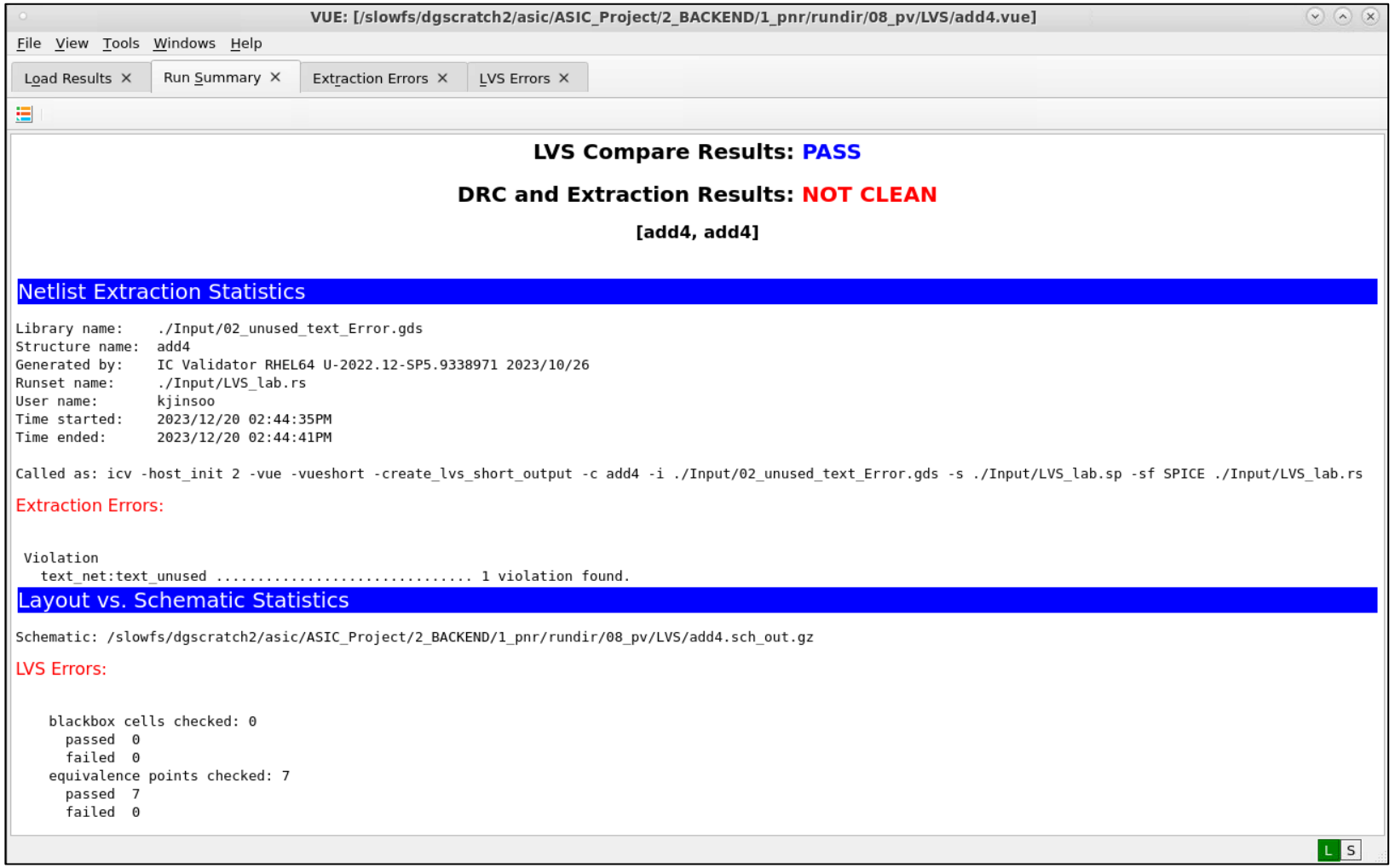


- **Similar to DRC results, LVS results can also be loaded into VUE**
- **Click on "ICV_VUE" button in ICVWB to open VUE window**
- **In the VUE window, load the <cellname>.vue file**
- **Run summary, Extraction Errors and LVS Errors will then be populated**

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Introduction to PV(Physical Verification) : Run LVS using ICV

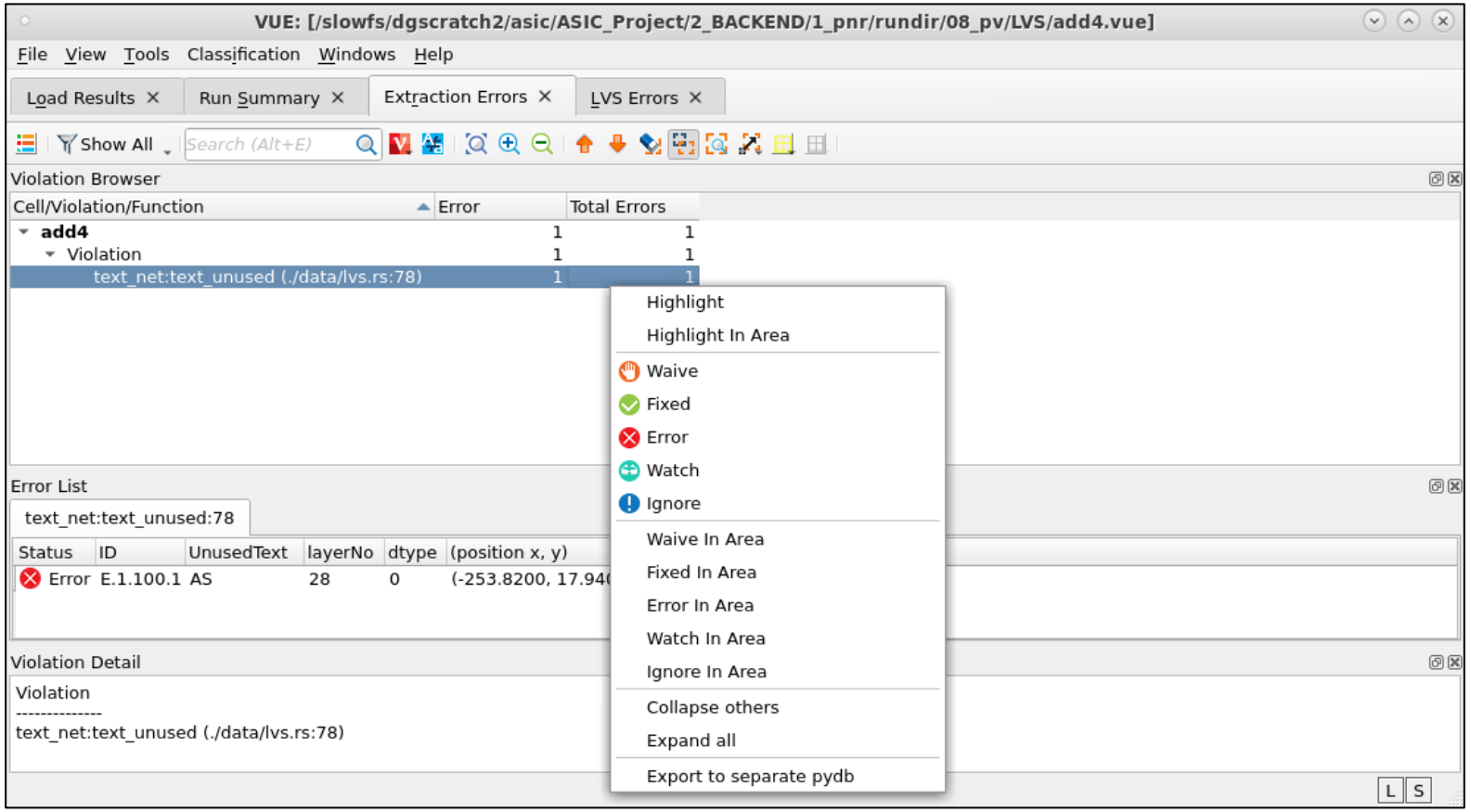
- **Load LVS Results in VUE (Run Summary)**
  - **The Run Summary tab provides details of extraction & comparison results as shown below**



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Introduction to PV(Physical Verification) : Run LVS using ICV

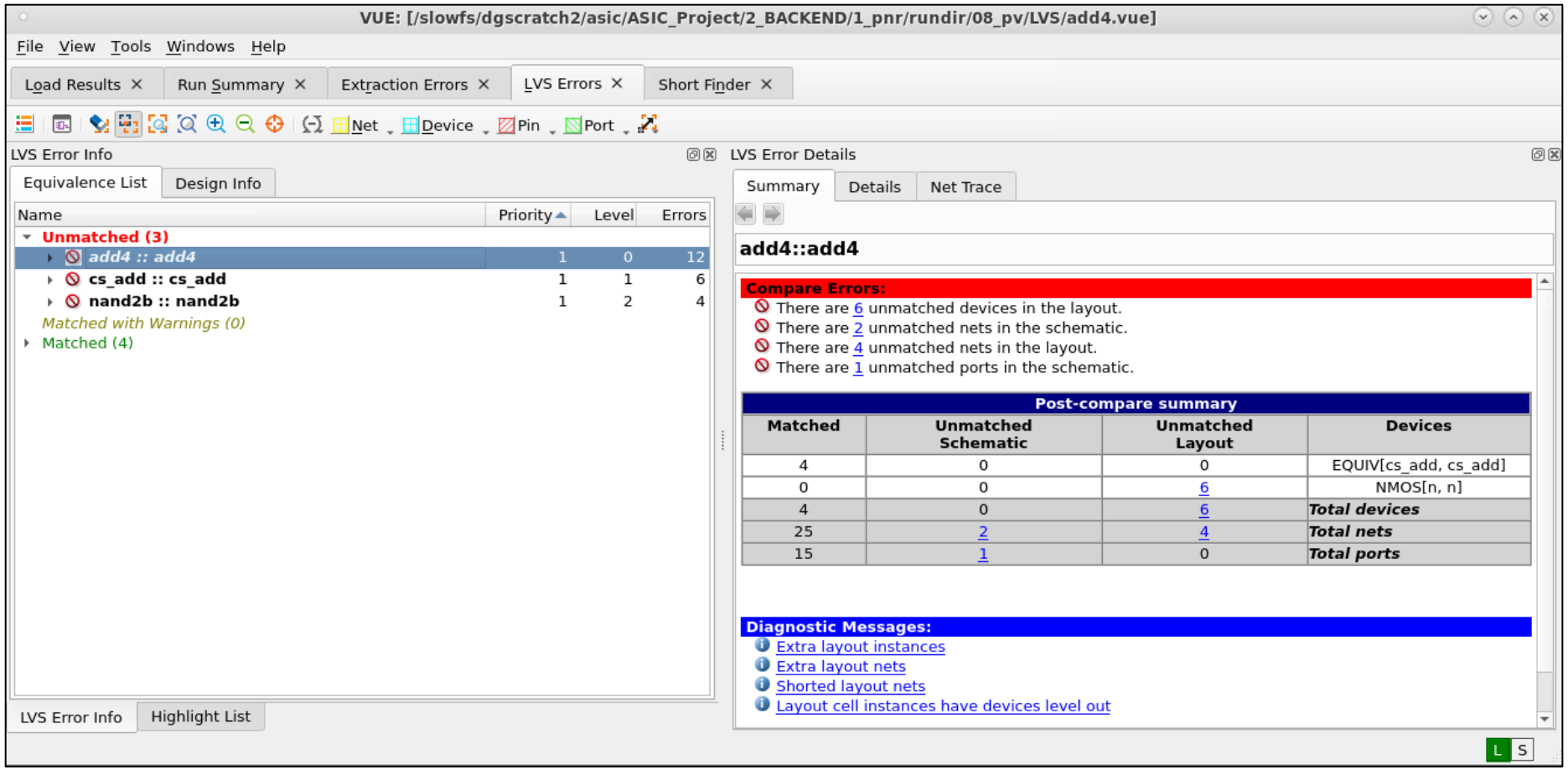
- **Load LVS Results in VUE (Extraction Errors)**
  - **The extraction errors tab provides details of violations found during extraction, like shorts, opens or if there is any missing device te rminals, etc**
  - **You can select and right-click violations to highlight them in ICVWB**



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Introduction to PV(Physical Verification) : Run LVS using ICV

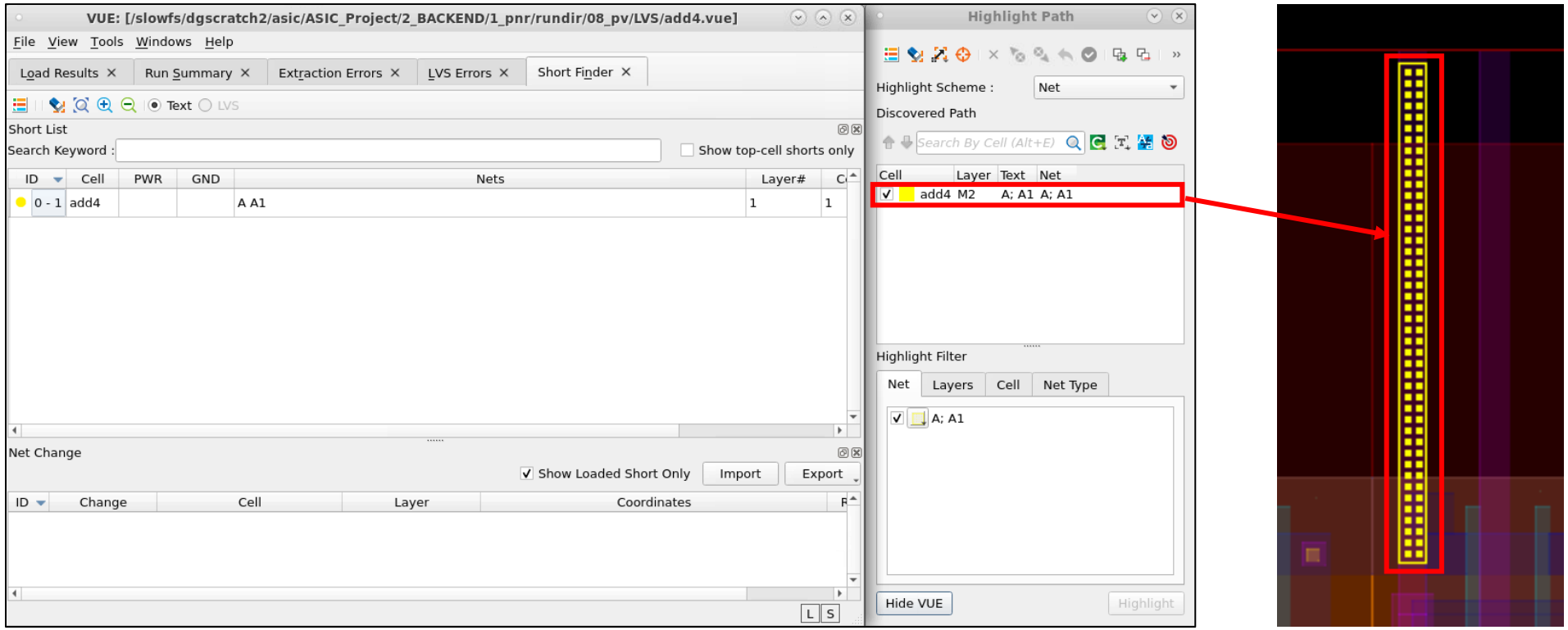
- **Load LVS Results in VUE (LVS Errors)**
  - **LVS Error tab provides details of violations observed during the comparison process**
  - **The violations are represented with respect to equivalence cell pairs**



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Introduction to PV(Physical Verification) : Run LVS using ICV

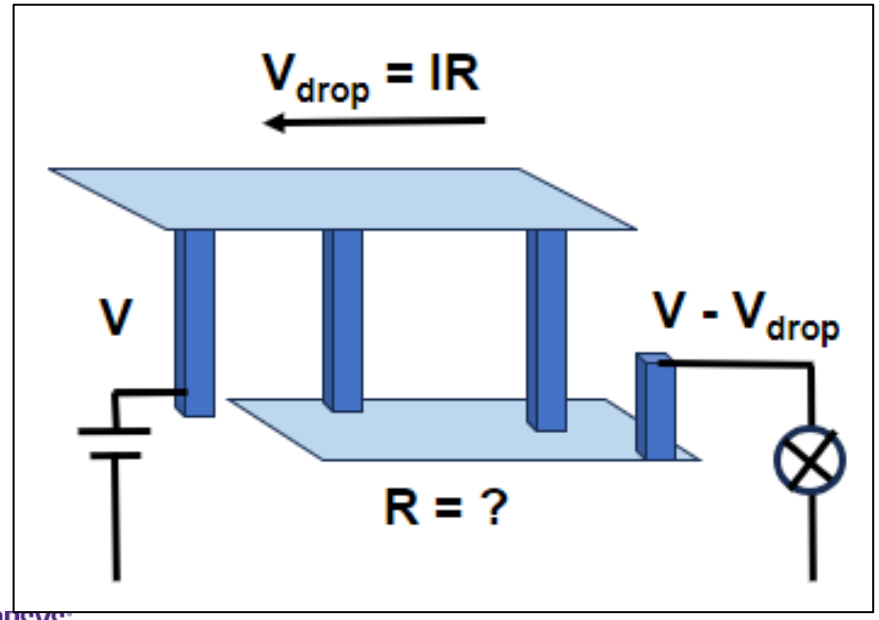
- **Load LVS Results in VUE (Extraction Errors)**
  - **Short Finder is an ICV utility which helps to debug shorts observed in design**
  - **This tab is enabled only when we use "-vueshort" command line option or enable "vue_short_finder = true" in the text_net runset fu nction**
  - **Double-clicking entries in the Short List will bring up the "Highlight Path" window**
  - **In the "Highlight Path window", the user can highlight polygons(s) associated with the path until the short is identified**



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Introduction to PV(Physical Verification) : Overview to IR DROP

- **What is IR Drop**
  - The voltage that gets to the internal circuitry is less than that applied to the chip (every metal layer offers resistance to the flo w of current)
  - IR Drop is defined as the average of the peak currents in the power network multiplied by the effective resistance from the P ower PADs to the Core area (Center) of the chip
  - IR Drop is a reduction in voltage that occurs on both Power & Ground networks
  - IR Drop Analysis ensures that PDN (Power Delivery Network) is robust, and that your system will function to specification
  - IR Drop is determined by the current flow & supply voltage
  - Distance between supply voltage and the component increases the IR Drop also increases



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Introduction to PV(Physical Verification) : Overview to IR DROP

- **IR Drop Analysis**
  - IR Drop Analysis will compute the actual IDD & ISS currents (these values are time-dependent)
  - IR Drip Analysis will compute Global IR drop which is important and more accurate, but cannot be compute separately for sm aller blocks, which may led to bigger IR Drop run time

  - **Local IR Drop**
    - IR Drop become a local phenomenon when a number of gates in close proximity switches at once
    - Local IR Drop can also be caused by a higher resistance to a specific portion of the Grid

  - **Global IR Drop**
    - IR Drop is a global phenomenon when activity in one region of a chip causes an IR Drop in other regions
    - Center of the chip usually has the largest IR Drop or the lowest supply voltage
    - Peak IR Drop is much larger than the Average IR Drop
    - Peak IR Drop happens in the worst-case switch patterns on the gates

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Introduction to PV(Physical Verification) : Overview to IR DROP

- **Types of IR Drop**
  - **Static IR Drop**
    - Static IR drop is average voltage drop for the design (The average current depends totally on the time period)
    - It is dependent on the RC of the power grid connecting the power supply to the respective STD cells
    - The average current depends totally on the time period
    - Gate-channel leakage current is the major reason for the static IR drop
  - **Dynamic IR Drop**
    - Dynamic IR drop is a drop in the voltage due to the high switching activity transistors
    - It happens when there is an increasing demand for current from the power supply due to switching activities of the chip
    - Dynamic IR drop depends on the switching time of the logic and is less dependent on the Clock Period
    - If many cells switch simultaneously at a high frequency, more current will be drawn through a PDN, causing fluctuating higher-voltage drop and is known as dynamic IR drop
    - ❖ ***PDN (Power Delivery Network)***

      In vlsi, PDN is a network of Metal Layers & Vias that connect the transistors to the main power supply

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Introduction to PV(Physical Verification) : Overview to IR DROP

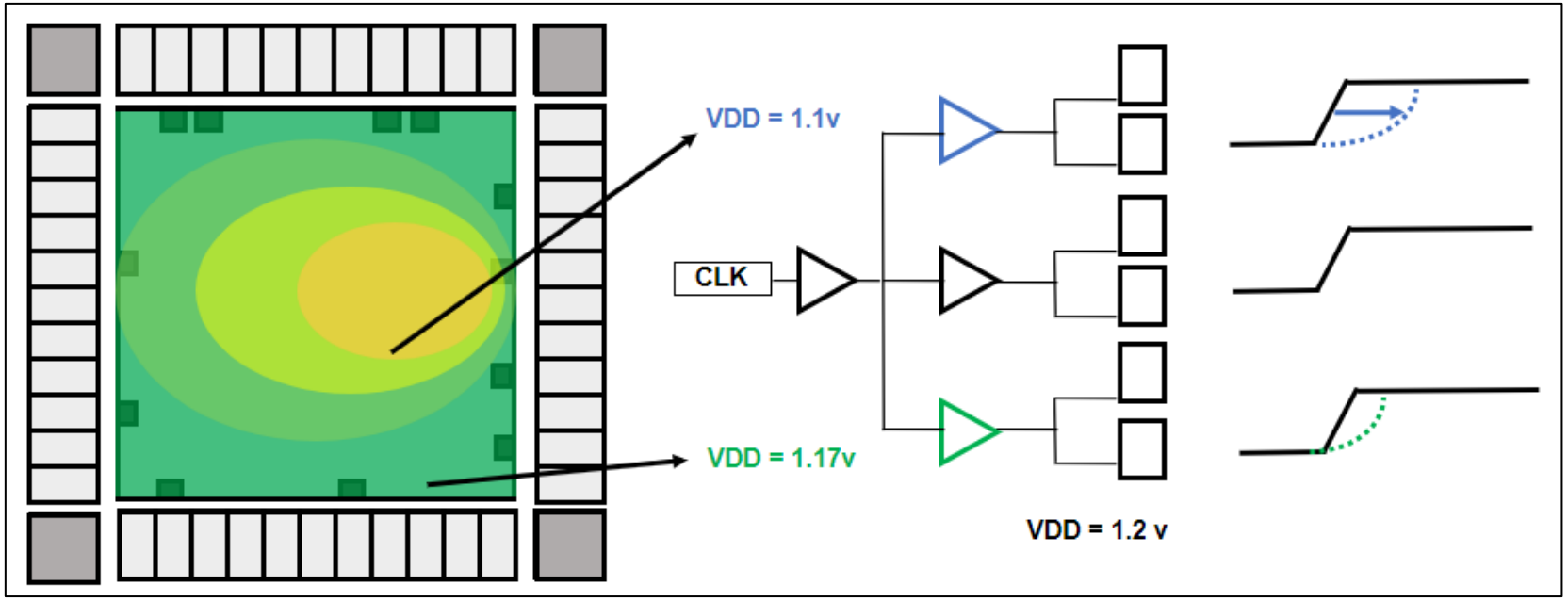
- **Reasons of IR Drop**
  - Improper placement of Power & Ground PADs
  - Wrong placement of Block
  - Bad Powerplan
  - Insufficient Core Ring & Power Strap width
  - Lesser no of Power Straps
  - Missing Vias in Powerplan
  - Insufficient number of Power PADs

- **Robustness Checks in IR Drop**
  - Open circuits
  - Missing or insufficient Vias
  - Current Density violations
  - Insufficient Power Rail

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Introduction to PV(Physical Verification) : Overview to IR DROP

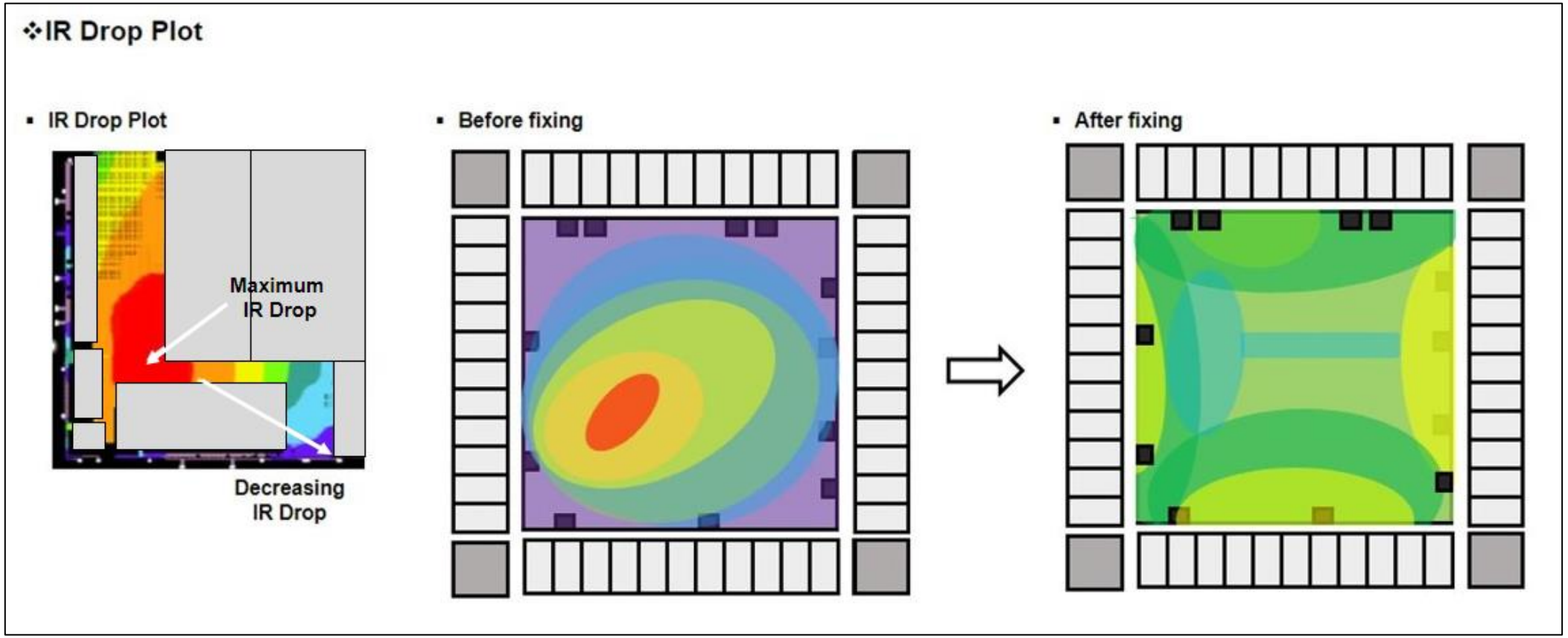
- **Impacts in IR Drop**
  - IR Drop Analysis confirms that the worst case voltage drop (Which is considered for the worst corner for timing) on a chip me ets IR Drop targets
  - Impacts in Timing
    - If the Voltage-Drop is too severe, the circuit will not get enough voltage (It resulting in the malfunction or timing failure in design)
    - If IR Drop increases Clock Skew, it will result in hold timing violation
    - If IR Drop increases Signal Skew, it will result in setup timing violation



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Introduction to PV(Physical Verification) : Overview to IR DROP

- **IR Drop Plot**
  - Power grid has a set of equipotential surface that form concentric circles centered in the middle of a block



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Introduction to PV(Physical Verification) : Overview to IR DROP

- **How to fix IR Drop**
  - Use different power grid tap points for clock buffers
  - Use smaller buffers
  - Rearrange blocks
  - More VDD pins
  - Connect bottom portion of grid to top portion
  - Distributing supplies symmetrically on the top
  - Lowering the resistance of supply & signal paths by making supply wires thicker in dimensions than signal wires (NDR rules)
  - Decap insertion after powerplan (This can solve dynamic IR Drop, at later stage of the design)

  ❖ Amount of Decap depends on
    - Acceptable ripple on VDD to VSS (typically 10% noise budget)
    - Switching activity of logic circuits
    - Current provided by power grid
    - Required frequency response (high frequency operation)

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WEEK 15 : LAB for DRC & Metal Fill

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PV(Physical Verification) : Introduction to pv.tcl

- Review and check the variables in each scripts
  - **Go to the Script Directory**

```
  < user@linux script >> pwd
/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/08_pv/script
  < user@linux script >> ls
01_design_setup_and_copy.tcl   04_fix_drc_after_run_drc.tcl          full_script
02_set_app_option_for_drc.tcl  05_set_app_option_for_metal_fill.tcl
03_run_drc.tcl                 06_create_metal_fill.tcl
```

  - **Review 01_design_setup_and_copy.tcl**

```
#made by kjinsoo
# module load
#module load icc2/2021.06-SP5
#module load icv/2022.12-SP5
# open icc2_shell
#icc2_shell -gui | tee -i ./10_pv.log

#set current step & before step
set step DRC
set before_step chip_finish
#source design_setup.tcl
source -e ./icc2_common_env/design_setup.tcl

#set_host_options -max_cores 16

#open lib
open_lib ../07_chip_finish/nlib/ORCA_${before_step}.nlib
#copy lib
copy_lib -from ../07_chip_finish/nlib/ORCA_${before_step}.nlib -to ./nlib/ORCA_${step}.nlib
#close & open current lib
close_lib -all
open_lib ./nlib/ORCA_${step}.nlib
#open block
open_block ./nlib/ORCA_${step}.nlib:ORCA.design
#link_block
link_block -rebind
```


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PV(Physical Verification) : Introduction to pv.tcl

- Review and check the variables in each scripts
  - **Review 02_set_app_option_for_drc.tcl**

```
####################
# ICV In-Design DRC
####################

#set app options before run DRC

set_app_options -list {signoff.check_drc.runset {./runset/saed32nm_1p9m_drc_rules.rs}} ;# Specifies the foun
dry runset that will be used while the tool invokes IC Validator to perform design rule checking on the curr
ent design
set_app_options -name signoff.check_drc.max_errors_per_rule -value 1000 ;# This application option specifies
 the maximum number of errors per rule to store in the error database that will be available to the Error Br
owser for viewing and reported in the IC Validator LAYOUT_ERRORS file
set_app_options -name signoff.check_drc.run_dir -value "z_icvDRC_run" ;# Specifies the path of the working d
irectory.
set_app_options -list {signoff.check_drc.read_design_views {*}} ;# If set, the cell instance data of each li
sted cell master is read from the design view. If the design view of a cell instance cannot be found, a warn
ing message is printed and the command reads cell instance data from the FRAME view instead (or convenience,
 all cells can be specified using the {*} wildcard)
```

  - **Review 03_run_drc.tcl**

```
#Run DRC

signoff_check_drc -check_all_runset_layers true
```


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PV(Physical Verification) : Introduction to pv.tcl

- Review and check the variables in each scripts
  - **Review 04_fix_drc_after_run_drc.tcl**

```
#Fix DRC Violation after Run DRC

#set app option

set_app_options -list {signoff.fix_drc.init_drc_error_db "z_icvDRC_run"} ;# Specifies the path of the direct
ory that contains the initial DRC error database generated by IC Validator

#fix drc
signoff_fix_drc
```

  - **Review 05_set_app_option_for_metal_fill.tcl**

```
##################
# run Metal Fill
##################

#set app_option for Metal Fill
set_app_options -list {signoff.create_metal_fill.runset {./runset/saed32nm_1p9m_mfill_rules.rs}} ;# The foun
dry runset will be used while the tool invokes IC Validator for metal fill insertion in the current design
set_app_options -list {signoff.create_metal_fill.read_design_views {*}} ;# If set, the cell instance data of
 each listed cell master is read from the design view. If the design view of a cell instance cannot be found
, a warning message is printed and the command reads cell instance data from the FRAME view instead (or conv
enience, all cells can be specified using the {*} wildcard)
```


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PV(Physical Verification) : Introduction to pv.tcl

- Review and check the variables in each scripts
  - **Review 06_create_metal_fill.tcl**

```
#create Metal fill
signoff_create_metal_fill \
        -timing_preserve_setup_slack_threshold 0.05

#save & exit

save_lib -all

exit
```


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PV(Physical Verification) : Use ICC2 (IC Compiler II tool)

It is also good to Read Design by source 10_pv.tcl, but when you proceed for the first time, it is recommended that you open the script and apply it one phrase at a time

1. **Invoke tool icc2_shell**
   - **Recommend the icc2/2021.06-SP5 version & icv/2022.12-SP5 version**

>> icc2_shell –gui | tee -i ./10_pv.log ;# (| tee -I ./01_read_design.log) is the command to output log

```
 < user@linux 08_pv >> icc2_shell -gui | tee -i ./10_pv.log



                            IC Compiler II (TM)

            Version S-2021.06-SP5 for linux64 - Jan 14, 2022 -SLE

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   Licensed Products in conformity with the applicable License Key for such
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                                infringers.

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           Inclusivity and Diversity" (Refer to article 000036315 at
                       https://solvnetplus.synopsys.com)

Loading user preference file /u/kjinsoo/.synopsys_icc2_gui/preferences.tcl
Load ICV ICCII menu file: /global/apps/icv_2022.12-SP5/etc/tcl-u/Icc2Menu.tcl
 + VUE INFO: Please click View->IC Validator VUE in LayoutWindow menu
           to launch VUE.

 + VUE INFO: Found a usable port: 2446

icc2_shell>
```


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PV(Physical Verification) : Use ICC2 (IC Compiler II tool)

2. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/08_pv/script/01_design_setup_and_copy.tcl

- **Set step & source design_setup.tcl**

```
icc2_shell> set step DRC
DRC
icc2_shell> set before_step chip_finish
chip_finish
icc2_shell> source -e ./icc2_common_env/design_setup.tcl
```

- **Open lib**

```
icc2_shell> open_lib ../07_chip_finish/nlib/ORCA_${before_step}.nlib
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/
07_chip_finish/nlib/ORCA_chip_finish.nlib' (FILE-007)
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/
00_create_ndm/ndm/EXPLORE_physical_only.ndm' (FILE-007)
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/
00_create_ndm/ndm/saed32hvt.ndm' (FILE-007)
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/
00_create_ndm/ndm/saed32io_fc.ndm' (FILE-007)
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/
00_create_ndm/ndm/saed32lvt.ndm' (FILE-007)
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/
00_create_ndm/ndm/saed32pll.ndm' (FILE-007)
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/
00_create_ndm/ndm/saed32rvt.ndm' (FILE-007)
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/
00_create_ndm/ndm/saed32sram.ndm' (FILE-007)
{ORCA_chip_finish.nlib}
```

Check that ndm is properly apllied each time you open the nlib

- **Copy ORCA_powerplan.nlib to ORCA_placement.nlib & open ORCA_placement.nlib**

```
copy_lib -from ../07_chip_finish/nlib/ORCA_${before_step}.nlib -to ./nlib/ORCA_${step}.nlib
Information: Copying from library "ORCA_chip_finish.nlib" to "./nlib/ORCA_DRC.nlib".
{ORCA_DRC.nlib}
icc2_shell>
icc2_shell> close_lib -all
Closing all libraries...
1
```


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PV(Physical Verification) : Use ICC2 (IC Compiler II tool)

2. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/08_pv/script/01_design_setup_and_copy.tcl

- **Open lib**

```
icc2_shell> open_lib ./nlib/ORCA_${step}.nlib
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/
08_pv/nlib/ORCA_DRC.nlib' (FILE-007)
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/
00_create_ndm/ndm/EXPLORE_physical_only.ndm' (FILE-007)
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/
00_create_ndm/ndm/saed32hvt.ndm' (FILE-007)
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/
00_create_ndm/ndm/saed32io_fc.ndm' (FILE-007)
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/
00_create_ndm/ndm/saed32lvt.ndm' (FILE-007)
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/
00_create_ndm/ndm/saed32pll.ndm' (FILE-007)
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/
00_create_ndm/ndm/saed32rvt.ndm' (FILE-007)
Information: Loading library file '/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/
00_create_ndm/ndm/saed32sram.ndm' (FILE-007)
{ORCA_DRC.nlib}
```

Check that ndm is properly apllied each time you open the nlib

- **Open block & link_block**

```
icc2_shell> open_block ./nlib/ORCA_${step}.nlib:ORCA.design
Information: User units loaded from library 'saed32hvt' (LNK-040)
Opening block 'ORCA_DRC.nlib:ORCA.design' in edit mode
{ORCA_DRC.nlib:ORCA.design}
icc2_shell> link_block
```

```
icc2_shell> link_block -rebind
Warning: Block 'ORCA_DRC.nlib:ORCA.design' is already linked. (LNK-067)
0
```


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PV(Physical Verification) : Use ICC2 (IC Compiler II tool)

3. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/08_pv/script/02_set_app_option_for_drc.tcl

- **Set_app_option before run drc**

```
icc2_shell> set_app_options -list {signoff.check_drc.runset {./runset/saed32nm_1p9m_drc_rules.rs}}
signoff.check_drc.runset ./runset/saed32nm_1p9m_drc_rules.rs
icc2_shell> set_app_options -name signoff.check_drc.max_errors_per_rule -value 1000
signoff.check_drc.max_errors_per_rule 1000
icc2_shell> set_app_options -name signoff.check_drc.run_dir -value "z_icvDRC_run"
signoff.check_drc.run_dir z_icvDRC_run
icc2_shell> set_app_options -list {signoff.check_drc.read_design_views {*}}
signoff.check_drc.read_design_views *
```

4. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/08_pv/script/03_run_drc.tcl

- **Run DRC(Design Rule Check) use ICV (IC Validator)**

```
icc2_shell> signoff_check_drc -check_all_runset_layers true

 The -check_all_runset_layers option needs to be run after routing layers only (default) option
 got DRC clean. The application option signoff.check_drc.read_design_view will be automatically
 enabled for this option.

Warning: Contact not found for layer: VIA1, width pair (0.050000, 0.056000)
Warning: Contact not found for layer: VIA2, width pair (0.056000, 0.056000)
Warning: Contact not found for layer: VIA3, width pair (0.056000, 0.056000)
Warning: Contact not found for layer: VIA4, width pair (0.056000, 0.056000)
Warning: Contact not found for layer: VIA5, width pair (0.056000, 0.056000)
Warning: Contact not found for layer: VIA6, width pair (0.056000, 0.056000)
Warning: Contact not found for layer: VIA7, width pair (0.056000, 0.056000)
Warning: Contact not found for layer: VIA8, width pair (0.056000, 0.160000)
Warning: Contact not found for layer: VIARDL, width pair (0.160000, 2.000000)

Start to run ICV ...
.....................................................................
.....................................................
```


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PV(Physical Verification) : Use ICC2 (IC Compiler II tool)

4. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/08_pv/script/03_run_drc.tcl

- **Run DRC(Design Rule Check) use ICV (IC Validator)**

```
ICV is done!
ICV messages:
WARNING: Could not open I1025_NS.design, using frame view instead.
WARNING: Could not open B12I1025_NS.design, using frame view instead.
WARNING: Could not open D8I1025_NS.design, using frame view instead.
WARNING: Could not open B16I1025_NS.design, using frame view instead.
WARNING: Could not open INVX2_RVT.design, using frame view instead.
For more details see /slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/08_pv/z_icvDRC_run/signoff_check_
drc.log
Overall engine Time=0:07:07 Highest command Mem=0.815 GB

--------------------------------------------------------------------------------
Rule: CO.E.11: Min enclosure of contact by PO at two opposite sides (if one side = 0, opposite >=0.01) must be 0.0
16                10 errors
Rule:             CO.E.12: CONT is not enclosed by M1              87 errors
Rule: CO.E.5: Minimum enclose by poly, at least two opposite sides if PO width > 0.1 must be 0.05
   10 errors
Rule: CO.E.9: Minimum enclosure of any contact by M1 must be 0.002um              1000 errors
  WARNING: Runset error limit exceeded for this rule, some errors omitted.
Rule:         CO.W.1: Exact contact size must be 0.042              10 errors
Rule: M1.S.10: Minimum M1 spacing of end of line (width <= 0.1) to neighbour line (width >= 0.8) must be 0.1
           4 errors
Rule:            M1.W.1: Minimum M1 width must be 0.05             178 errors
Rule: M1.W.2.1: Minimum width of thin wire with length less than 0.1, connected to fat wire with width larges than
 0.15, must be 0.06                7 errors
Rule: M1.W.2.3: Minimum width of thin wire with length less than 0.7, connected to fat wire with width larges than
 1.5, must be 0.15                32 errors
Rule:              M1.W.2: Maximum M1 width must be 5              13 errors
Rule: M2.W.2.1: Minimum width of thin wire with length less than 0.1, connected to fat wire with width larges than
 0.15, must be 0.06                1 error
Rule: M3.S.2: Minimum spacing when either metal line is wider than 0.15 must be 0.06                 9 errors
Rule: NWELL.S.1: Minimum spacing between wells at same potential must be 0.23                1 error
Rule: R.M5.W.2.2: Minimum width of thin wire with length less than 0.17, connected to fat wire with width larges t
han 0.3, must be 0.07             30 errors
--------------------------------------------------------------------------------

signoff_check_drc is finished.

Create error data signoff_check_drc.err ...
```

DRC Results

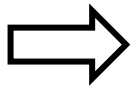
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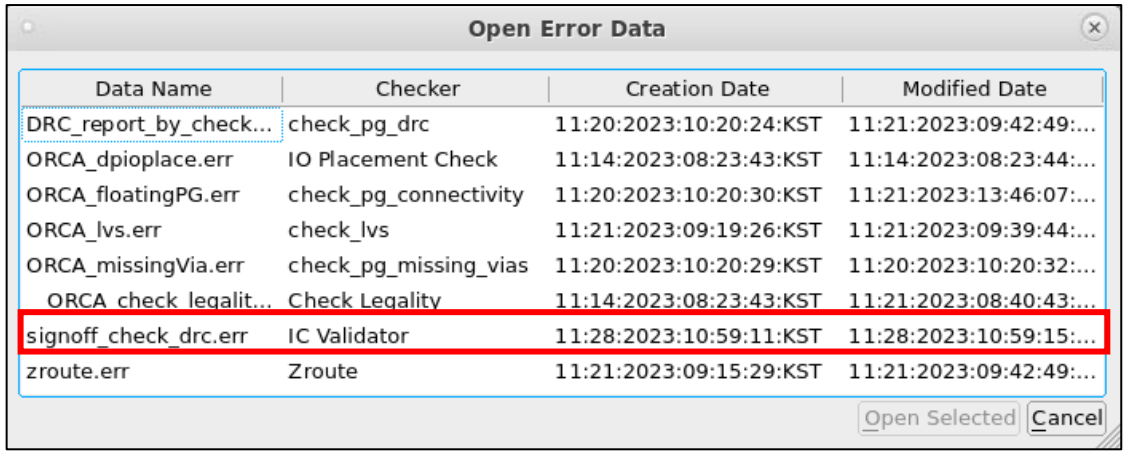
PV(Physical Verification) : Use ICC2 (IC Compiler II tool)

- **Check DRC Error after Run DRC**

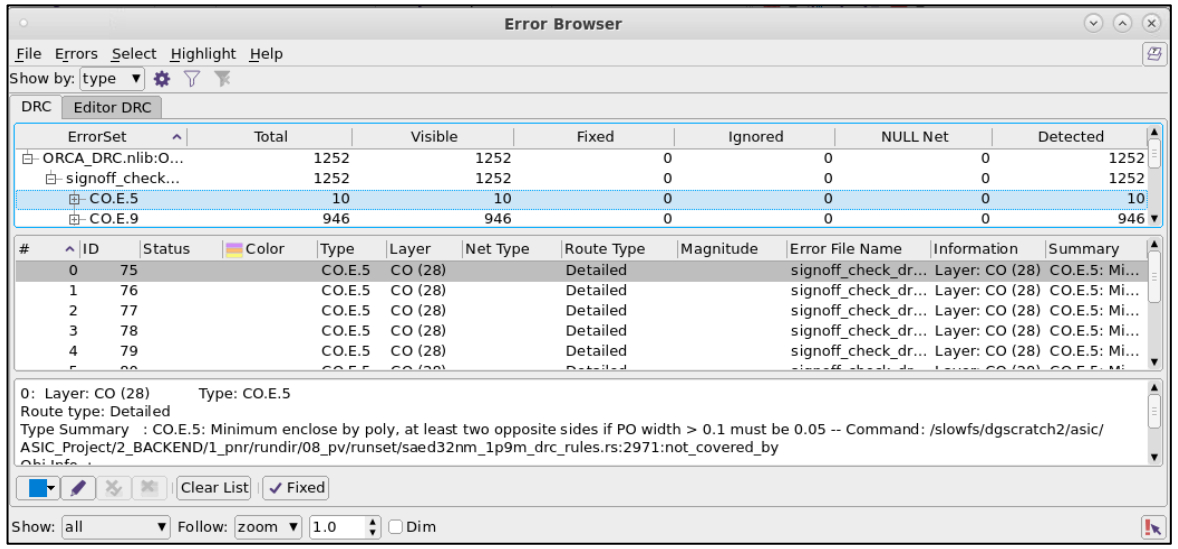
1. **Open Error Browser & open Error Data**







2. **Analyze DRC Errors**



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PV(Physical Verification) : Use ICC2 (IC Compiler II tool)

- **Check DRC Error after Run DRC**

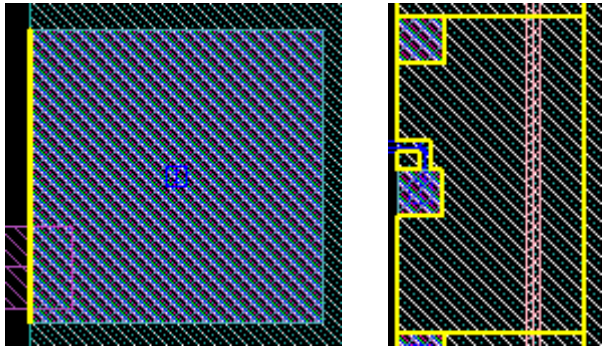
  2. Analyze DRC Errors (Example of DRC Errors)

  - **CO.E & CO.W Error**

    ❖ This type of Errors are the width of the PO (Base Metal Layer) and the M1 contact related Error inside the Macros

    ❖ It is a DRC Error that occurs inside the Macro, it cannot be modified in the ICC2 Tool

  - **Ex)**



CO.E.9

CO.E.12

  - **(M1.W1, M1.W2, M1.W2.1, M1,W2.3) Error**

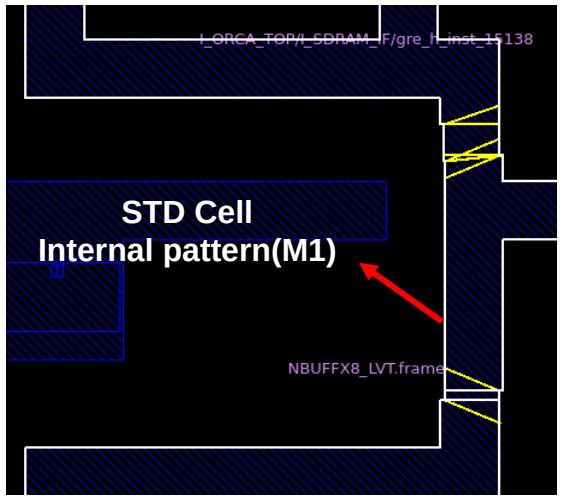
    ❖ This type of Errors are caused by pin or internal pattern of M1 layer inside STD Cells & Macros

    ❖ It is a DRC Error that occurs inside the STD cell & Macro, it cannot be modified in the ICC2 Tool

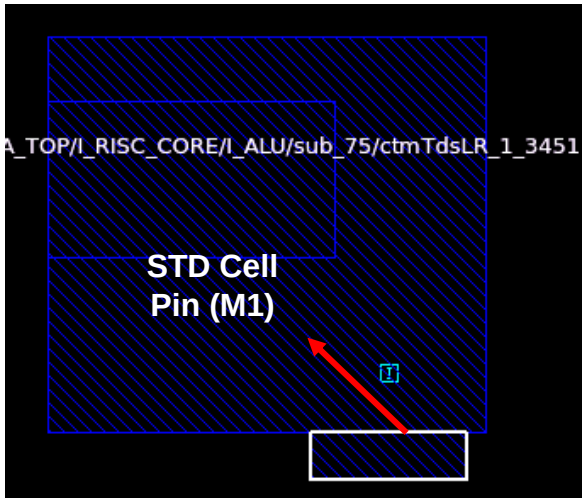
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PV(Physical Verification) : Use ICC2 (IC Compiler II tool)

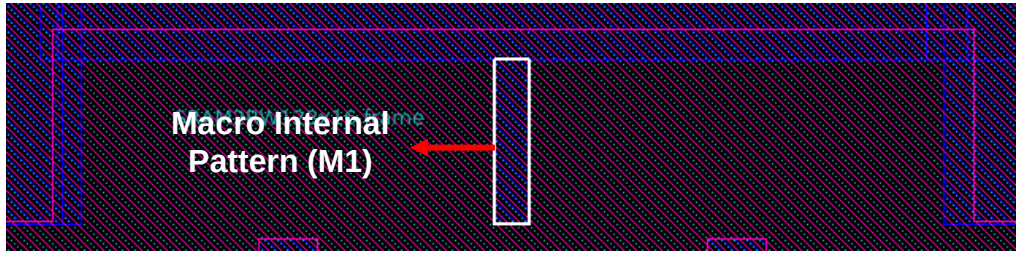
- **Check DRC Error after Run DRC**
  - **2. Analyze DRC Errors (Example of DRC Errors)**
  - **(M1.W1, M1.W2, M1.W2.1, M1,W2.3) Error**
  - **Ex)**



M1.W1



M1.W2.1



M1.W2.3

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PV(Physical Verification) : Use ICC2 (IC Compiler II tool)

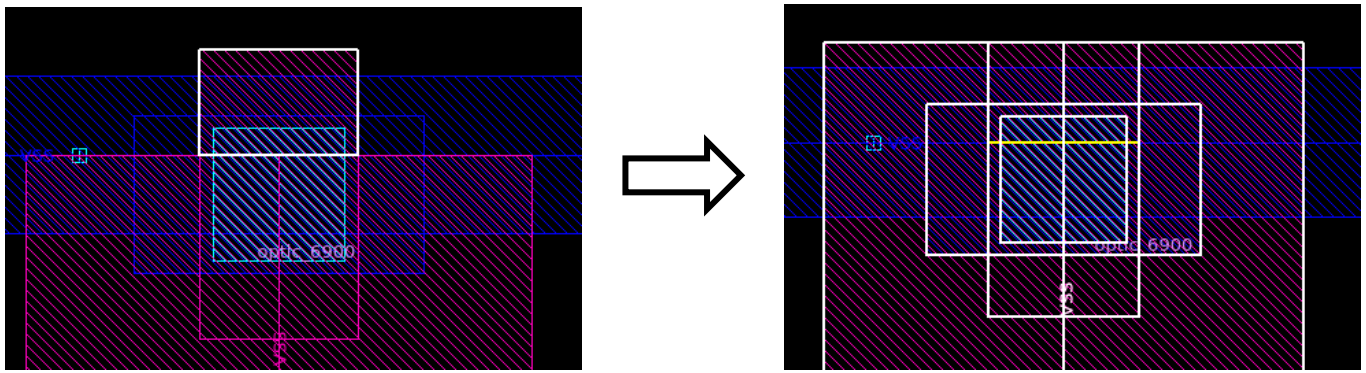
- **Check DRC Error after Run DRC**

  2. Analyze DRC Errors (Example of DRC Errors)

  - **M2.W2.1 Error**

```
0:  Layer: M2 (13)          Type: M2.W.2.1
Net type: Ground;           Route type: Strap/Std Cell Pin Connection
Type Summary   : M2.W.2.1: Minimum width of thin wire with length less than 0.1, connected to fat wire with width larges than 0.15, must be 0.06
-- Command: /slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/08_pv/runset/saed32nm_1p9m_drc_rules.rs:1466:or
Obj Info  :
          Net: VSS
```

  - **Fix Error (Manual)**



❖ The minimum width of a thin wire with a length of less than 0.1 connected to a fat wire larger than 0.15 must be 0.06, thus increasing the length of the VSS M2 wire connected to the VIA

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PV(Physical Verification) : Use ICC2 (IC Compiler II tool)

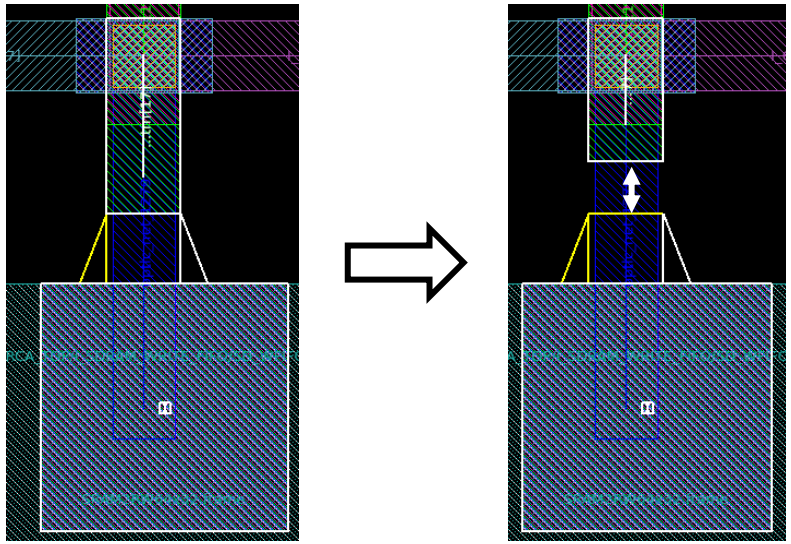
- **Check DRC Error after Run DRC**

  2. Analyze DRC Errors (Example of DRC Errors)

  - **M4.S.2 Error**

```
2:  Layer: M4 (17)-M4 (17)              Type: M4.S.2
Route type: Detailed
Type Summary   : M4.S.2: Minimum spacing when either metal line is wider than 0.15 must be 0.06 -- Command: /slowfs/dgscratch2/asic/
ASIC_Project/2_BACKEND/1_pnr/rundir/08_pv/runset/saed32nm_1p9m_drc_rules.rs:1813:external2
Obj Info  : distance = 0.0560 um -- E2E
        Violated required spacing, actual (0.056 um)
```

  - **Fix Error (Manual)**



❖ Spread the spacing between M4 pin and M4 routing by 0.06 or more

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PV(Physical Verification) : Use ICC2 (IC Compiler II tool)

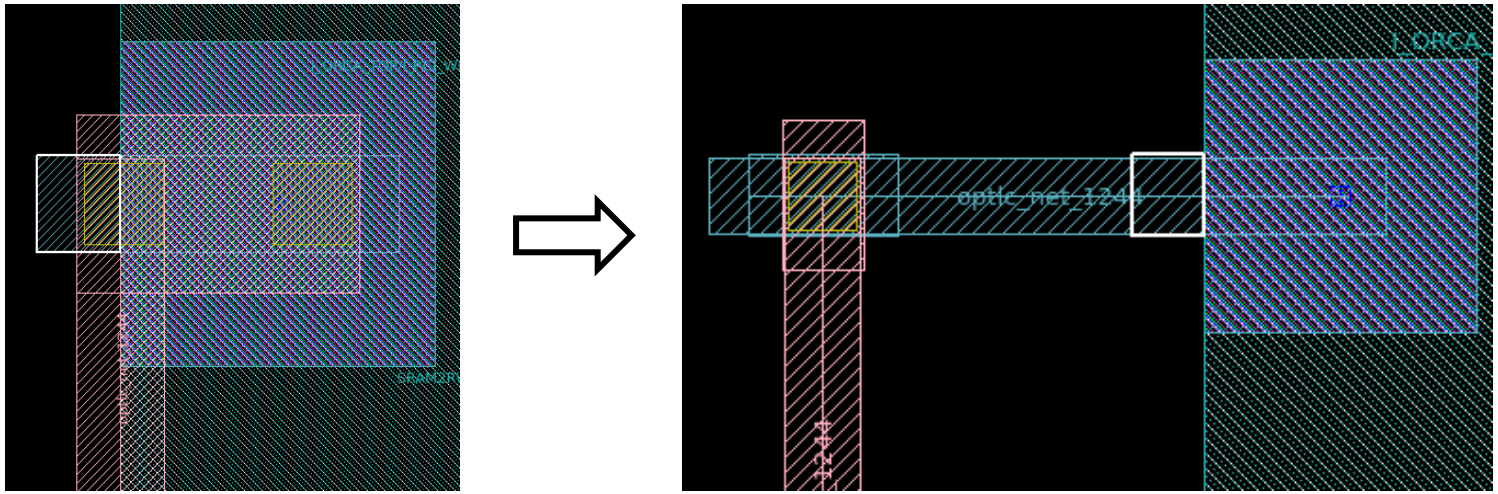
- **Check DRC Error after Run DRC**

  2. Analyze DRC Errors (Example of DRC Errors)

  - **R.M5.W.2.1 Error**

```
0:  Layer: M5 (19)          Type: R.M5.W.2.1
Net type: Signal;           Route type: Detailed
Type Summary   : R.M5.W.2.1: Minimum width of thin wire with length less than 0.1, connected to fat wire with width larges than 0.15, must be
0.06 -- Command: /slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/08_pv/runset/saed32nm_1p9m_drc_rules.rs:2022:or
Obj Info  :
          Net: optlc_net_1244
```

  - **Fix Error (Manual)**



❖ Redraw the M5 wire and VIA according to the summary value

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PV(Physical Verification) : Use ICC2 (IC Compiler II tool)

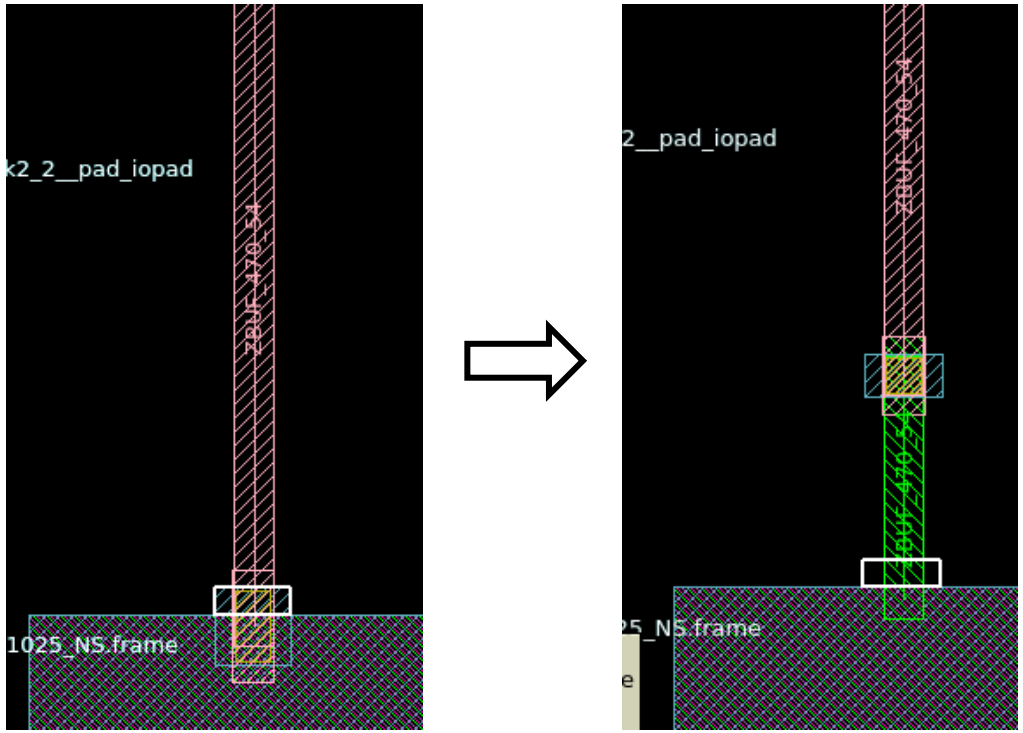
- **Check DRC Error after Run DRC**

  2. Analyze DRC Errors (Example of DRC Errors)

  - **R.M5.W.2.1 Error**

```
0:  Layer: M5 (19)          Type: R.M5.W.2.2
Net type: Signal;           Route type: Detailed
Type Summary   : R.M5.W.2.2: Minimum width of thin wire with length less than 0.17, connected to fat wire with width larges than 0.3, must be
0.07 -- Command: /slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/08_pv/runset/saed32nm_1p9m_drc_rules.rs:2040:or
Obj Info  :
          Net: ZBUF_470_54
```

  - **Fix Error (Manual)**



❖ The VIA is directly attached to the pin and draws M4 to prevent Width Errors from occurring, and connects it to the pin

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PV(Physical Verification) : Use ICC2 (IC Compiler II tool)

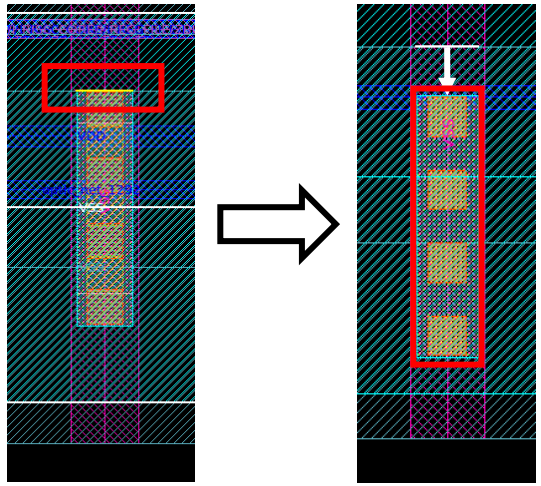
- **Check DRC Error after Run DRC**

  2. Analyze DRC Errors (Example of DRC Errors)

  - **R.M5.W.2.1 Error**

```
3:  Layer: M5 (19)          Type: R.M5.W.2.2
Net type: Ground;           Route type: Ring/Strap
Type Summary   : R.M5.W.2.2: Minimum width of thin wire with length less than 0.17, connected to fat wire with width larges than 0.3, must be
0.07 -- Command: /slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/08_pv/runset/saed32nm_1p9m_drc_rules.rs:2040:or
Obj Info  :
          Net: VSS
```

  - **Fix Error (Manual)**



❖ Move the location of the M2 to M7 VIA so that it does not touch the interface of the M5 Macro RIng

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PV(Physical Verification) : Use ICC2 (IC Compiler II tool)

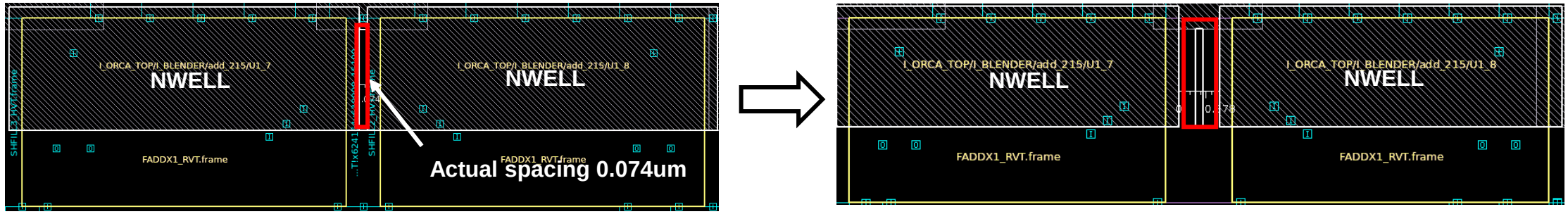
- **Check DRC Error after Run DRC**

  2. Analyze DRC Errors (Example of DRC Errors)

  - **NWELL.S.1 Error**

```
0:  Layer:    Type: NWELL.S.1
Type Summary   : NWELL.S.1: Minimum spacing between wells at same potential must be 0.23 -- Command: /slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/
rundir/08_pv/runset/saed32nm_1p9m_drc_rules.rs:356:external1
Obj Info  : distance = 0.0740 um -- E2E
        Violated required spacing, actual (0.074 um)
```

  - **Fix Error (Manual)**



- ❖ This Error is a spacing Error between NWELL
- ❖ Actual spacing of NWELL is 0.074um, and the spacing must be secured by at least 0.23um
- ❖ Since the location of the cell has been changed, reroute for the changed cells

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PV(Physical Verification) : Use ICC2 (IC Compiler II tool)

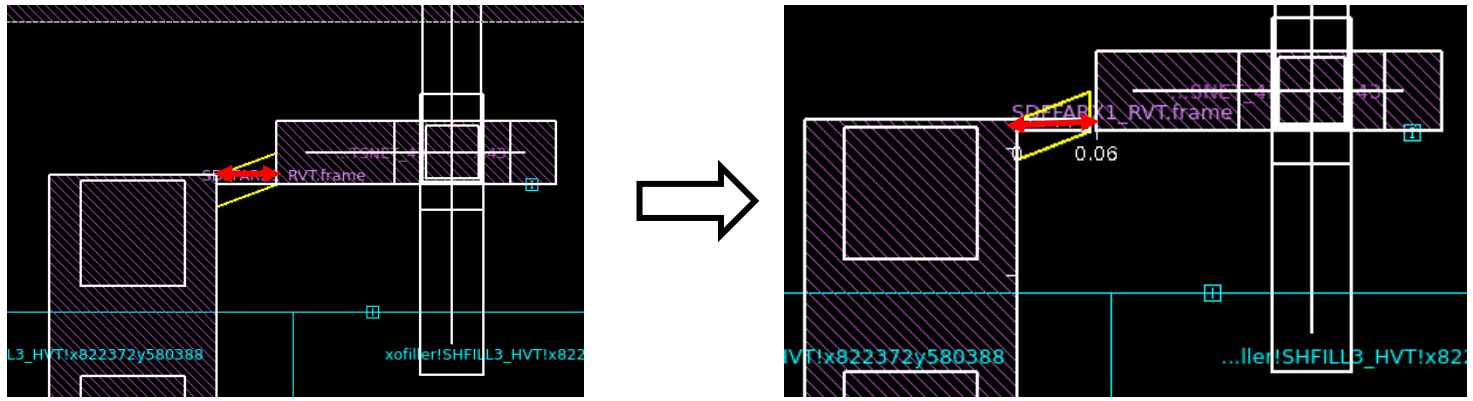
- **Check DRC Error after Run DRC**

  2. Analyze DRC Errors (Example of DRC Errors)

  - **M3.S.2 Error**

```
0:  Layer: M3 (15)-M3 (15)              Type: M3.S.2
Net type: Ground;          Route type: Detailed/Strap
Type Summary   : M3.S.2: Minimum spacing when either metal line is wider than 0.15 must be 0.06 -- Command: /slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/
1_pnr/rundir/08_pv/runset/saed32nm_1p9m_drc_rules.rs:1634:external2
Obj Info  : distance = 0.0560 um -- E2E
          Violated required spacing, actual (0.056 um)
           Net: VSS
```

  - **Fix Error (Manual)**



❖ Make the spacing between the M3 VIA and the M3 Layer more than 0.06um

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PV(Physical Verification) : Use ICC2 (IC Compiler II tool)

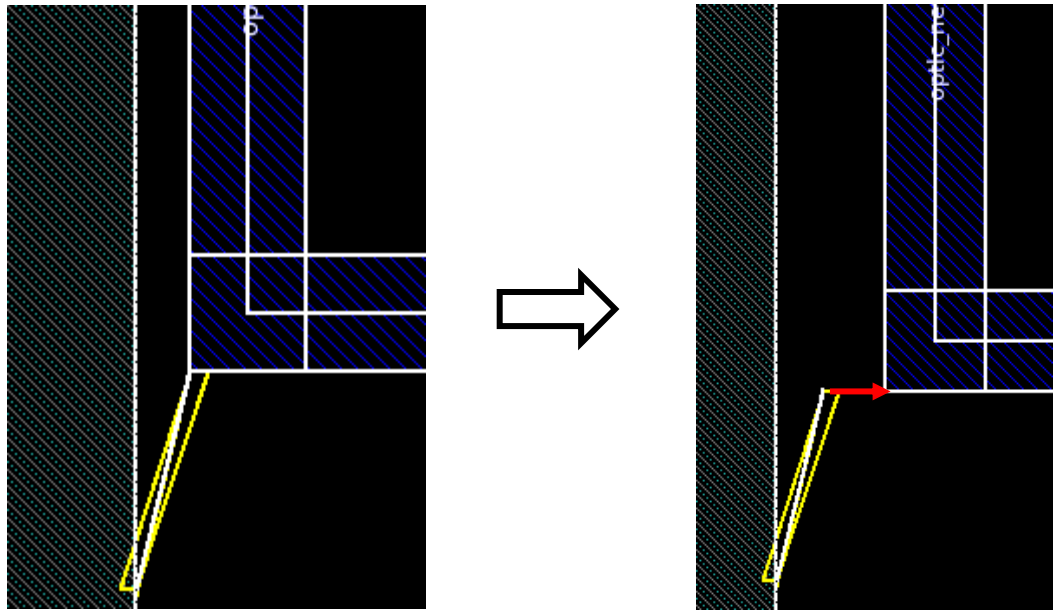
- **Check DRC Error after Run DRC**

  2. Analyze DRC Errors (Example of DRC Errors)

  - **M3.S.2 Error**

```
2:  Layer: M1 (11)-M1 (11)              Type: M1.S.10
Net type: Signal;           Route type: Detailed
Type Summary   : M1.S.10: Minimum M1 spacing of end of line (width <= 0.1) to neighbour line (width >= 0.8) must be 0.1 -- Command: /slowfs/dgscratch2/asic/
ASIC_Project/2_BACKEND/1_pnr/rundir/08_pv/runset/saed32nm_1p9m_drc_rules.rs:1182:external2
Obj Info  : distance = 0.0980 um -- E2E
          Violated required spacing, actual (0.098 um)
           Net: optlc_net_1303
```

  - **Fix Error (Manual)**



❖ Move M1 routing to the rihgt to secure spacing

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PV(Physical Verification) : Use ICC2 (IC Compiler II tool)

5. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/08_pv/script/04_fix_drc_after_run_drc.tcl

- **Fix DRC Errors automatically use ICV (IC Validator)**

```
icc2_shell> set_app_options -list {signoff.fix_drc.init_drc_error_db "z_icvDRC_run"}
signoff.fix_drc.init_drc_error_db z_icvDRC_run
icc2_shell> signoff_fix_drc
Information: Running the advanced guidance based ADR flow.
Reading ICV Error data...
Information: Metal2 is vertical. Switching all the directional adr advanced guidances.
Total of 38 error(s) targeted for Auto-Fix.
Current working directory is /slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/08_pv
Successfully loaded the route guidances
Cell Min-Routing-Layer = M1
Cell Max-Routing-Layer = M6
Found antenna rule mode 4, diode mode 2:
        layer M1:  max ratio 1000, diode ratio {0.06 0 400 40000}
        layer M2:  max ratio 1000, diode ratio {0.06 0 400 40000}
        layer M3:  max ratio 1000, diode ratio {0.06 0 400 40000}
        layer M4:  max ratio 1000, diode ratio {0.06 0 400 40000}
        layer M5:  max ratio 1000, diode ratio {0.06 0 400 40000}
        layer M6:  max ratio 1000, diode ratio {0.06 0 400 40000}
        layer M7:  max ratio 1000, diode ratio {0.06 0 400 40000}
        layer M8:  max ratio 1000, diode ratio {0.06 0 400 40000}
        layer M9:  max ratio 1000, diode ratio {0.06 0 8000 50000}
        layer MRDL:  max ratio 1000, diode ratio {0 0 1 0 0}
        layer CO: , diode ratio {0 0 1 0 0}
        layer VIA1:  max ratio 20, diode ratio {0.06 0 200 1000}
        layer VIA2:  max ratio 20, diode ratio {0.06 0 200 1000}
        layer VIA3:  max ratio 20, diode ratio {0.06 0 200 1000}
        layer VIA4:  max ratio 20, diode ratio {0.06 0 200 1000}
        layer VIA5:  max ratio 20, diode ratio {0.06 0 200 1000}
        layer VIA6:  max ratio 20, diode ratio {0.06 0 200 1000}
        layer VIA7:  max ratio 20, diode ratio {0.06 0 200 1000}
        layer VIA8:  max ratio 20, diode ratio {0.06 0 200 1000}
        layer VIARDL:  max ratio 20, diode ratio {0 0 1 0 0}
```

Antenna Rule

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PV(Physical Verification) : Use ICC2 (IC Compiler II tool)

5. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/08_pv/script/04_fix_drc_after_run_drc.tcl

- **Fix DRC Errors automatically use ICV (IC Validator)**

DRC AUTOFIX summary

```
signoff_check_drc is finished.

Create error data signoff_check_drc.err ...
Closing block 'ORCA_DRC.nlib:ORCA_ADR.design'
Reading ICV Error data...
Information: Metal2 is vertical. Switching all the directional adr advanced guidances.
Total of 0 error(s) targeted for Auto-Fix.
No more targeted error found.
Signoff DRC Auto Fixing has completed in 1 repair loops.

Input DRC Error Database.......... /slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/08_pv/z_icvDRC_run
Limit to Cell..................... ORCA
Maximum Errors/Command............ 1000

SIGNOFF AUTOFIX DRC SUMMARY:

       DRC NAME        :   TOTAL DRC   : PROCESSED DRC :  TARGETED DRC : REMAINING TARGETED DRC :   FIX RATE   : COMMENT
CO.E.11: Min enclosure  :          10   :           10   :            0   :                  0   :          --   : 10 ignored: <#2>
CO.E.12: CONT is not en :          87   :           87   :            0   :                  0   :          --   : 41 ignored: <#2>;46 ignored: <#3>
CO.E.5: Minimum enclose :          10   :           10   :            0   :                  0   :          --   : 10 ignored: <#2>
CO.E.9: Minimum enclosu :        1141   :         1000   :            0   :                  0   :          --   : 1000 ignored: <#1>
CO.W.1: Exact contact s :          10   :           10   :            0   :                  0   :          --   : 10 ignored: <#2>
M1.S.10: Minimum M1 spa :           4   :            4   :            4   :                  0   :       100.0% :
M1.W.1: Minimum M1 widt :         178   :          178   :            0   :                  0   :          --   : 178 ignored: <#3>
M1.W.2.1: Minimum width :           7   :            7   :            1   :                  0   :       100.0% : 6 ignored: <#3>
M1.W.2.3: Minimum width :          32   :           32   :            0   :                  0   :          --   : 32 ignored: <#3>
M1.W.2: Maximum M1 widt :          13   :           13   :            0   :                  0   :          --   : 13 ignored: <#2>
M2.W.2.1: Minimum width :           1   :            1   :            0   :                  0   :          --   : 1 ignored: <#3>
M3.S.2: Minimum spacing :           9   :            9   :            0   :                  0   :          --   : 9 ignored: <#3>
R.M5.W.2.2: Minimum wid :          30   :           30   :            2   :                  0   :       100.0% : 28 ignored: <#3>
TOTAL                   :               :                :            7   :                  0   :       100.0% :

NOTE:
Reasons : Detail Info
     #1 : the amount of errors exceed the limitation of maximum errors per rule
     #2 : size of error shapes greater than threshold
     #3 : shapes associated with error are not Zroute modifiable
```

Reason of DRC Error ignore list

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PV(Physical Verification) : Use ICC2 (IC Compiler II tool)

6. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/08_pv/script/05_set_app_option_for_metal_fill.tcl

- **Set_app_option before create Metal Fill**

```
icc2_shell> set_app_options -list {signoff.create_metal_fill.runset {./runset/saed32nm_1p9m_mfill_rules.rs}}
signoff.create_metal_fill.runset ./runset/saed32nm_1p9m_mfill_rules.rs
icc2_shell>
icc2_shell> set_app_options -list {signoff.create_metal_fill.read_design_views {*}}
signoff.create_metal_fill.read_design_views *
```

7. Icc2_shell> source $WORK_DIR/2_BACKEND/1_pnr/rundir/08_pv/script/06_create_metal_fill.tcl

- **Create Metal Fill use ICV (IC Validator)**

```
icc2_shell> signoff_create_metal_fill \
? -timing_preserve_setup_slack_threshold 0.05
Info:: Layer m11 is not defined for app option space_to_nets
Info:: Layer m12 is not defined for app option space_to_nets
Info:: Layer m13 is not defined for app option space_to_nets
Info:: Layer m14 is not defined for app option space_to_nets
Info:: Layer m15 is not defined for app option space_to_nets
Info:: Layer m11 is not defined for app option space_to_clock_nets
Info:: Layer m12 is not defined for app option space_to_clock_nets
```

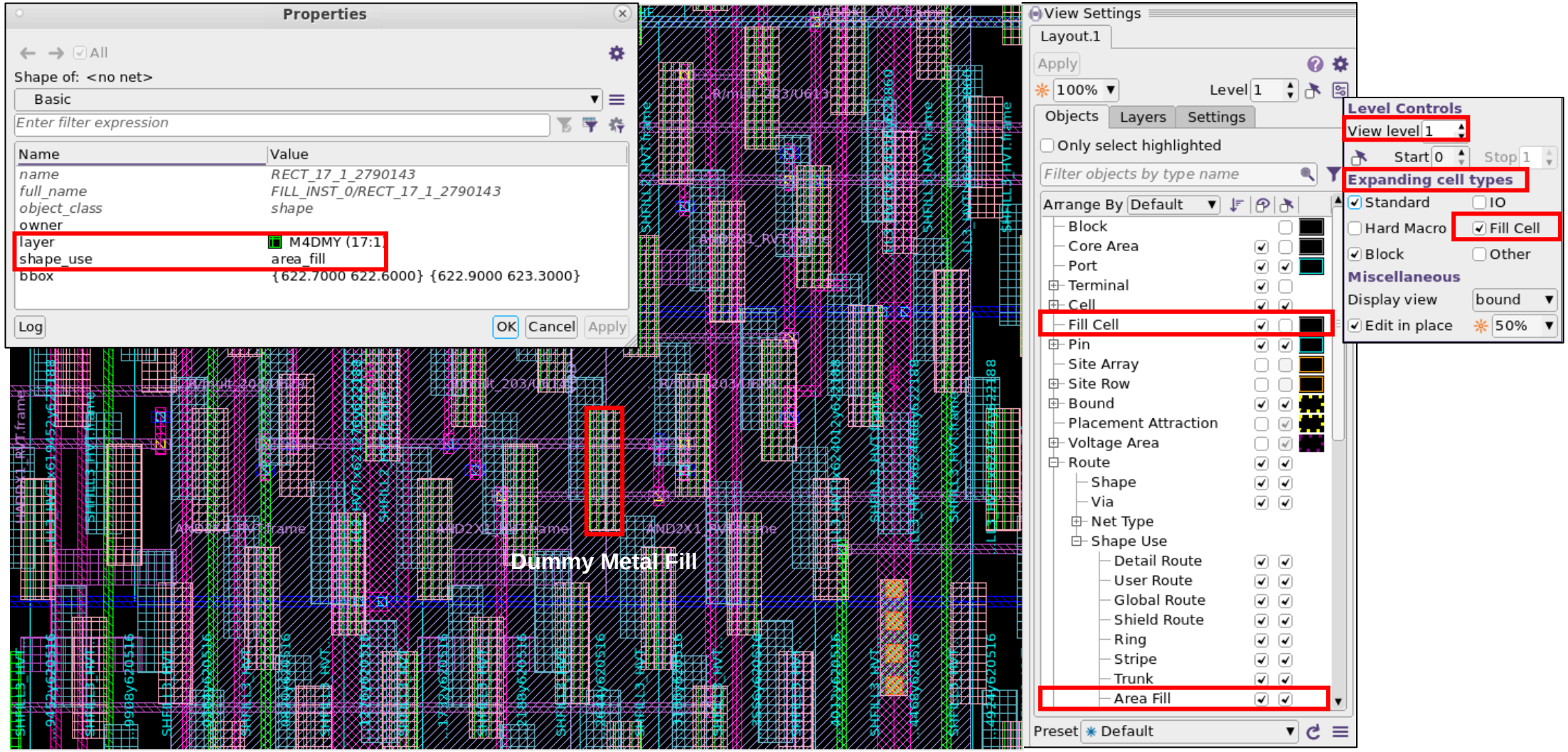
```
Start to run ICV ...
.....................................................................
.....................................................................
Updated tech file has been written at: /slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/08_pv/signoff_fill_run/icv_run_2/tech.tf

ICV is done!
ICV messages:
WARNING: Could not open I1025_NS.design, using frame view instead.
WARNING: Could not open B12I1025_NS.design, using frame view instead.
WARNING: Could not open D8I1025_NS.design, using frame view instead.
WARNING: Could not open B16I1025_NS.design, using frame view instead.
WARNING: Could not open INVX2_RVT.design, using frame view instead.
Run status:
Fills created successfully. For more details, see /slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/08_pv/signoff_fill_run/icv_run_2/signoff_create_me
tal_fill.log
Overall engine Time=0:03:31 Highest command Mem=2.745 GB
1
```


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PV(Physical Verification) : Use ICC2 (IC Compiler II tool)

- **Check Metal Fill**



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WEEK 15 : LAB for LVS

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PV(Physical Verification) : Introduction to LVS_lab.tcl

- Review and check the variables in each scripts
  - **Go to the Script Directory**

```
  < user@linux script >> pwd
/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/08_pv/LVS/script
  < user@linux script >> ls
LVS_lab.tcl
```

  - **Review LVS_lab.tcl**

```
#made by kjinsoo
# module load
#module load icv/2022.12-SP5
#module load icvwb/2021.06-SP2
# open ICV wokrbench
#icvwb &

########################
#case 1 text_short Error
########################
#run icv
icv -host_init 2 -vue -vueshort -create_lvs_short_output -c add4 -i ./Input/01_text_short_Error.gds
-s ./Input/LVS_lab.sp -sf SPICE ./Input/LVS_lab.rs

#save layout after fix Error (remove A1 text)
layout save -format gds ./Input/01_text_short_Error.gds -rename ./Input/01_text_short_Error_fix.gds
-libname add4_merged.gds__merging

#fix & rerun icv
icv -host_init 2 -vue -vueshort -create_lvs_short_output -c add4 -i ./Input/01_text_short_Error_fix.
gds -s ./Input/LVS_lab.sp -sf SPICE ./Input/LVS_lab.rs
```


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PV(Physical Verification) : Introduction to LVS_lab.tcl

- Review and check the variables in each scripts
  - **Review LVS_lab.tcl**

```
##############################
#case 2 used unused text Error
##############################

#run icv
icv -host_init 2 -vue -vueshort -create_lvs_short_output -c add4 -i ./Input/02_unused_text_Error.gds -s ./Input/LVS_lab.sp -sf SPICE ./Input/LVS_lab.rs

#save layout after fix Error (remove AS text)
layout save -format gds ./Input/02_unused_text_Error.gds -rename ./Input/02_unused_text_Error_fix.gds -libname add4_merged.gds__merging

#fix & rerun icv
icv -host_init 2 -vue -vueshort -create_lvs_short_output -c add4 -i ./Input/02_unused_text_Error_fix.gds -s ./Input/LVS_lab.sp -sf SPICE ./Input/LVS_lab.rs

##############################
#case 3 device extraction Error
##############################

#run icv
icv -host_init 2 -vue -vueshort -create_lvs_short_output -c add4 -i ./Input/03_device_extraction_Error.gds -s ./Input/LVS_lab.sp -sf SPICE ./Input/LVS_lab.rs

#save layout after fix Error (remove 5:0 layer)
layout save -format gds ./Input/03_device_extraction_Error.gds -rename ./Input/03_device_extraction_Error_fix.gds -libname add4_merged.gds__merging

#fix & rerun icv
icv -host_init 2 -vue -vueshort -create_lvs_short_output -c add4 -i ./Input/03_device_extraction_Error_fix.gds -s ./Input/LVS_lab.sp -sf SPICE ./Input/LVS_lab.rs

##############################
#case 4 unmatched device Error
##############################

#run icv
icv -host_init 2 -vue -vueshort -create_lvs_short_output -c add4 -i ./Input/04_unmatched_device_Error.gds -s ./Input/LVS_lab.sp -sf SPICE ./Input/LVS_lab.rs

#save layout after fix Error (remove 5:0 layer)
layout save -format gds ./Input/04_unmatched_device_Error.gds -rename ./Input/04_unmatched_device_Error_fix.gds -libname add4_merged.gds__merging

#fix & rerun icv
icv -host_init 2 -vue -vueshort -create_lvs_short_output -c add4 -i ./Input/04_unmatched_device_Error_fix.gds -s ./Input/LVS_lab.sp -sf SPICE ./Input/LVS_lab.rs
```


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WEEK 15 : LAB for LVS (Case 1 : Text short Error)

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PV(Physical Verification) : Use ICV (IC Validator tool)

1. **Execute the following ICV command (case 1)**
   - **Recommend the ICVWB 2021.06-SP2 version & icv/2022.12-SP5 version**

```
  < user@linux LVS >> icv -host_init 2 -vue -vueshort -create_lvs_short_output -c add4 -i ./Input/
01_text_short_Error.gds -s ./Input/LVS_lab.sp -sf SPICE ./Input/LVS_lab.rs

                                IC Validator

          Version U-2022.12-SP5 for linux64 - Oct 26, 2023 cl#9338971

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Time Started: 2023-12-19 10:09:26
Running on: us01odcvde31334.internal.synopsys.com 3.10.0-1160.31.1.el7.x86_64

-------------

Called as: icv -host_init 2 -vue -vueshort -create_lvs_short_output -c add4 -i ./Input/01_text_sho
rt_Error.gds -s ./Input/LVS_lab.sp -sf SPICE ./Input/LVS_lab.rs
```


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PV(Physical Verification) : Use ICV (IC Validator tool)

1. **Execute the following ICV command (case 1)**
   - **Recommend the ICVWB 2021.06-SP2 version & icv/2022.12-SP5 version**

```
Parsing runset "./Input/LVS_lab.rs"

./data/lvs.rs:115: warning: ICV LVS device settings currently use append mode. Use command line op
tion '-D ICV_REVERT_DEVICE_SETTING_APPEND_MODE' or runset instruction '#define ICV_REVERT_DEVICE_S
ETTING_APPEND_MODE' to revert LVS device setting back to overwrite mode in this release.
Loaded runset from cache: /u/kjinsoo/.icvrscache/5d5f3409cdf10cabb098ba5b99d3eec5-B03F5003-72BE-43
13-9278-FE01D266E969-us01odcvde31334-20399.rscache
Cached Runset Compile Time=0:00:01 User=0.47 Sys=0.05 Mem=0.434 GB

Virtual Machine Init Time=0:00:00 User=0.00 Sys=0.00 Mem=0.434 GB


Host startup begins.
Host startup done: 2 successes, 0 failures.
        us01odcvde31334.internal.synopsys.com: 2 total cores, maximum 2 commands in parallel.
Selecting "us01odcvde31334.internal.synopsys.com" as Primary host

System Startup Time=0:00:01 User=0.00 Sys=0.02 Mem=0.434 GB


Command execution begins.  Details recorded in the summary and log files:
    /slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/08_pv/LVS/run_details/add4.sum
    /slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/08_pv/LVS/run_details/LVS_lab.dp.l
og

Running ...
  [0:00:01][0%]   Total: 2 CPUs on 1 host
  [0:00:01][1%]
  [0:00:01][2%]
```

```
-------------
IC Validator Run: Time=0:00:06

IC Validator Machine Memory Report
us01odcvde31334.internal.synopsys.com : Average = 0.054 GB, Peak = 0.054 GB

Overall Disk Usage Disk=0.008 GB
Network Disk Usage Peak=0.008 GB
Group File Disk Usage Peak=0.001 GB
Overall engine Time=0:00:06 Highest command Mem=0.034 GB

Overall Master Mem=0.437 GB

Time Ended: 2023-12-19 10:09:33

IC Validator is done.
```


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PV(Physical Verification) : Use ICV (IC Validator tool)

2. **After the run completes, check the add4.LAYOUT_ERRORS file (text short Error)**
   - **Recommend the ICVWB 2021.06-SP2 version & icv/2022.12-SP5 version**

```
                    LAYOUT ERRORS RESULTS: ERRORS

                  ##### ####  ####   ###  ####   ###
                  #     #   # #   # #   # #   # #
                  ####  ####  ####  #   # ####   ###
                  #     #  #  #  #  #   # #  #      #
                  ##### #   # #   #  ###  #   # ####

================================================================================

Library name:    ./Input/01_text_short_Error.gds
Structure name:  add4
Generated by:    IC Validator RHEL64 U-2022.12-SP5.9338971 2023/10/26
Runset name:     ./Input/LVS_lab.rs
User name:       kjinsoo
Time started:    2023/12/19 10:09:29AM
Time ended:      2023/12/19 10:09:33AM

Called as: icv -host_init 2 -vue -vueshort -create_lvs_short_output -c add4 -i ./Input/01_text_shor
t_Error.gds -s ./Input/LVS_lab.sp -sf SPICE ./Input/LVS_lab.rs

                        ERROR SUMMARY

 Violation
   text_net:text_short ............................... 1 violation found.




                        ERROR DETAILS


--------------------------------------------------------------------------------
Violation
--------------------------------------------------------------------------------

./data/lvs.rs:78:text_net:text_short
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
Structure NetId Used Text layerNo dtype (position x, y)      TextOn TextFrom TextExplodedFrom
                     Text layerNo dtype (position x, y)      TextOn TextFrom TextExplodedFrom
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
add4      12     *   A    30      0     (-249.7600, 44.3150) M2     LAYER
                     A1   30      0     (-249.5000, 81.0000) M2     LAYER
```

LVS Error caused by a A and A1 not havin g the same text

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PV(Physical Verification) : Use ICV (IC Validator tool)

2. **After the run completes, check the add4.LVS_ERRORS file (unmatched device)**
   - **Recommend the ICVWB 2021.06-SP2 version & icv/2022.12-SP5 version**

```
+------------------------------------------------------------------------+
|                    ICV_Compare LVS Comparison Report                   |
+------------------------------------------------------------------------+

ICV_Compare (R) Hierarchical Layout Vs. Schematic
        RHEL64 U-2022.12-SP5.9338971 2023/10/26
Copyright (C) Synopsys, Inc. All rights reserved.


------------------------------------------------------------------------
LVS error file     = add4.LVS_ERRORS
Layout error file  = add4.LAYOUT_ERRORS
Schematic netlist  = /slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/08_pv/LVS/add4.sch_out.gz
Layout netlist     = /slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/08_pv/LVS/add4.net.gz
Runset file        = ./data/lvs.rs
Working directory  = /slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/08_pv/LVS
Compare directory  = run_details/compare
Compare start time = 2023-12-20 13:59:02



------------------------------------------------------------------------
Final comparison result:PASS

                      #####   ##   ##### #####
                      #   #  #  #  #     #
                      ##### ###### ##### #####
                      #     #    #     #     #
                      #     #    # ##### #####


TOP equivalence point:
                [add4, add4]

Comparison summary

    7 Successful equivalence points
    0 Failed equivalence points
```

```
TOP equivalence point:
                [add4, add4]

Comparison summary

    7 Successful equivalence points
    0 Failed equivalence points

TOP-level Post-compare summary (* = unmatched devices, nets or ports):

        Matched      Unmatched    Unmatched        Instance types
                     schematic    layout           [schematic, layout]
        ---------    ---------    ---------        --------------------
                4            0            0        EQUIV[cs_add, cs_add]
        ---------    ---------    ---------        --------------------
                4            0            0        Total instances

               19            0            0        Total nets

               16            0            0        Total ports



Schematic and layout agree at all user-intended equivalent points involved
in compare.
```


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PV(Physical Verification) : Use ICV (IC Validator tool)

3. **check the results with the help of VUE and the short finder tool (open ICVWB)**
   - **Recommend the ICVWB 2021.06-SP2 version & icv/2022.12-SP5 version**

```
 < user@linux LVS >> icvwb &
[1] 32113
 < user@linux LVS >>
                                    ICVWB

          Version S-2021.06-SP2 for linux64 - Nov 22, 2021 1010135

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                      https://solvnetplus.synopsys.com)
```

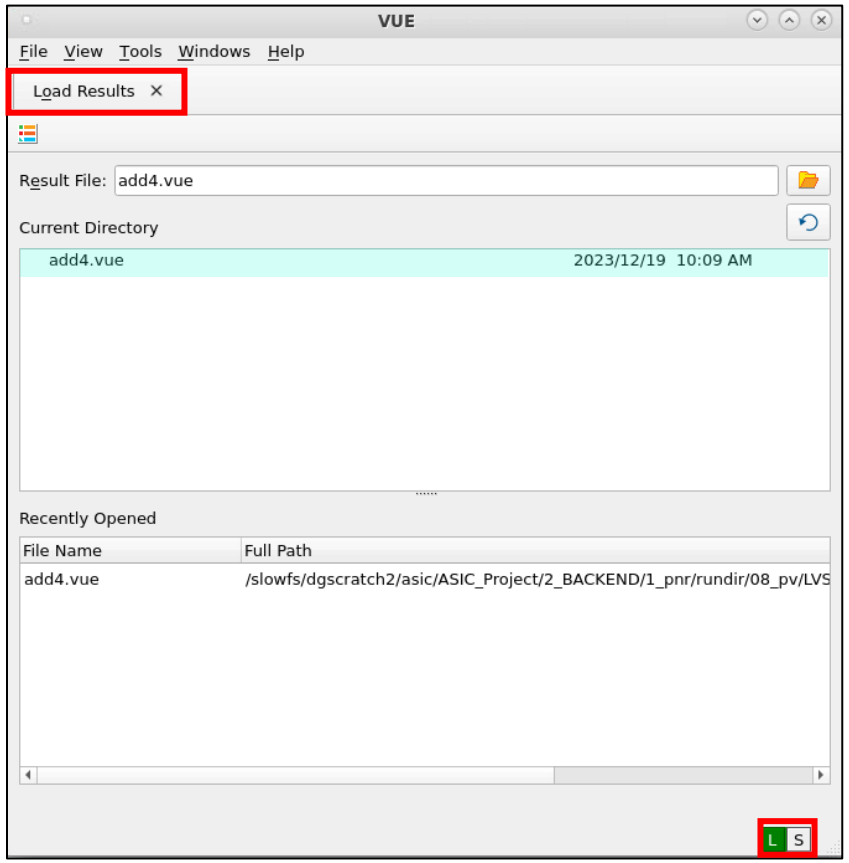
4. **Click "ICV_VUE" button (A VUE window will appear that is linked to icbwb)**

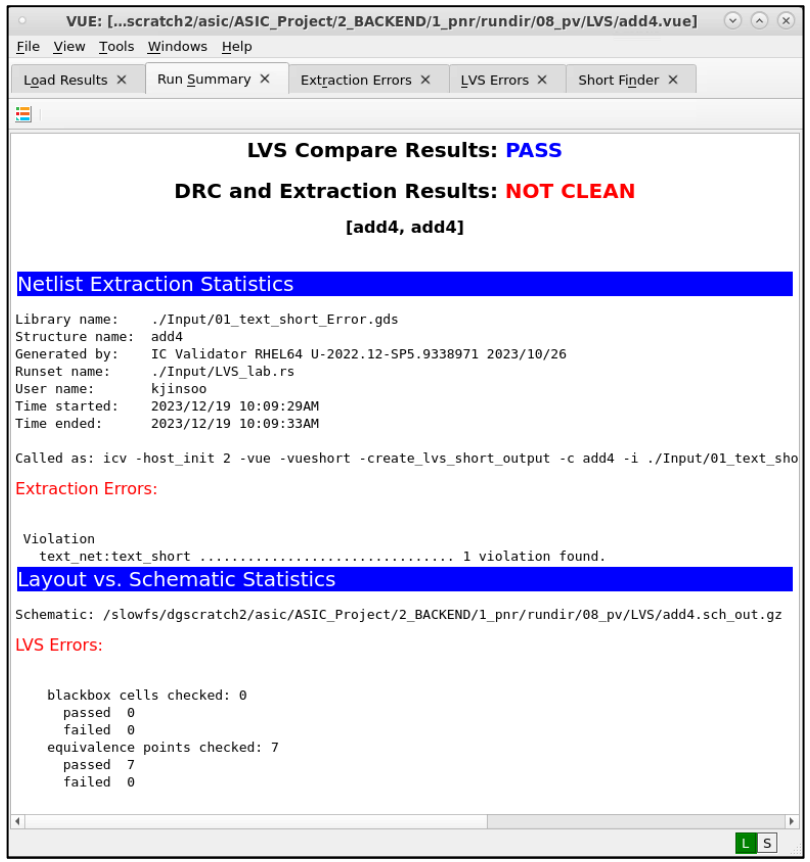


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PV(Physical Verification) : Use ICV (IC Validator tool)

5. Inside of the "Load Results" tab, click on the add4.vue file that was generated in your LVS run directory

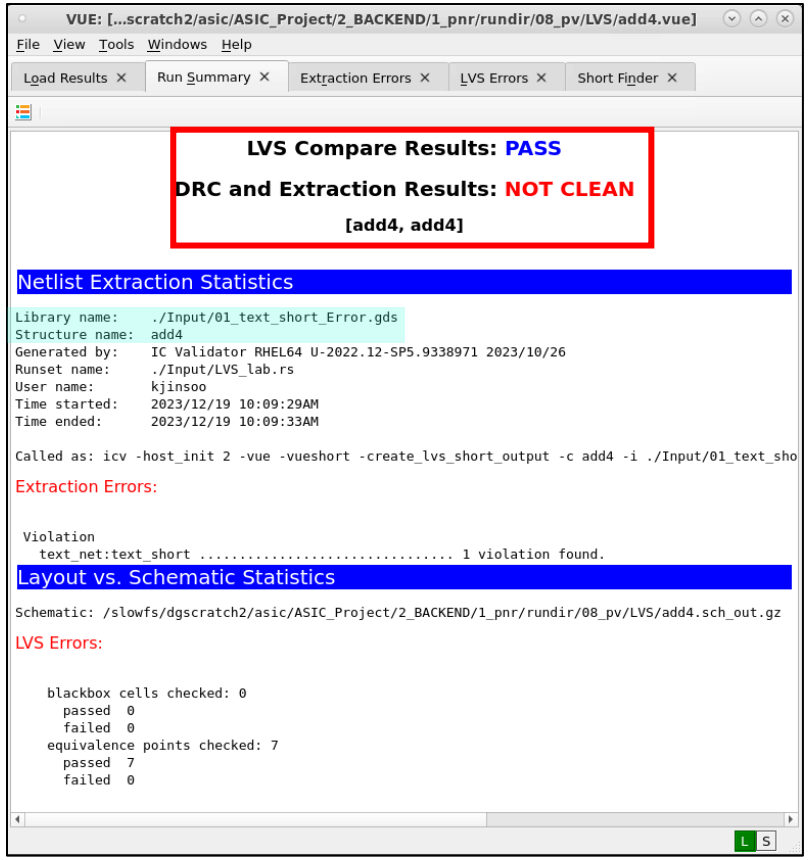




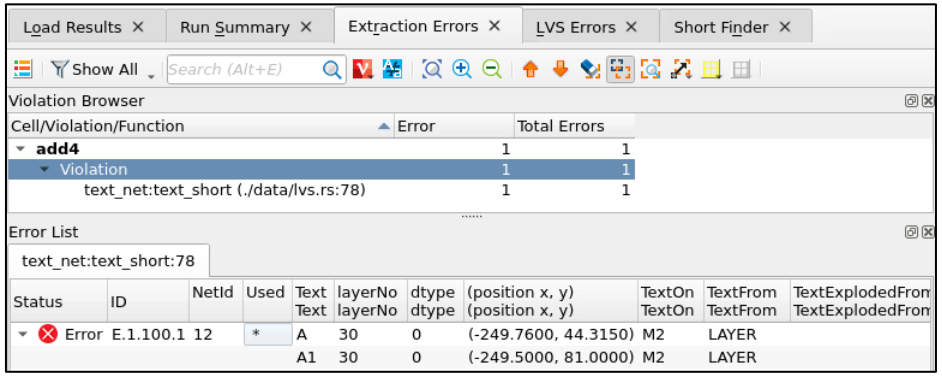
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PV(Physical Verification) : Use ICV (IC Validator tool)

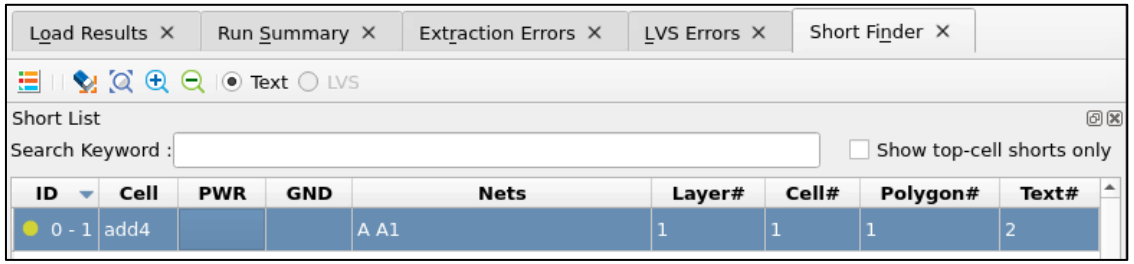
6. Check Each Tabs in VUE



The Run Summary window indicates the LVS comparison is successful but had extraction Errors



LVS Errors tab, it shows that all the equivalence points are matched

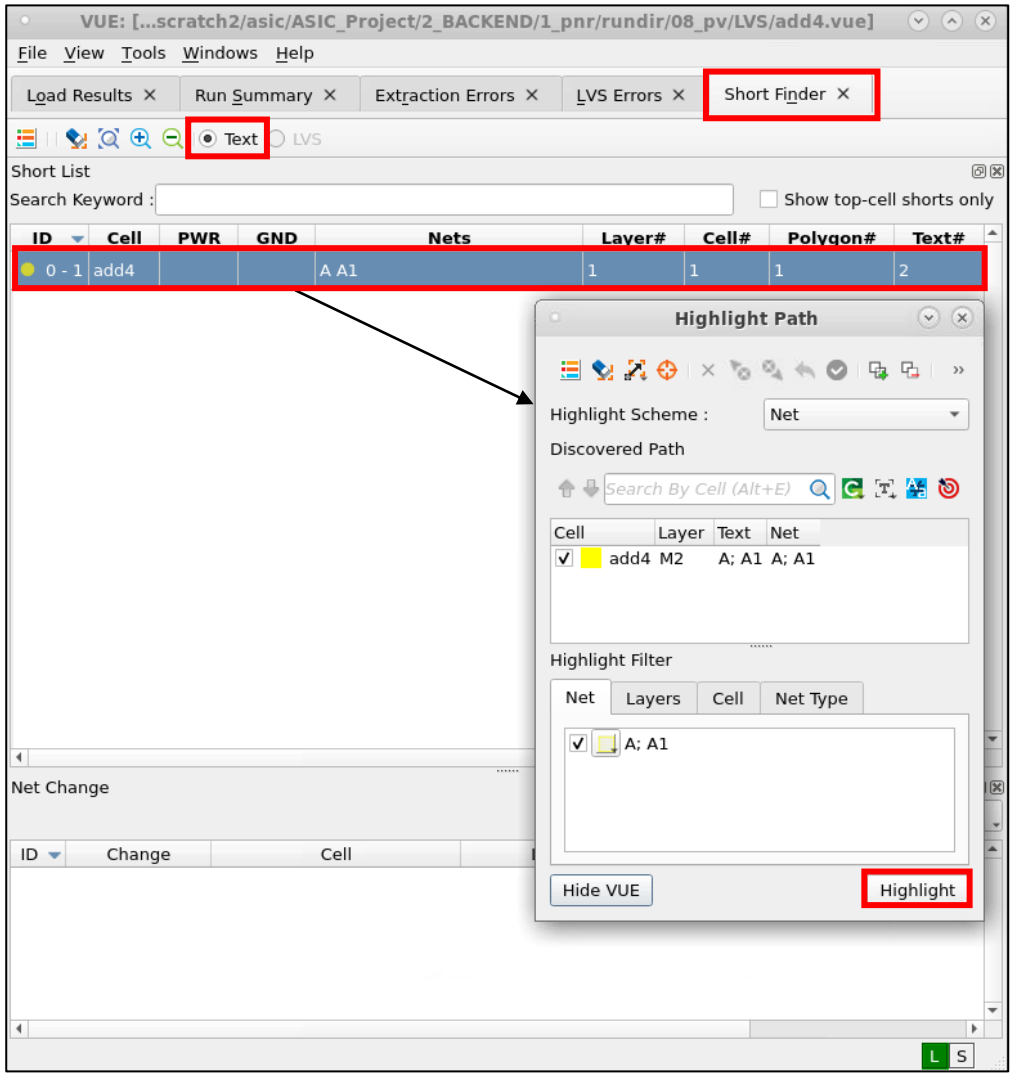


Short Finder tab helps user to debug shorts

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PV(Physical Verification) : Use ICV (IC Validator tool)

7. Analyze for two types of shorts using Short Finder tab & ICVWB

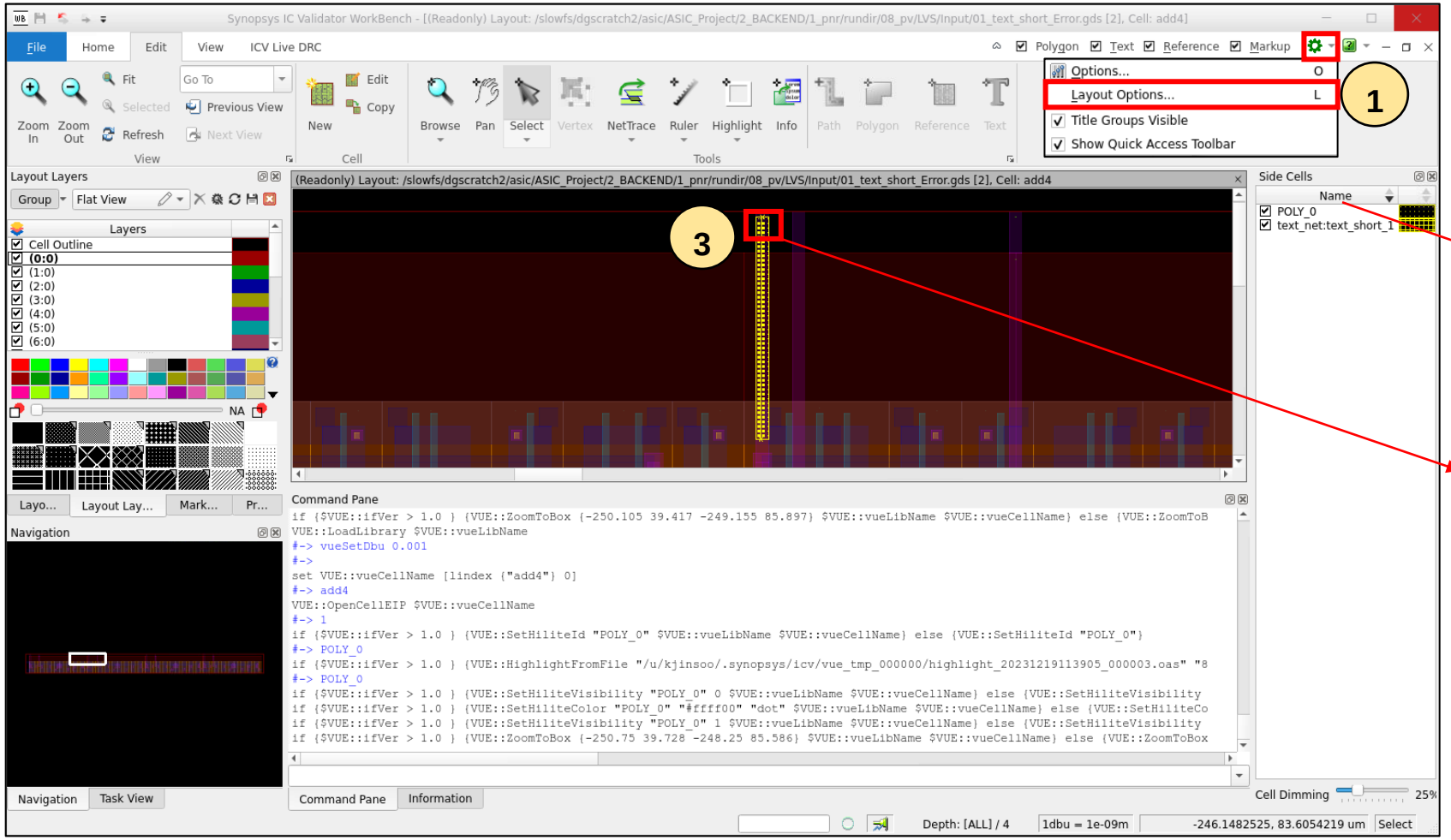


1. Short Finder tab helps user to debug & analyze shorts
2. This tab allows debug for two types of shorts Text & LVS (Since there are o nly text shorts in this design, it is the only se)
3. If the "Highlight Path" window is not visible, double clock on an entry from t he Shrot list
4. Inside the "Highlight Path" window, click on "Highlight" (This will highlight a nd zoom to the shortest path of the short in ICVWB)

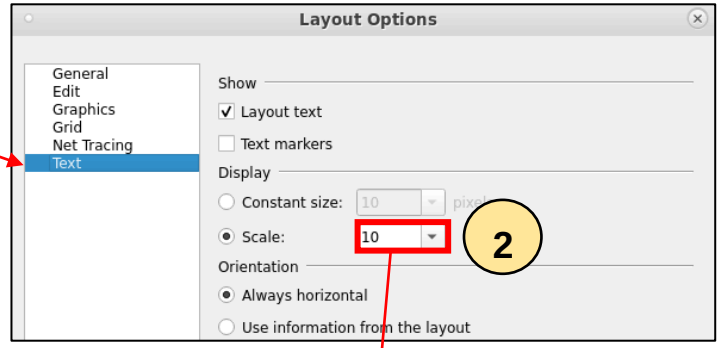
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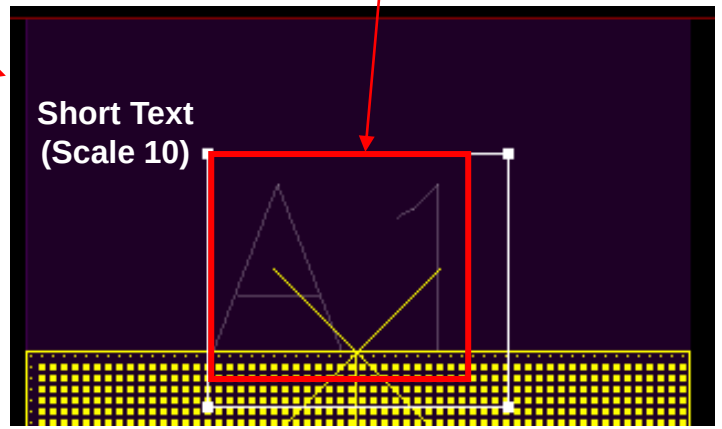
PV(Physical Verification) : Use ICV (IC Validator tool)

7. **Analyze for two types of shorts using Short Finder tab & ICVWB**
   - **In the design, you can check the path shown in yellow dotted lines (If you zoom into the path, you can observe two text [A and A1] on the same Metal)**
   - **If you are having trouble seeing text in the design, navigate to the Layout options window**



Inside of this window you can use the "Contact size" or "Scale" settings to adjust the text size in ICV WB

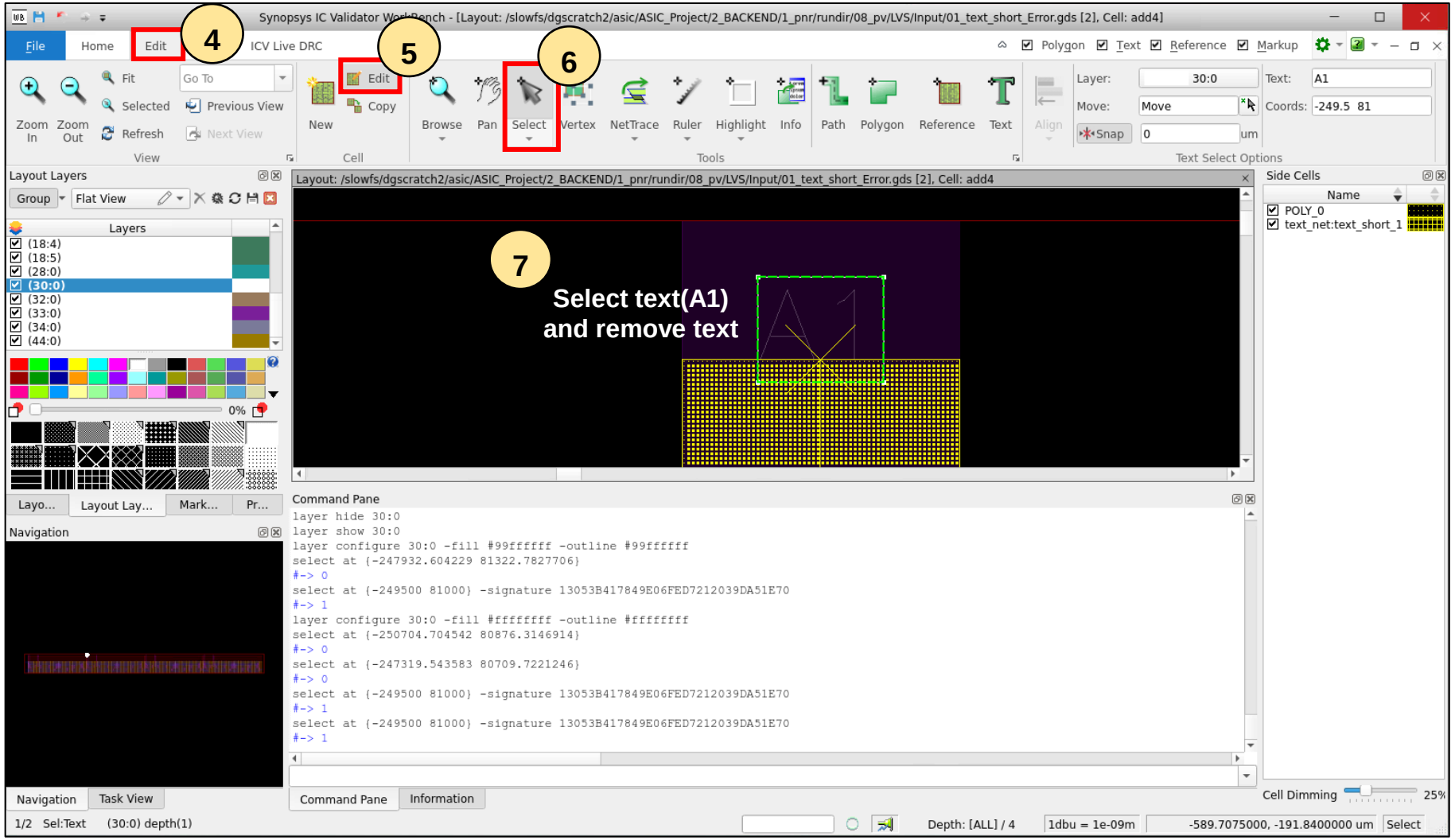




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PV(Physical Verification) : Use ICV (IC Validator tool)

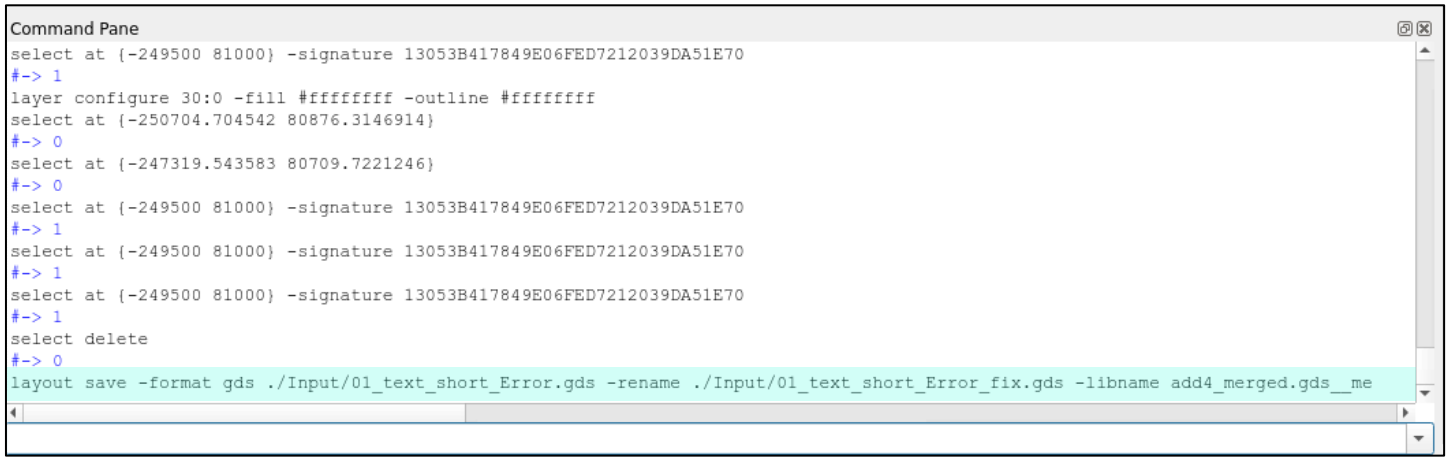
8. **Remove A1 text & save layout (gds format)**
   - **If you remove A1 based on the design net names, then design will be extraction clean (Ensure the cell is in "Edit" mode to select and delete text)**



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PV(Physical Verification) : Use ICV (IC Validator tool)

8. **Remove A1 text & save layout (gds format)**
   - **Save layout after remove text**



   - **Check gds file after save layout**

```
  < user@linux Input >> pwd
/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/08_pv/LVS/Input
  < user@linux Input >> ls
01_text_short_Error_fix.gds  02_unused_text_Error.gds           LVS_lab.rs
01_text_short_Error.gds      04_unmatched_device_Error_fix.gds  LVS_lab.sp
```


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PV(Physical Verification) : Use ICV (IC Validator tool)

9. **Execute the following ICV command**
   - **Use 01_text_short_Error_fix.gds**

```
  < user@linux LVS >> icv -host_init 2 -vue -vueshort -create_lvs_short_output -c add4 -i ./Input/
01_text_short_Error_fix.gds -s ./Input/LVS_lab.sp -sf SPICE ./Input/LVS_lab.rs

                                IC Validator

         Version U-2022.12-SP5 for linux64 - Oct 26, 2023 cl#9338971

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Time Started: 2023-12-19 13:52:44
Running on: us01odcvde31334.internal.synopsys.com 3.10.0-1160.31.1.el7.x86_64

-------------

Called as: icv -host_init 2 -vue -vueshort -create_lvs_short_output -c add4 -i ./Input/01_text_sho
rt_Error_fix.gds -s ./Input/LVS_lab.sp -sf SPICE ./Input/LVS_lab.rs
```



```
IC Validator Machine Memory Report
us01odcvde31334.internal.synopsys.com : Average = 0.166 GB, Peak = 0.166 GB

Overall Disk Usage Disk=0.009 GB
Network Disk Usage Peak=0.009 GB
Group File Disk Usage Peak=0.000 GB
Overall engine Time=0:00:07 Highest command Mem=0.031 GB

Overall Master Mem=0.438 GB

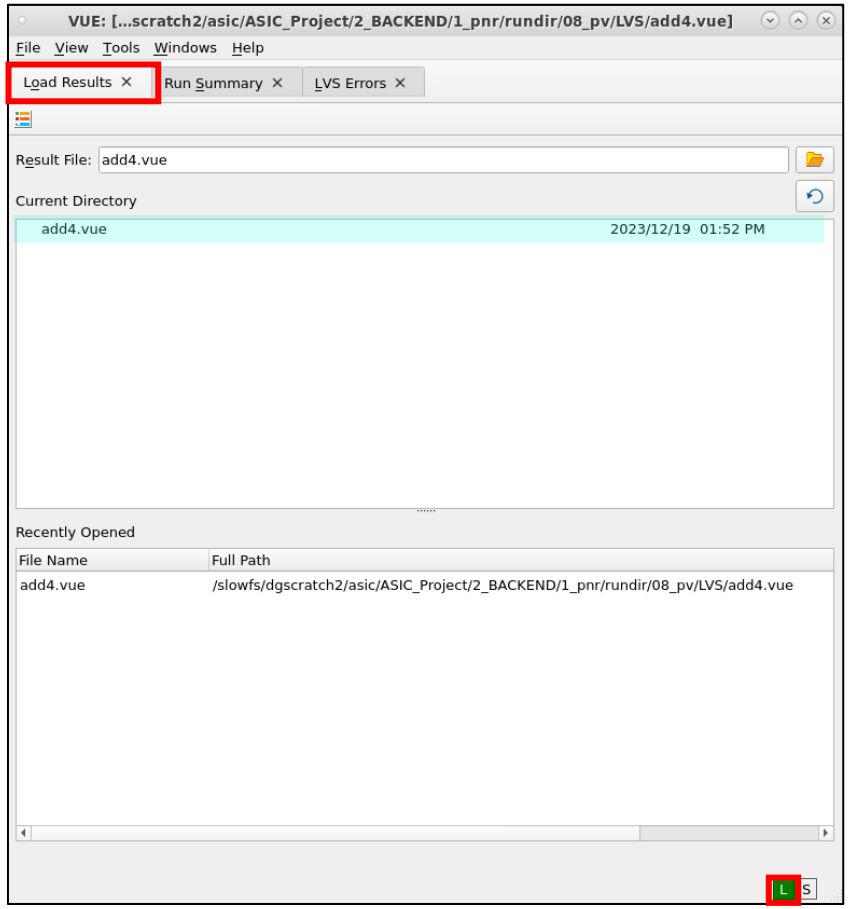
Time Ended: 2023-12-19 13:52:52

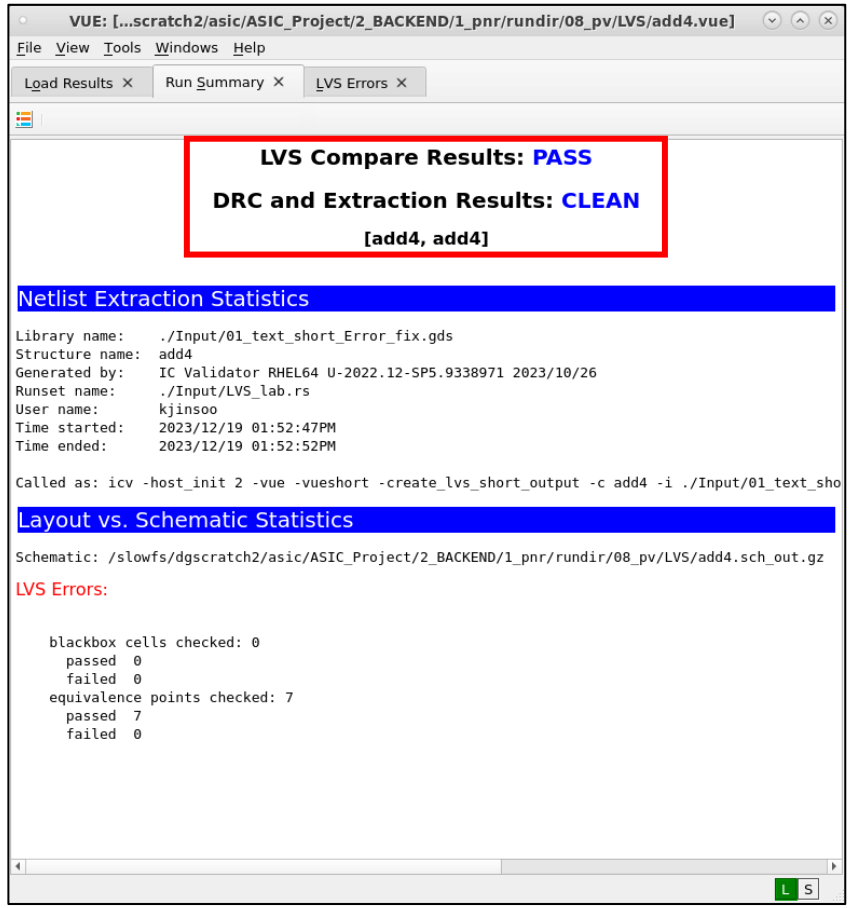
IC Validator is done.
```


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PV(Physical Verification) : Use ICV (IC Validator tool)

10. Inside of the "Load Results" tab, click on the add4.vue file that was generated in your LVS run directory





Check LVS Compare Results & DRC and Extraction Results

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WEEK 15 : LAB for LVS (Case 2 : Unused text Error)

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PV(Physical Verification) : Use ICV (IC Validator tool)

1. **Execute the following ICV command (case 2)**
   - **Recommend the ICVWB 2021.06-SP2 version & icv/2022.12-SP5 version**

```
  < user@linux LVS >> icv -host_init 2 -vue -vueshort -create_lvs_short_output -c add4 -i ./Input/
02_unused_text_Error.gds -s ./Input/LVS_lab.sp -sf SPICE ./Input/LVS_lab.rs

                                IC Validator

          Version U-2022.12-SP5 for linux64 - Oct 26, 2023 cl#9338971

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Time Started: 2023-12-19 14:06:06
Running on: us01odcvde31334.internal.synopsys.com 3.10.0-1160.31.1.el7.x86_64

-------------

Called as: icv -host_init 2 -vue -vueshort -create_lvs_short_output -c add4 -i ./Input/02_unused_t
ext_Error.gds -s ./Input/LVS_lab.sp -sf SPICE ./Input/LVS_lab.rs
```


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PV(Physical Verification) : Use ICV (IC Validator tool)

1. **Execute the following ICV command (case 2)**
   - **Recommend the ICVWB 2021.06-SP2 version & icv/2022.12-SP5 version**

```
Parsing runset "./Input/LVS_lab.rs"

./data/lvs.rs:115: warning: ICV LVS device settings currently use append mode. Use command line op
tion '-D ICV_REVERT_DEVICE_SETTING_APPEND_MODE' or runset instruction '#define ICV_REVERT_DEVICE_S
ETTING_APPEND_MODE' to revert LVS device setting back to overwrite mode in this release.
Loaded runset from cache: /u/kjinsoo/.icvrscache/5d5f3409cdf10cabb098ba5b99d3eec5-B03F5003-72BE-43
13-9278-FE01D266E969-us01odcvde31334-20399.rscache
Cached Runset Compile Time=0:00:01 User=0.47 Sys=0.05 Mem=0.434 GB

Virtual Machine Init Time=0:00:00 User=0.00 Sys=0.00 Mem=0.434 GB


Host startup begins.
Host startup done: 2 successes, 0 failures.
        us01odcvde31334.internal.synopsys.com: 2 total cores, maximum 2 commands in parallel.
Selecting "us01odcvde31334.internal.synopsys.com" as Primary host

System Startup Time=0:00:01 User=0.00 Sys=0.02 Mem=0.434 GB


Command execution begins.  Details recorded in the summary and log files:
    /slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/08_pv/LVS/run_details/add4.sum
    /slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/08_pv/LVS/run_details/LVS_lab.dp.l
og

Running ...
  [0:00:01][0%]   Total: 2 CPUs on 1 host
  [0:00:01][1%]
  [0:00:01][2%]
```

```
-------------
IC Validator Run: Time=0:00:06

IC Validator Machine Memory Report
us01odcvde31334.internal.synopsys.com : Average = 0.054 GB, Peak = 0.054 GB

Overall Disk Usage Disk=0.008 GB
Network Disk Usage Peak=0.008 GB
Group File Disk Usage Peak=0.001 GB
Overall engine Time=0:00:06 Highest command Mem=0.034 GB

Overall Master Mem=0.437 GB

Time Ended: 2023-12-19 10:09:33

IC Validator is done.
```


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PV(Physical Verification) : Use ICV (IC Validator tool)

2. **After the run completes, check the add4.LAYOUT_ERRORS file (unused text Error)**
   - **Recommend the ICVWB 2021.06-SP2 version & icv/2022.12-SP5 version**

```
                    LAYOUT ERRORS RESULTS: ERRORS

                 ##### ####  ####   ###  ####   ###
                 #     #   # #   # #   # #   # #
                 ####  ####  ####  #   # ####   ###
                 #     #  #  #  #  #   # #  #      #
                 ##### #   # #   #  ###  #   # ####

================================================================================

Library name:    ./Input/02_unused_text_Error.gds
Structure name:  add4
Generated by:    IC Validator RHEL64 U-2022.12-SP5.9338971 2023/10/26
Runset name:     ./Input/LVS_lab.rs
User name:       kjinsoo
Time started:    2023/12/19 02:06:09PM
Time ended:      2023/12/19 02:06:13PM

Called as: icv -host_init 2 -vue -vueshort -create_lvs_short_output -c add4 -i ./Input/02_unused_tex
t_Error.gds -s ./Input/LVS_lab.sp -sf SPICE ./Input/LVS_lab.rs

                        ERROR SUMMARY

 Violation
   text_net:text_unused ............................. 1 violation found.




                        ERROR DETAILS


--------------------------------------------------------------------------------
Violation
--------------------------------------------------------------------------------

./data/lvs.rs:78:text_net:text_unused
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
Structure UnusedText layerNo dtype (position x, y)
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
add4      AS         28      0     (-253.8200, 17.9400)
```

LVS Error caused by unused text

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PV(Physical Verification) : Use ICV (IC Validator tool)

2. **After the run completes, check the add4.LVS_ERRORS file (unmatched device)**
   - **Recommend the ICVWB 2021.06-SP2 version & icv/2022.12-SP5 version**

```
+------------------------------------------------------------------------+
|                    ICV_Compare LVS Comparison Report                   |
+------------------------------------------------------------------------+

ICV_Compare (R) Hierarchical Layout Vs. Schematic
        RHEL64 U-2022.12-SP5.9338971 2023/10/26
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------------------------------------------------------------------------
LVS error file     = add4.LVS_ERRORS
Layout error file  = add4.LAYOUT_ERRORS
Schematic netlist  = /slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/08_pv/LVS/add4.sch_out.gz
Layout netlist     = /slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/08_pv/LVS/add4.net.gz
Runset file        = ./data/lvs.rs
Working directory  = /slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/08_pv/LVS
Compare directory  = run_details/compare
Compare start time = 2023-12-20 13:59:02



------------------------------------------------------------------------
Final comparison result:PASS

                      #####   ##   ##### #####
                      #   #  #  #  #     #
                      ##### ###### ##### #####
                      #     #    #     #     #
                      #     #    # ##### #####


TOP equivalence point:
                [add4, add4]

Comparison summary

    7 Successful equivalence points
    0 Failed equivalence points
```

```
TOP equivalence point:
                [add4, add4]

Comparison summary

    7 Successful equivalence points
    0 Failed equivalence points

TOP-level Post-compare summary (* = unmatched devices, nets or ports):

        Matched      Unmatched    Unmatched        Instance types
                     schematic    layout           [schematic, layout]
        ---------    ---------    ---------        --------------------
                4            0            0        EQUIV[cs_add, cs_add]
        ---------    ---------    ---------        --------------------
                4            0            0        Total instances

               19            0            0        Total nets

               16            0            0        Total ports



Schematic and layout agree at all user-intended equivalent points involved
in compare.
```


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PV(Physical Verification) : Use ICV (IC Validator tool)

3. **check the results with the help of VUE and the short finder tool (open ICVWB)**
   - **Recommend the ICVWB 2021.06-SP2 version & icv/2022.12-SP5 version**

```
 < user@linux LVS >> icvwb &
[1] 32113
 < user@linux LVS >>
                                    ICVWB

          Version S-2021.06-SP2 for linux64 - Nov 22, 2021 1010135

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                      https://solvnetplus.synopsys.com)
```

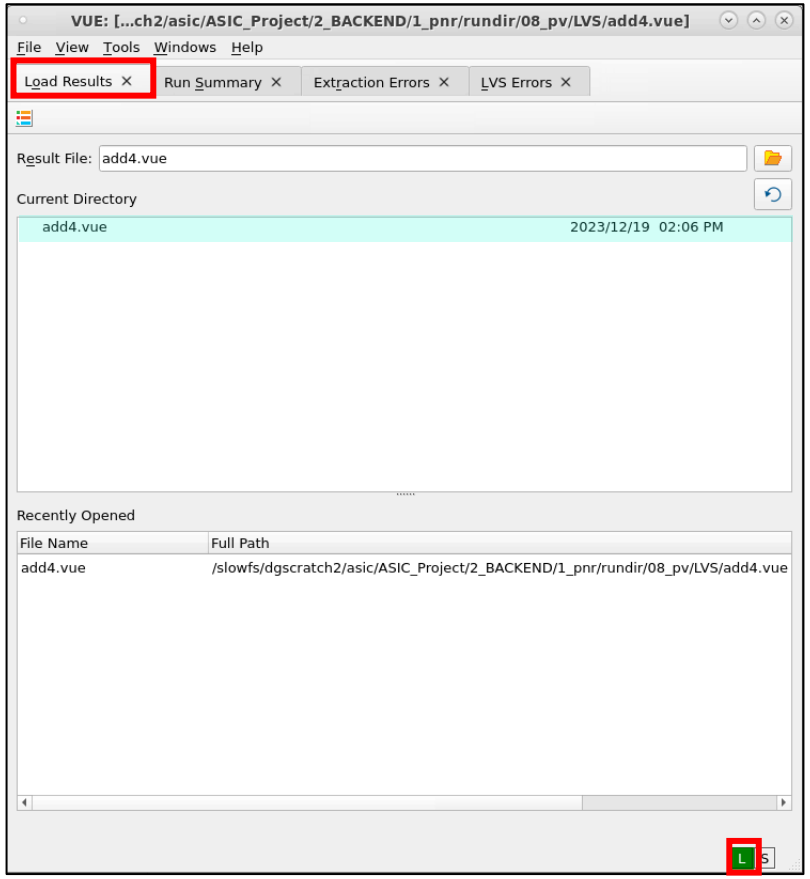
4. **Click "ICV_VUE" button (A VUE window will appear that is linked to icbwb)**

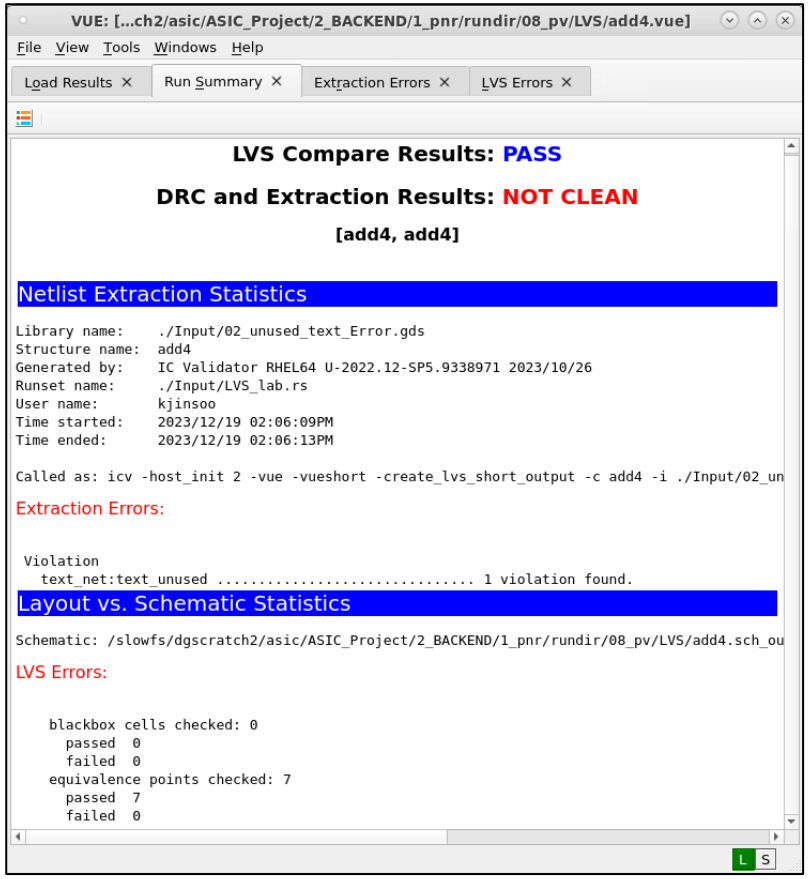


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PV(Physical Verification) : Use ICV (IC Validator tool)

5. **Inside of the "Load Results" tab, click on the add4.vue file that was generated in your LVS run directory**

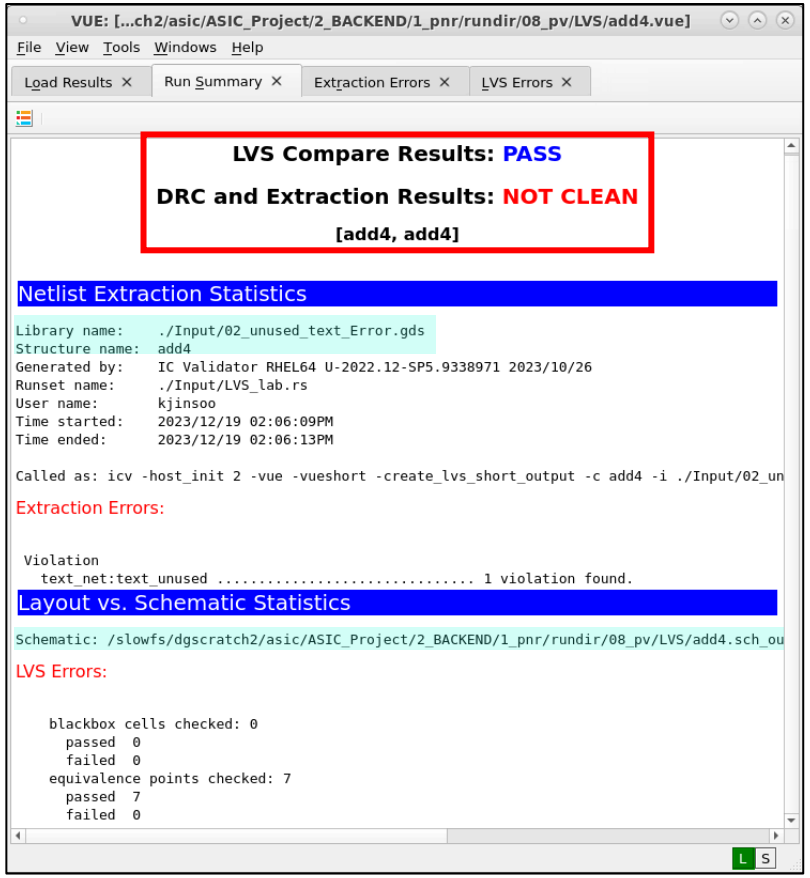




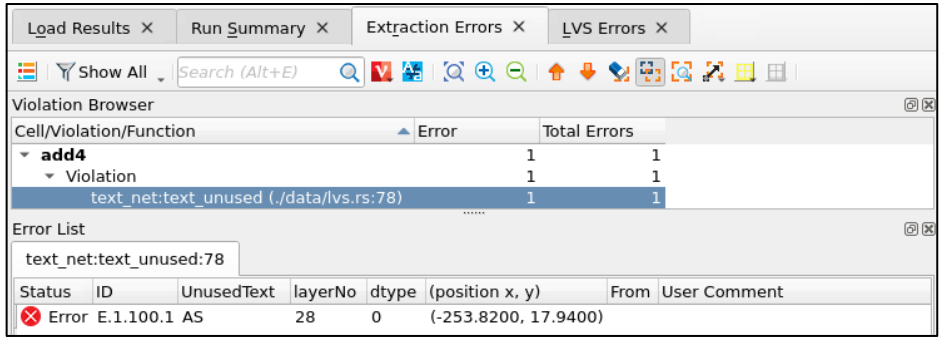
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PV(Physical Verification) : Use ICV (IC Validator tool)

6. Check Each Tabs in VUE



The Run Summary window indicates the LVS comparison is successful but had extraction Errors

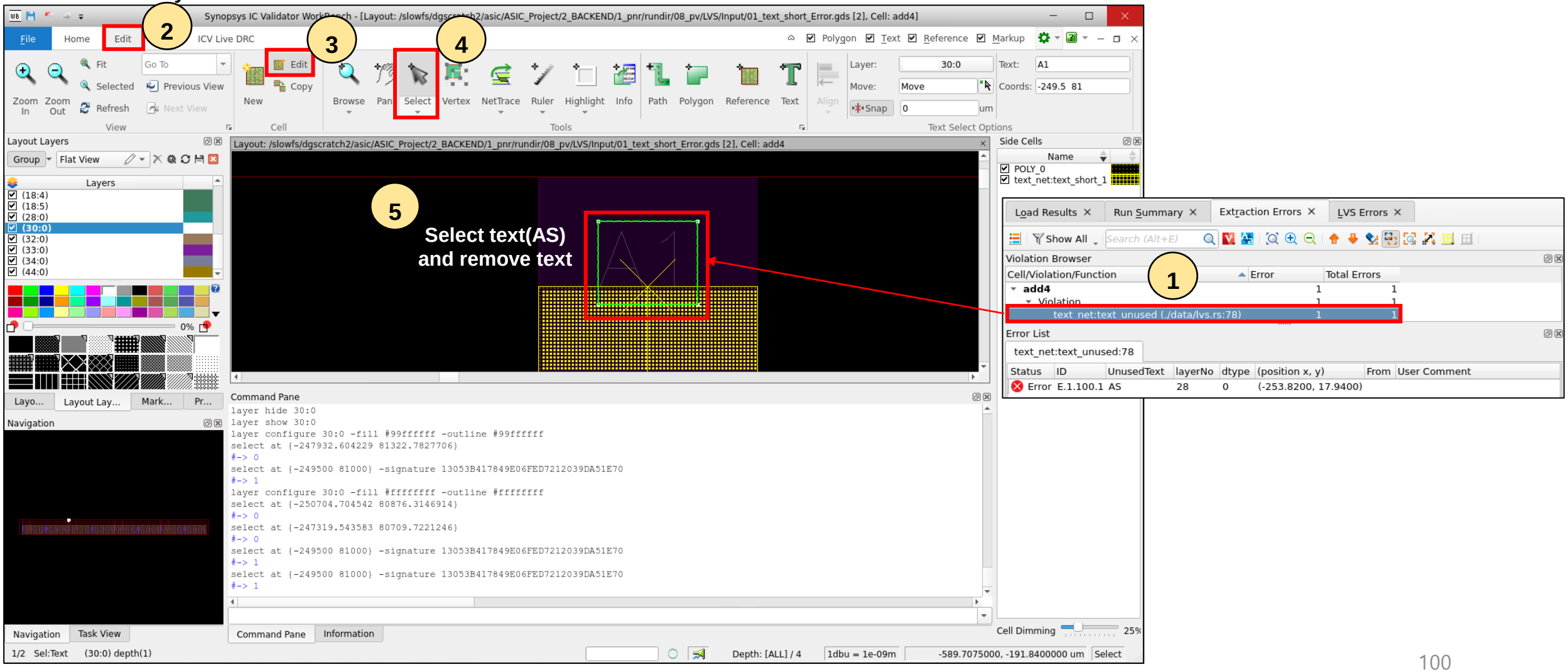


LVS Errors tab, it shows that all the equivalence points are matched

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PV(Physical Verification) : Use ICV (IC Validator tool)

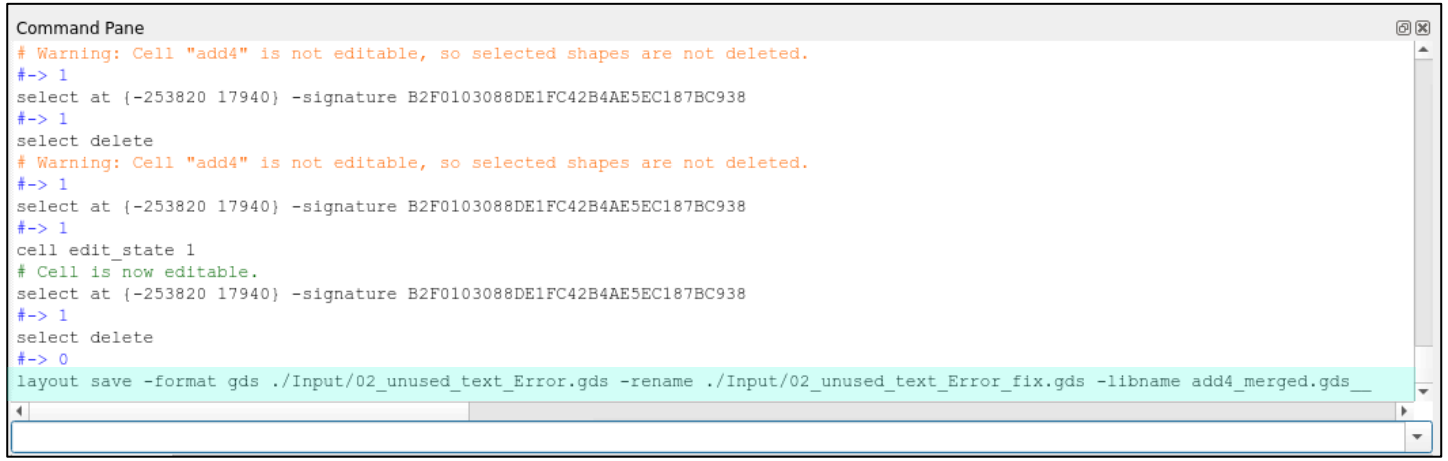
7. **Remove AS text & save layout (gds format)**
   - **When you right-click on the first violation and click highlight, you will observe a text object [AS] is on the design which is not connected t o any net**



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PV(Physical Verification) : Use ICV (IC Validator tool)

7. **Remove A1 text & save layout (gds format)**
   - **Save layout after remove text**



   - **Check gds file after save layout**

```
  < user@linux Input >> pwd
/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/08_pv/LVS/Input
  < user@linux Input >> ls
01_text_short_Error_fix.gds   02_unused_text_Error.gds           LVS_lab.sp
01_text_short_Error.gds       04_unmatched_device_Error_fix.gds
02_unused_text_Error_fix.gds  LVS_lab.rs
```


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PV(Physical Verification) : Use ICV (IC Validator tool)

8. **Execute the following ICV command**
   - **Use 01_text_short_Error_fix.gds**

```
  < user@linux LVS >> icv -host_init 2 -vue -vueshort -create_lvs_short_output -c add4 -i ./Input/
02_unused_text_Error_fix.gds -s ./Input/LVS_lab.sp -sf SPICE ./Input/LVS_lab.rs

                                IC Validator

          Version U-2022.12-SP5 for linux64 - Oct 26, 2023 cl#9338971

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Time Started: 2023-12-19 14:55:21
Running on: us01odcvde31334.internal.synopsys.com 3.10.0-1160.31.1.el7.x86_64

-------------

Called as: icv -host_init 2 -vue -vueshort -create_lvs_short_output -c add4 -i ./Input/02_unused_t
ext_Error_fix.gds -s ./Input/LVS_lab.sp -sf SPICE ./Input/LVS_lab.rs
```



```
IC Validator Machine Memory Report
us01odcvde31334.internal.synopsys.com : Average = 0.166 GB, Peak = 0.166 GB

Overall Disk Usage Disk=0.009 GB
Network Disk Usage Peak=0.009 GB
Group File Disk Usage Peak=0.000 GB
Overall engine Time=0:00:07 Highest command Mem=0.031 GB

Overall Master Mem=0.438 GB

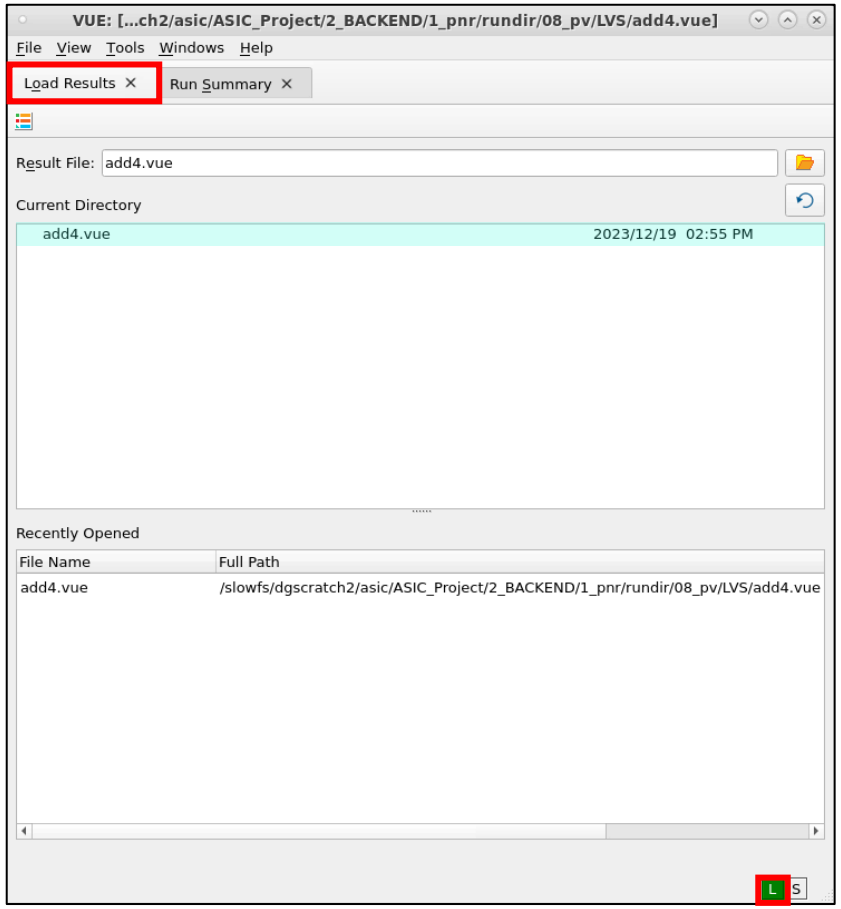
Time Ended: 2023-12-19 13:52:52

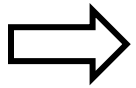
IC Validator is done.
```

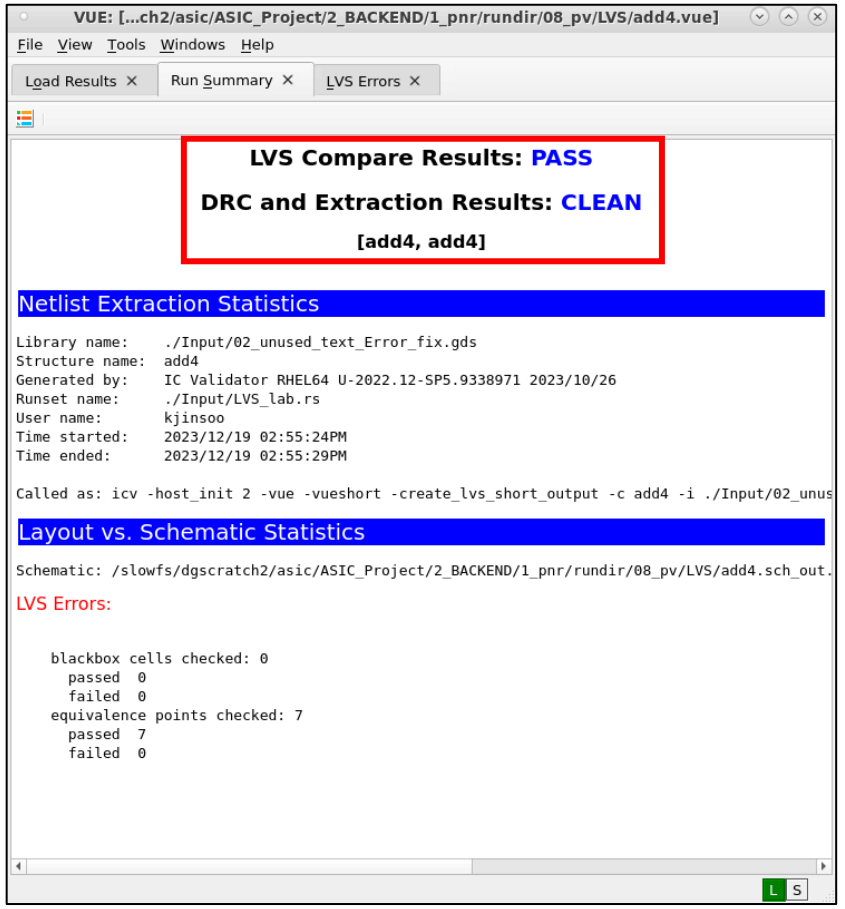

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PV(Physical Verification) : Use ICV (IC Validator tool)

9. Inside of the "Load Results" tab, click on the add4.vue file that was generated in your LVS run directory







Check LVS Compare Results & DRC and Extraction Results

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WEEK 15 : LAB for LVS (Case 3 : Device extraction Error)

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PV(Physical Verification) : Use ICV (IC Validator tool)

1. **Execute the following ICV command (case 3)**
   - **Recommend the ICVWB 2021.06-SP2 version & icv/2022.12-SP5 version**

```
  < user@linux LVS >> icv -host_init 2 -vue -vueshort -create_lvs_short_output -c add4 -i ./Input/
03_device_extraction_Error.gds -s ./Input/LVS_lab.sp -sf SPICE ./Input/LVS_lab.rs

                                IC Validator

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Time Started: 2023-12-20 09:55:50
Running on: us01odcvde31334.internal.synopsys.com 3.10.0-1160.31.1.el7.x86_64

-------------

Called as: icv -host_init 2 -vue -vueshort -create_lvs_short_output -c add4 -i ./Input/03_device_e
xtraction_Error.gds -s ./Input/LVS_lab.sp -sf SPICE ./Input/LVS_lab.rs
```


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PV(Physical Verification) : Use ICV (IC Validator tool)

1. **Execute the following ICV command (case 3)**
   - **Recommend the ICVWB 2021.06-SP2 version & icv/2022.12-SP5 version**

```
Parsing runset "./Input/LVS_lab.rs"

./data/lvs.rs:115: warning: ICV LVS device settings currently use append mode. Use command line op
tion '-D ICV_REVERT_DEVICE_SETTING_APPEND_MODE' or runset instruction '#define ICV_REVERT_DEVICE_S
ETTING_APPEND_MODE' to revert LVS device setting back to overwrite mode in this release.
Loaded runset from cache: /u/kjinsoo/.icvrscache/5d5f3409cdf10cabb098ba5b99d3eec5-B03F5003-72BE-43
13-9278-FE01D266E969-us01odcvde31334-20399.rscache
Cached Runset Compile Time=0:00:01 User=0.47 Sys=0.05 Mem=0.434 GB

Virtual Machine Init Time=0:00:00 User=0.00 Sys=0.00 Mem=0.434 GB


Host startup begins.
Host startup done: 2 successes, 0 failures.
        us01odcvde31334.internal.synopsys.com: 2 total cores, maximum 2 commands in parallel.
Selecting "us01odcvde31334.internal.synopsys.com" as Primary host

System Startup Time=0:00:01 User=0.00 Sys=0.02 Mem=0.434 GB


Command execution begins.  Details recorded in the summary and log files:
    /slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/08_pv/LVS/run_details/add4.sum
    /slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/08_pv/LVS/run_details/LVS_lab.dp.l
og

Running ...
  [0:00:01][0%]   Total: 2 CPUs on 1 host
  [0:00:01][1%]
  [0:00:01][2%]
```

```
-------------
IC Validator Run: Time=0:00:06

IC Validator Machine Memory Report
us01odcvde31334.internal.synopsys.com : Average = 0.054 GB, Peak = 0.054 GB

Overall Disk Usage Disk=0.008 GB
Network Disk Usage Peak=0.008 GB
Group File Disk Usage Peak=0.001 GB
Overall engine Time=0:00:06 Highest command Mem=0.034 GB

Overall Master Mem=0.437 GB

Time Ended: 2023-12-19 10:09:33

IC Validator is done.
```


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PV(Physical Verification) : Use ICV (IC Validator tool)

2. **After the run completes, check the add4.LAYOUT_ERRORS file (device extraction Error)**
   - **Recommend the ICVWB 2021.06-SP2 version & icv/2022.12-SP5 version**

```
                    LAYOUT ERRORS RESULTS: ERRORS

                  ##### ####  ####   ###  ####   ###
                  #     #   # #   # #   # #   # #
                  ####  ####  ####  #   # ####   ###
                  #     #  #  #  #  #   # #  #      #
                  ##### #   # #   #  ###  #   # ####

==========================================================================

Library name:    ./Input/03_device_extraction_Error.gds
Structure name:  add4
Generated by:    IC Validator RHEL64 U-2022.12-SP5.9338971 2023/10/26
Runset name:     ./Input/LVS_lab.rs
User name:       kjinsoo
Time started:    2023/12/20 10:01:39AM
Time ended:      2023/12/20 10:01:43AM

Called as: icv -host_init 2 -vue -vueshort -create_lvs_short_output -c add4 -i ./Input/03_device_e
xtraction_Error.gds -s ./Input/LVS_lab.sp -sf SPICE ./Input/LVS_lab.rs

                        ERROR SUMMARY

 Violation
   nmos[n]:device extraction errors ................... 1 violation found.




                        ERROR DETAILS


--------------------------------------------------------------------------
Violation
--------------------------------------------------------------------------

./data/lvs.rs:94:nmos[n]:device extraction errors
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
Structure  Pin     Error Type         (position x, y)
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
add4       SRC|DRN TOO_MANY_TERMINALS (-335.4975, 11.2825)
```

The device has too many Terminals, caus ing an extraction Errors

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PV(Physical Verification) : Use ICV (IC Validator tool)

2. **After the run completes, check the add4.LVS_ERRORS file (device extraction Error)**
   - **Recommend the ICVWB 2021.06-SP2 version & icv/2022.12-SP5 version**

```
+------------------------------------------------------------------------+
|                    ICV_Compare LVS Comparison Report                   |
+------------------------------------------------------------------------+

ICV_Compare (R) Hierarchical Layout Vs. Schematic
        RHEL64 U-2022.12-SP5.9338971 2023/10/26
Copyright (C) Synopsys, Inc. All rights reserved.


------------------------------------------------------------------------
LVS error file     = add4.LVS_ERRORS
Layout error file  = add4.LAYOUT_ERRORS
Schematic netlist  = /slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/08_pv/LVS/add4.sch_out.gz
Layout netlist     = /slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/08_pv/LVS/add4.net.gz
Runset file        = ./data/lvs.rs
Working directory  = /slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/08_pv/LVS
Compare directory  = run_details/compare
Compare start time = 2023-12-20 13:59:02



------------------------------------------------------------------------
Final comparison result:PASS

                      #####   ##   ##### #####
                      #   #  #  #  #     #
                      ##### ###### ##### #####
                      #     #    #     #     #
                      #     #    # ##### #####


TOP equivalence point:
                [add4, add4]

Comparison summary

    7 Successful equivalence points
    0 Failed equivalence points
```

```
TOP equivalence point:
                [add4, add4]

Comparison summary

    7 Successful equivalence points
    0 Failed equivalence points

TOP-level Post-compare summary (* = unmatched devices, nets or ports):

        Matched      Unmatched    Unmatched        Instance types
                     schematic    layout           [schematic, layout]
        ---------    ---------    ---------        --------------------
                4            0            0        EQUIV[cs_add, cs_add]
        ---------    ---------    ---------        --------------------
                4            0            0        Total instances

               19            0            0        Total nets

               16            0            0        Total ports



Schematic and layout agree at all user-intended equivalent points involved
in compare.
```


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PV(Physical Verification) : Use ICV (IC Validator tool)

3. **check the results with the help of VUE and the short finder tool (open ICVWB)**
   - **Recommend the ICVWB 2021.06-SP2 version & icv/2022.12-SP5 version**

```
 < user@linux LVS >> icvwb &
[1] 32113
 < user@linux LVS >>
                                    ICVWB

          Version S-2021.06-SP2 for linux64 - Nov 22, 2021 1010135

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                      https://solvnetplus.synopsys.com)
```

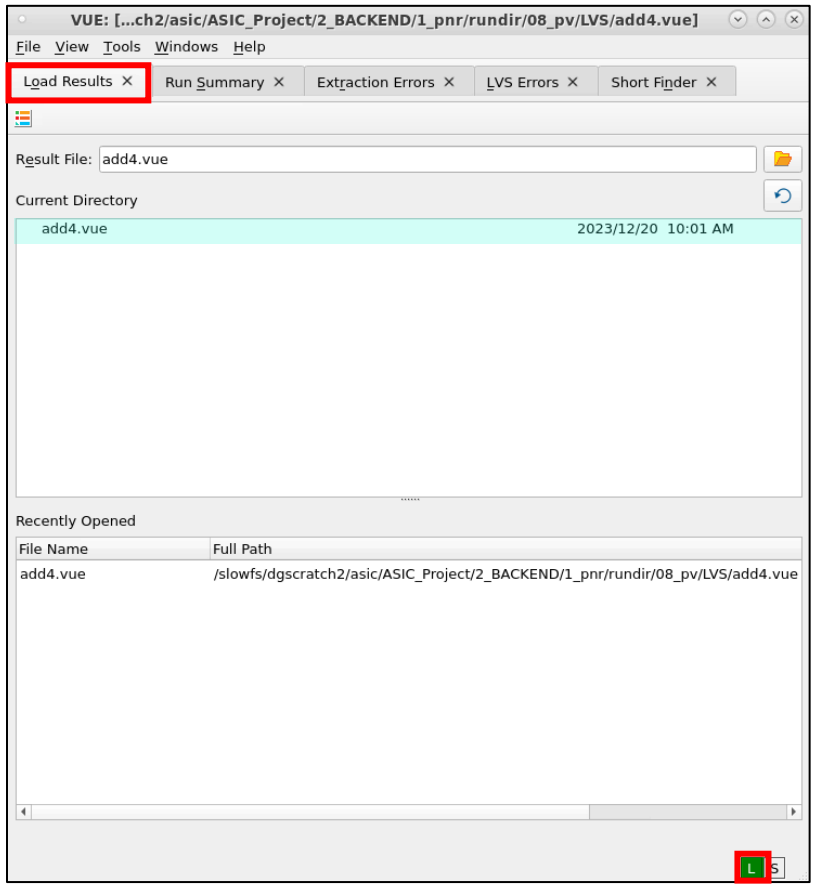
4. **Click "ICV_VUE" button (A VUE window will appear that is linked to icbwb)**

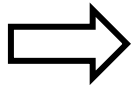


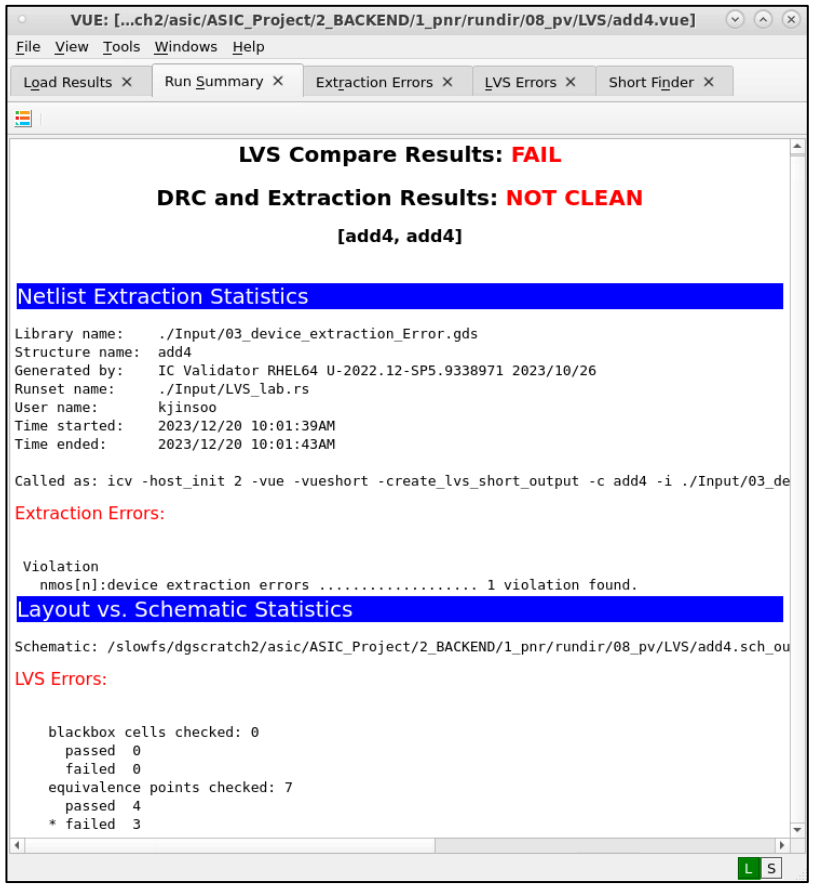
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PV(Physical Verification) : Use ICV (IC Validator tool)

5. Inside of the "Load Results" tab, click on the add4.vue file that was generated in your LVS run directory



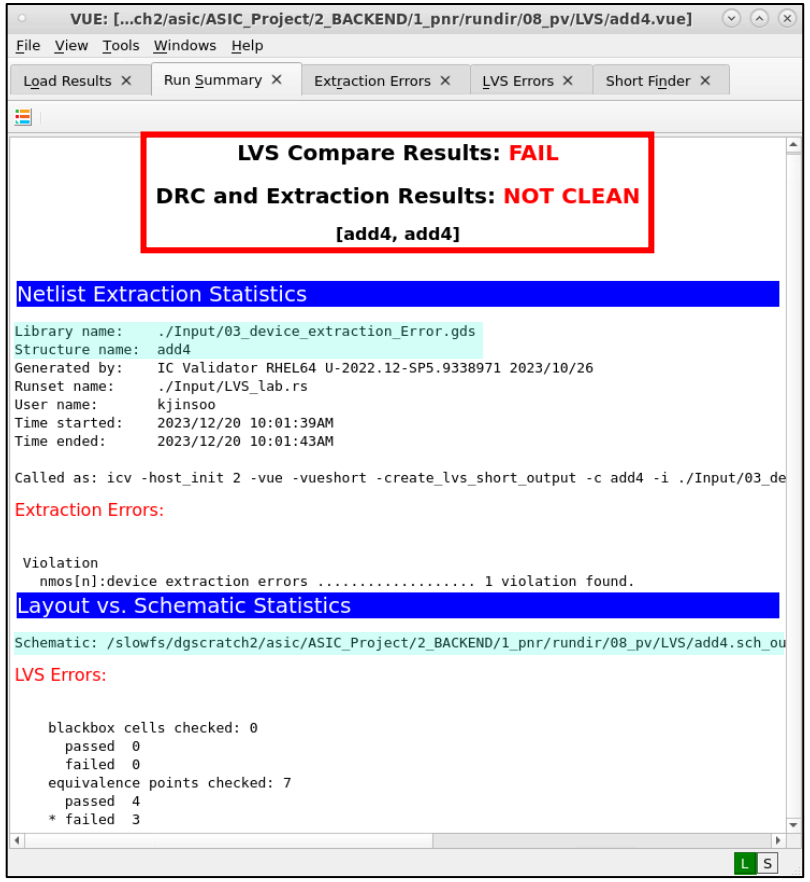




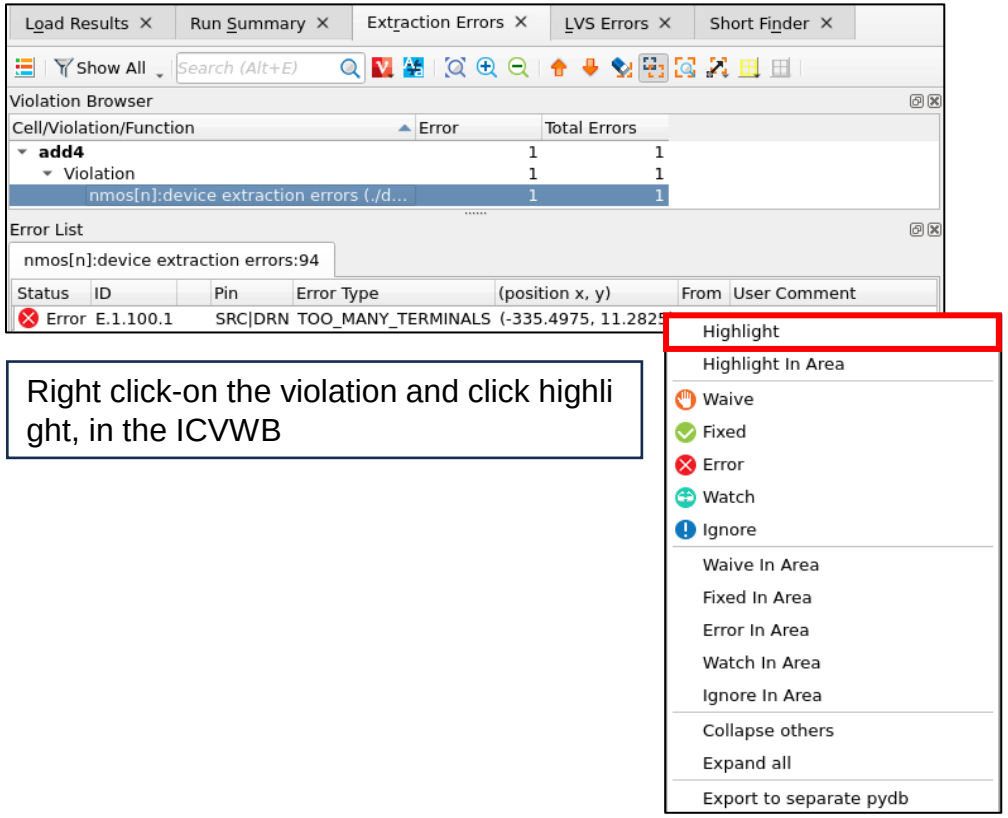
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PV(Physical Verification) : Use ICV (IC Validator tool)

6. Check Each Tabs in VUE



The Run Summary window indicates the LVS comparison is FAIL also had extraction Errors

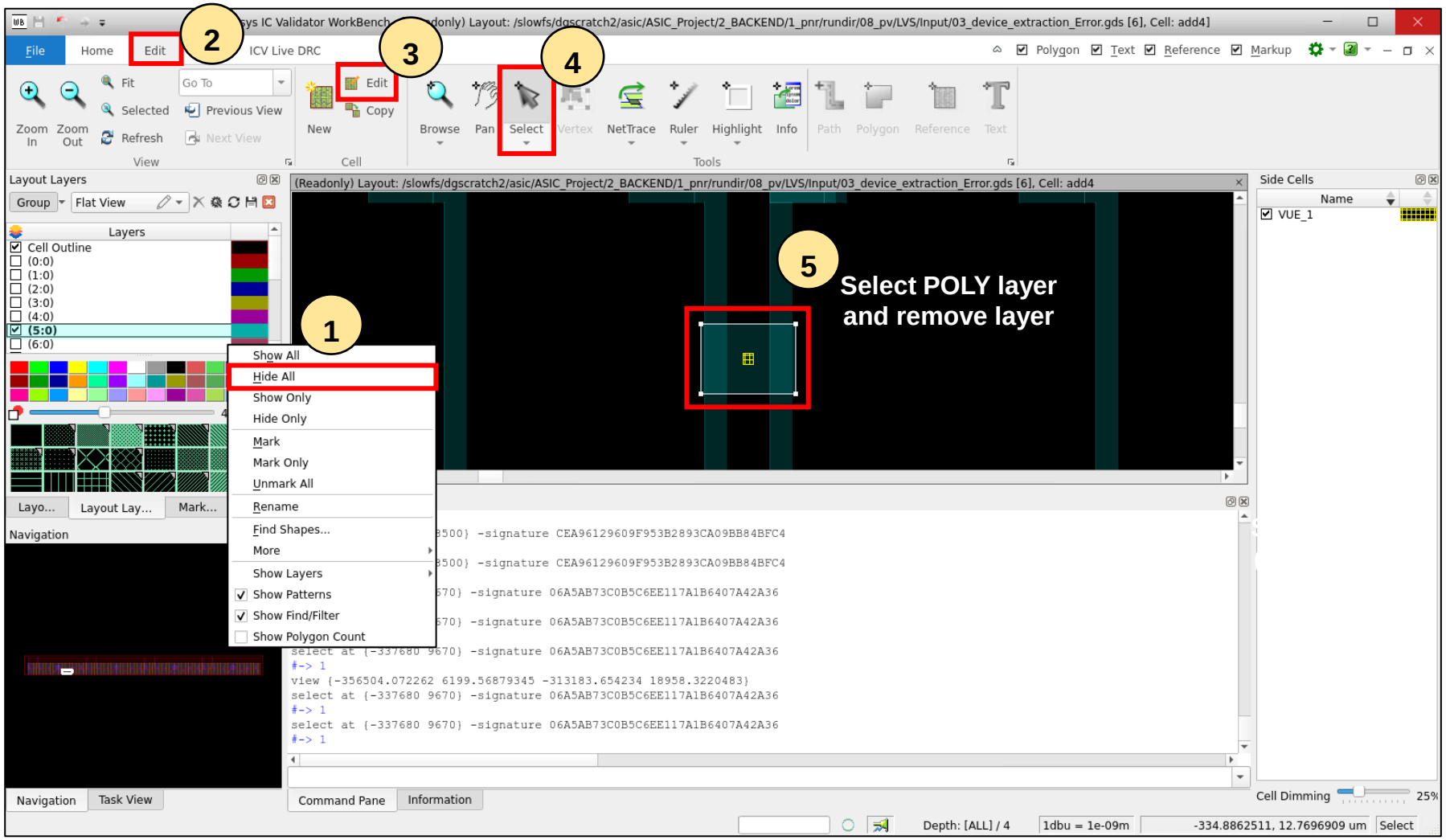


Right click-on the violation and click highlight, in the ICVWB

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PV(Physical Verification) : Use ICV (IC Validator tool)

7. **Remove POLY layer (5 : 0) & save layout (gds format)**
   - **In the ICVWB window deselect all of the layers so that only the POLY layer is visible (Layer 5 : 0)**
   - **Remove the shape that is causing a short and creating additional terminals**



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PV(Physical Verification) : Use ICV (IC Validator tool)

7. **Remove A1 text & save layout (gds format)**
   - **Save layout after remove text**



   - **Check gds file after save layout**

```
  < user@linux Input >> pwd
/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/08_pv/LVS/Input
  < user@linux Input >> ls
01_text_short_Error_fix.gds         03_device_extraction_Error.gds
01_text_short_Error.gds             04_unmatched_device_Error_fix.gds
02_unused_text_Error_fix.gds        LVS_lab.rs
02_unused_text_Error.gds            LVS_lab.sp
03_device_extraction_Error_fix.gds
```


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PV(Physical Verification) : Use ICV (IC Validator tool)

8. **Execute the following ICV command**
   - **Use 01_text_short_Error_fix.gds**

```
  < user@linux LVS >> icv -host_init 2 -vue -vueshort -create_lvs_short_output -c add4 -i ./Input/
03_device_extraction_Error_fix.gds -s ./Input/LVS_lab.sp -sf SPICE ./Input/LVS_lab.rs

                                IC Validator

          Version U-2022.12-SP5 for linux64 - Oct 26, 2023 cl#9338971

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Time Started: 2023-12-20 10:39:56
Running on: us01odcvde31334.internal.synopsys.com 3.10.0-1160.31.1.el7.x86_64

-------------

Called as: icv -host_init 2 -vue -vueshort -create_lvs_short_output -c add4 -i ./Input/03_device_e
xtraction_Error_fix.gds -s ./Input/LVS_lab.sp -sf SPICE ./Input/LVS_lab.rs
```



```
IC Validator Machine Memory Report
us01odcvde31334.internal.synopsys.com : Average = 0.166 GB, Peak = 0.166 GB

Overall Disk Usage Disk=0.009 GB
Network Disk Usage Peak=0.009 GB
Group File Disk Usage Peak=0.000 GB
Overall engine Time=0:00:07 Highest command Mem=0.031 GB

Overall Master Mem=0.438 GB

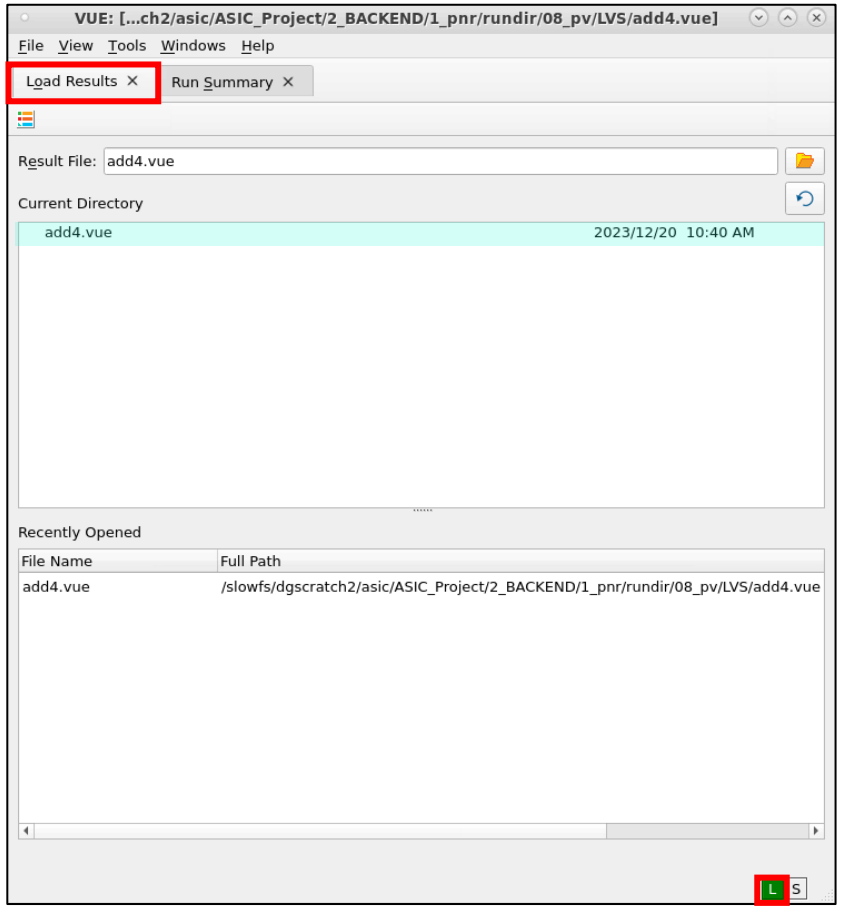
Time Ended: 2023-12-19 13:52:52

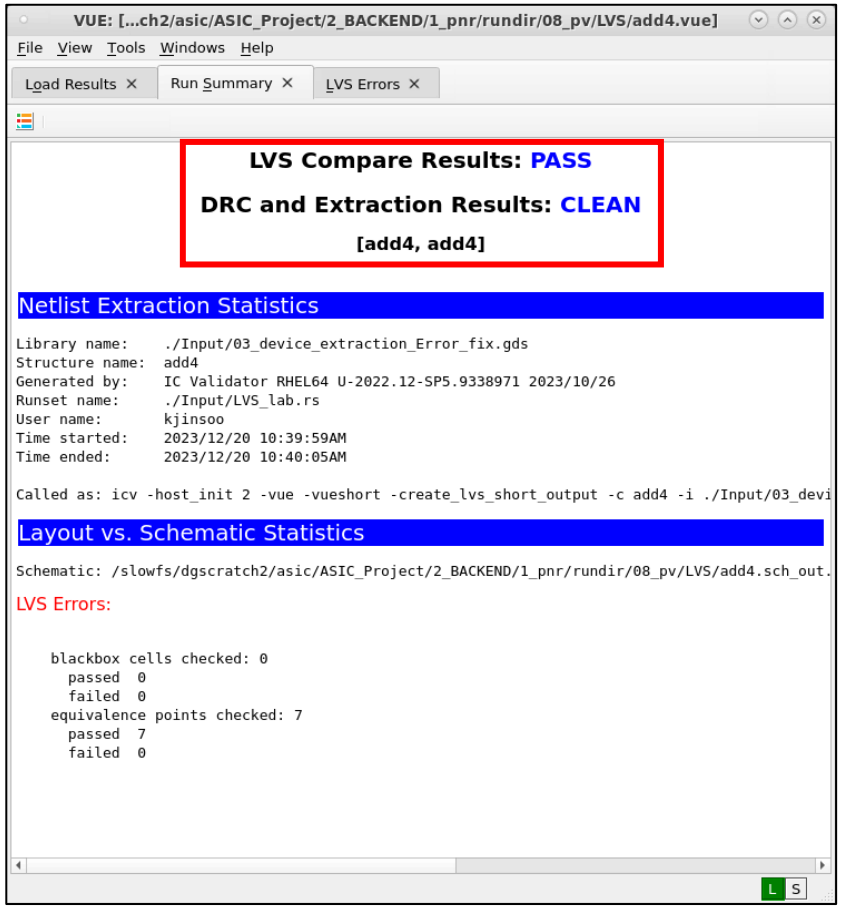
IC Validator is done.
```


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PV(Physical Verification) : Use ICV (IC Validator tool)

9. Inside of the "Load Results" tab, click on the add4.vue file that was generated in your LVS run directory





Check LVS Compare Results & DRC and Extraction Results

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WEEK 15 : LAB for LVS (Case 4 : Unmatched device Error)

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PV(Physical Verification) : Use ICV (IC Validator tool)

1. **Execute the following ICV command (case 4)**
   - **Recommend the ICVWB 2021.06-SP2 version & icv/2022.12-SP5 version**

```
  < user@linux LVS >> icv -host_init 2 -vue -vueshort -create_lvs_short_output -c add4 -i ./Input/
04_unmatched_device_Error.gds -s ./Input/LVS_lab.sp -sf SPICE ./Input/LVS_lab.rs

                                IC Validator

          Version U-2022.12-SP5 for linux64 - Oct 26, 2023 cl#9338971

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                        https://solvnetplus.synopsys.com)

Time Started: 2023-12-20 10:45:16
Running on: us01odcvde31334.internal.synopsys.com 3.10.0-1160.31.1.el7.x86_64

-------------

Called as: icv -host_init 2 -vue -vueshort -create_lvs_short_output -c add4 -i ./Input/04_unmatche
d_device_Error.gds -s ./Input/LVS_lab.sp -sf SPICE ./Input/LVS_lab.rs
```


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PV(Physical Verification) : Use ICV (IC Validator tool)

1. **Execute the following ICV command (case 4)**
   - **Recommend the ICVWB 2021.06-SP2 version & icv/2022.12-SP5 version**

```
Parsing runset "./Input/LVS_lab.rs"

./data/lvs.rs:115: warning: ICV LVS device settings currently use append mode. Use command line op
tion '-D ICV_REVERT_DEVICE_SETTING_APPEND_MODE' or runset instruction '#define ICV_REVERT_DEVICE_S
ETTING_APPEND_MODE' to revert LVS device setting back to overwrite mode in this release.
Loaded runset from cache: /u/kjinsoo/.icvrscache/5d5f3409cdf10cabb098ba5b99d3eec5-B03F5003-72BE-43
13-9278-FE01D266E969-us01odcvde31334-20399.rscache
Cached Runset Compile Time=0:00:01 User=0.47 Sys=0.05 Mem=0.434 GB

Virtual Machine Init Time=0:00:00 User=0.00 Sys=0.00 Mem=0.434 GB


Host startup begins.
Host startup done: 2 successes, 0 failures.
        us01odcvde31334.internal.synopsys.com: 2 total cores, maximum 2 commands in parallel.
Selecting "us01odcvde31334.internal.synopsys.com" as Primary host

System Startup Time=0:00:01 User=0.00 Sys=0.02 Mem=0.434 GB


Command execution begins.  Details recorded in the summary and log files:
    /slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/08_pv/LVS/run_details/add4.sum
    /slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/08_pv/LVS/run_details/LVS_lab.dp.l
og

Running ...
  [0:00:01][0%]   Total: 2 CPUs on 1 host
  [0:00:01][1%]
  [0:00:01][2%]
```

```
-------------
IC Validator Run: Time=0:00:06

IC Validator Machine Memory Report
us01odcvde31334.internal.synopsys.com : Average = 0.054 GB, Peak = 0.054 GB

Overall Disk Usage Disk=0.008 GB
Network Disk Usage Peak=0.008 GB
Group File Disk Usage Peak=0.001 GB
Overall engine Time=0:00:06 Highest command Mem=0.034 GB

Overall Master Mem=0.437 GB

Time Ended: 2023-12-19 10:09:33

IC Validator is done.
```


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PV(Physical Verification) : Use ICV (IC Validator tool)

2. **After the run completes, check the add4.LAYOUT_ERRORS file (unmatched device Error)**
   - **Recommend the ICVWB 2021.06-SP2 version & icv/2022.12-SP5 version**

```
                  LAYOUT ERRORS RESULTS: CLEAN

                   #### #     #####   ##   #   #
                  #    #     #      #  #  ##  #
                  #    #     ####  ###### # # #
                  #    #     #     #    # #  ##
                   #### ##### ##### #    # #   #

================================================================================

Library name:     ./Input/03_device_extraction_Error_fix.gds
Structure name:   add4
Generated by:     IC Validator RHEL64 U-2022.12-SP5.9338971 2023/10/26
Runset name:      ./Input/LVS_lab.rs
User name:        kjinsoo
Time started:     2023/12/20 10:39:59AM
Time ended:       2023/12/20 10:40:05AM

Called as: icv -host_init 2 -vue -vueshort -create_lvs_short_output -c add4 -i ./Input/03_device_e
xtraction_Error_fix.gds -s ./Input/LVS_lab.sp -sf SPICE ./Input/LVS_lab.rs
```

After the run completes, observe the add4.LAYOUT_ERRORS file, extracted layout is clea n

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PV(Physical Verification) : Use ICV (IC Validator tool)

2. **After the run completes, check the add4.LVS_ERRORS file (unmatched device Error)**
   - **Recommend the ICVWB 2021.06-SP2 version & icv/2022.12-SP5 version**

```
+------------------------------------------------------------------------+
|                    ICV_Compare LVS Comparison Report                   |
+------------------------------------------------------------------------+

ICV_Compare (R) Hierarchical Layout Vs. Schematic
        RHEL64 U-2022.12-SP5.9338971 2023/10/26
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------------------------------------------------------------------------
LVS error file     = add4.LVS_ERRORS
Layout error file  = add4.LAYOUT_ERRORS
Schematic netlist  = /slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/08_pv/LVS/add4.sc
h_out.gz
Layout netlist     = /slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/08_pv/LVS/add4.ne
t.gz
Runset file        = ./data/lvs.rs
Working directory  = /slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/08_pv/LVS
Compare directory  = run_details/compare
Compare start time = 2023-12-20 11:05:08



------------------------------------------------------------------------
Final comparison result:FAIL

                      ######   ##   ##### #
                      #       #  #    #   #
                      #####  ######   #   #
                      #      #    #   #   #
                      #      #    # ##### #####


TOP equivalence point:
                [add4, add4]

Comparison summary

    6 Successful equivalence points
  * 1 Failed equivalence points
        1 First priority errors
        0 Second priority errors
```

```
> invb != invb (level = 2)

Error summary:

          1 Unmatched schematic instance
          1 Unmatched schematic net
          2 Unmatched layout instances
          3 Unmatched layout nets

          1 Matched instance
          3 Matched nets

Port summary:

          1 Unmatched schematic port
          1 Unmatched layout port
          3 Matched ports


Post-compare summary (* = unmatched devices, nets or ports):

       Matched      Unmatched    Unmatched        Instance types
                    schematic    layout           [schematic, layout]
       ---------    ---------    ---------        --------------------
              1            0            0        PMOS[p, p]
              0            1            2      * NMOS[n, n]
       ---------    ---------    ---------        --------------------
              1            1            2      * Total instances

              3            1            3      * Total nets

              3            1            1      * Total ports
```

You can check the occurrence of an error due t o a unmatched device between layout and sch ematic

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PV(Physical Verification) : Use ICV (IC Validator tool)

3. **check the results with the help of VUE and the short finder tool (open ICVWB)**
   - **Recommend the ICVWB 2021.06-SP2 version & icv/2022.12-SP5 version**

```
 < user@linux LVS >> icvwb &
[1] 32113
 < user@linux LVS >>
                                    ICVWB

          Version S-2021.06-SP2 for linux64 - Nov 22, 2021 1010135

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```

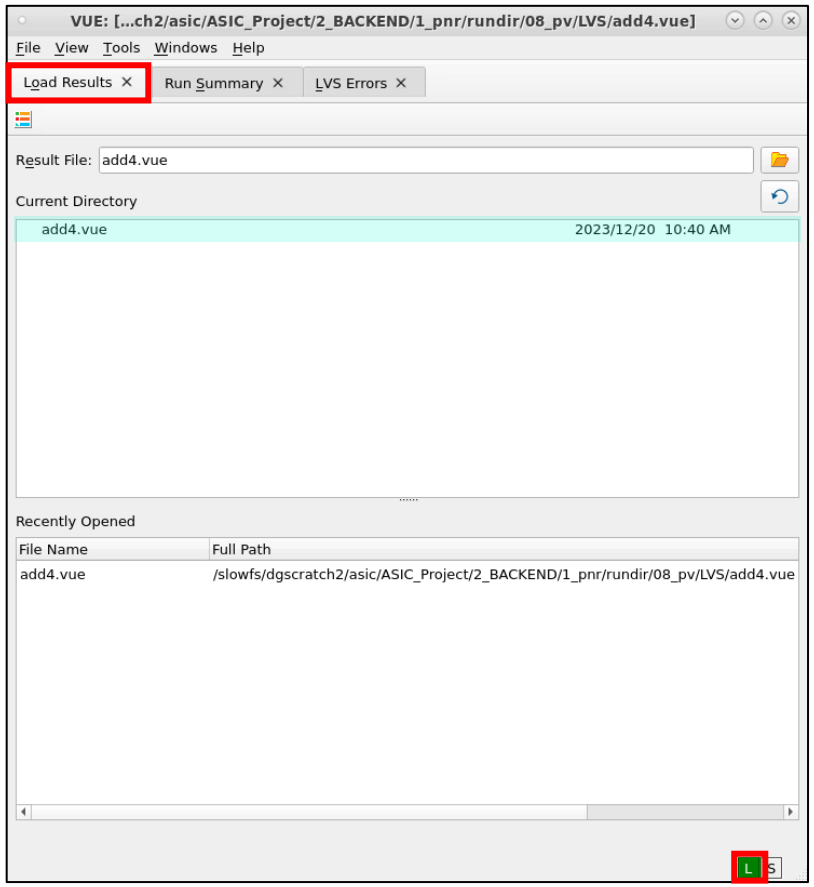
4. **Click "ICV_VUE" button (A VUE window will appear that is linked to ICVWB)**

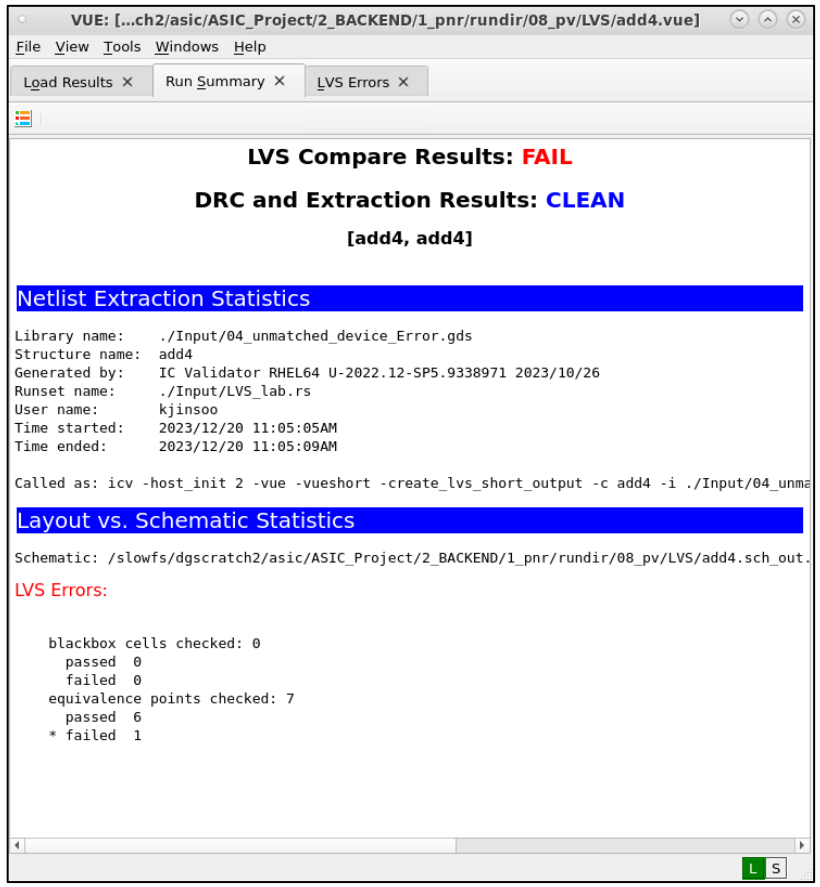


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PV(Physical Verification) : Use ICV (IC Validator tool)

5. Inside of the "Load Results" tab, click on the add4.vue file that was generated in your LVS run directory

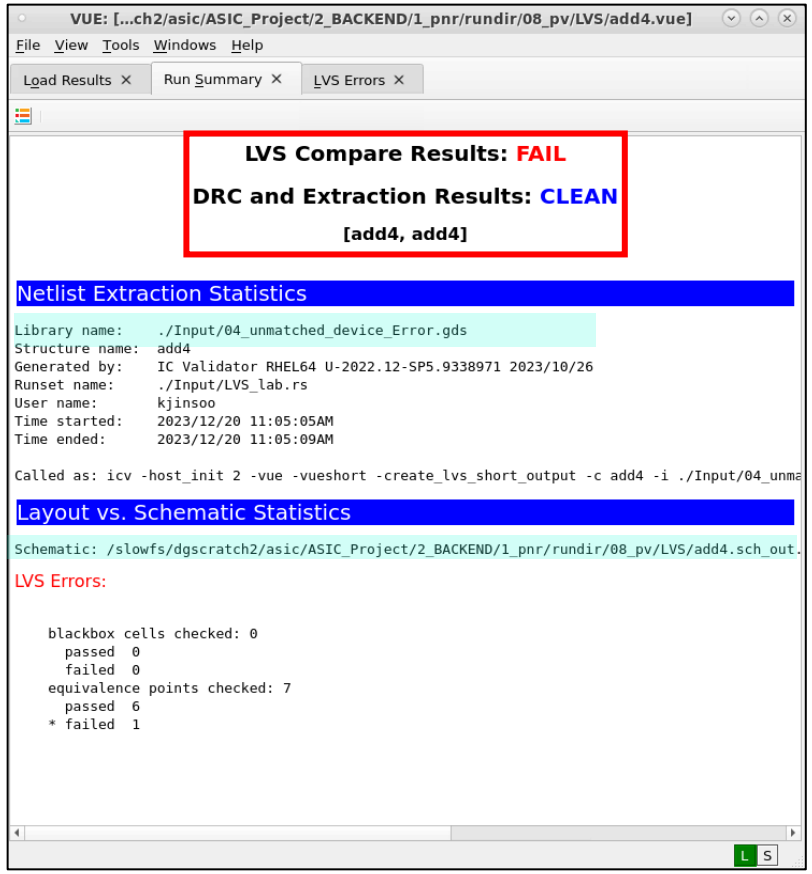


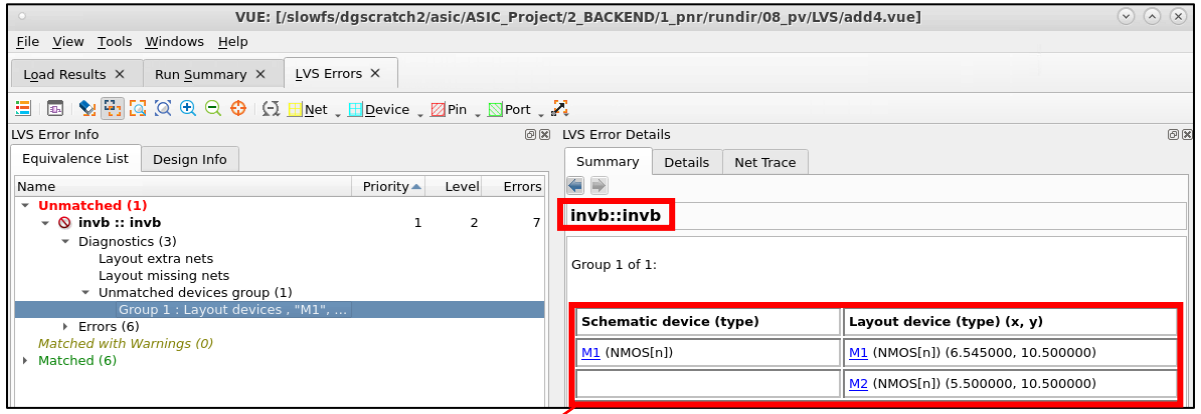


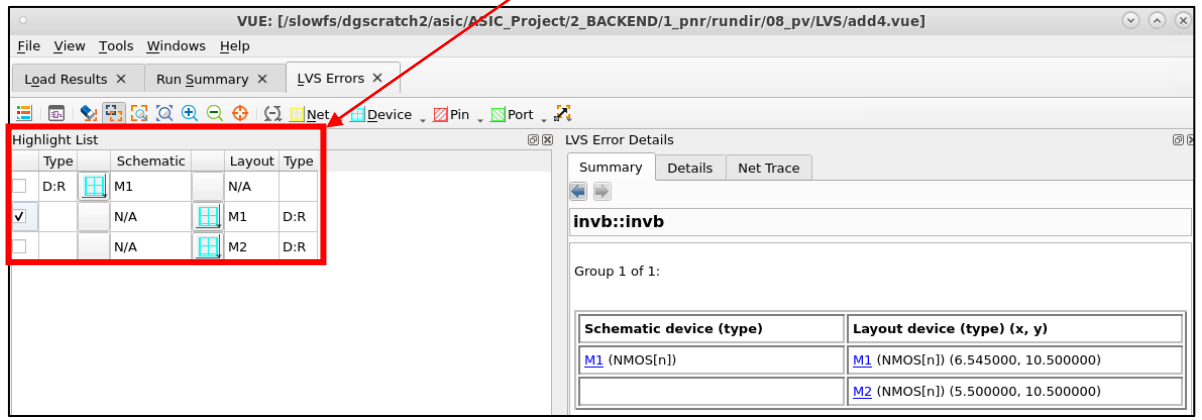
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PV(Physical Verification) : Use ICV (IC Validator tool)

6. Check Each Tabs in VUE







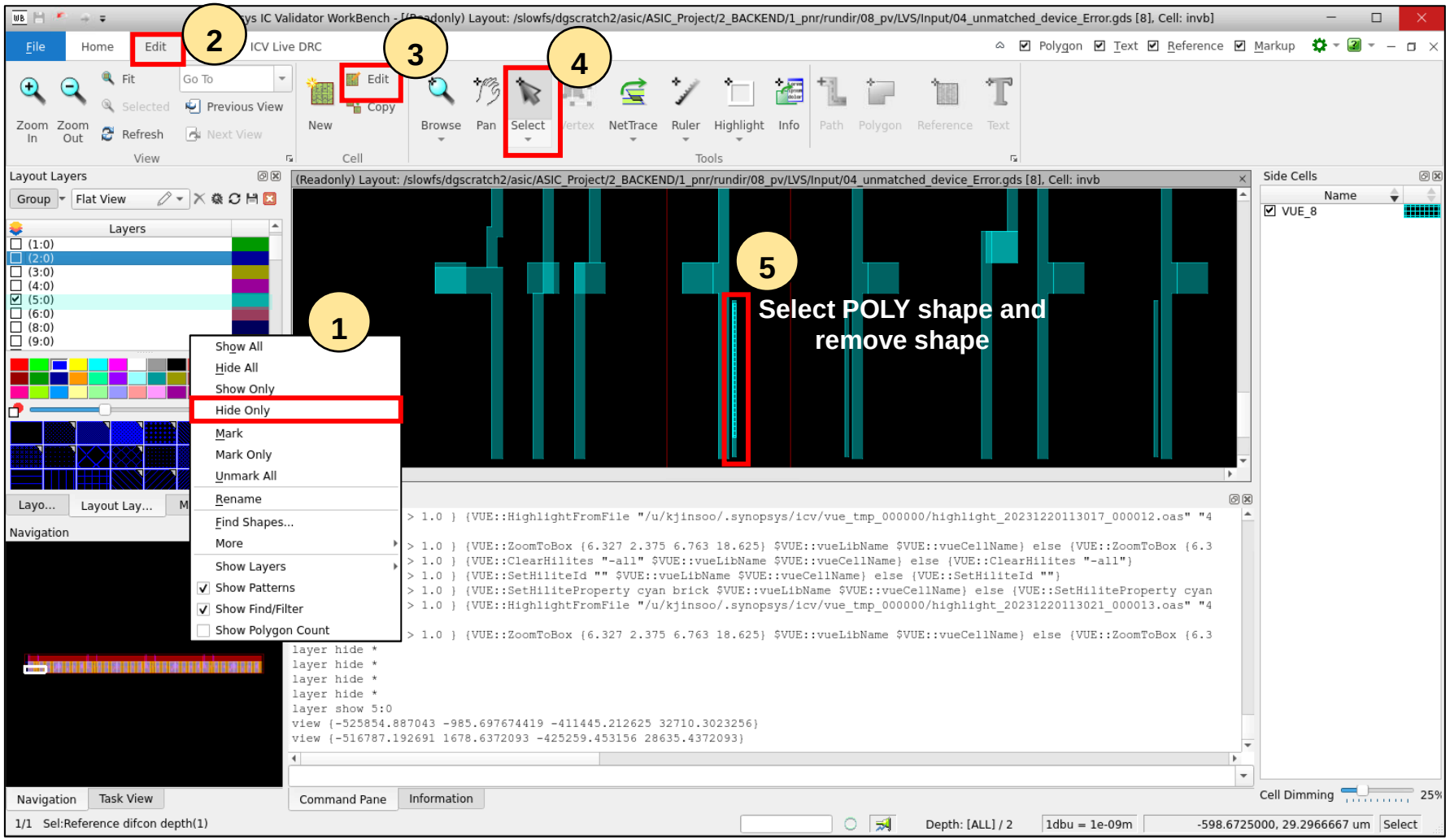
The Run Summary window indicates the LVS comparison is fail but extraction is clean

Check the violation in the LVS Errors tab (It shows that there is an extra NMOS device in the "Invb"cell)

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PV(Physical Verification) : Use ICV (IC Validator tool)

7. **Remove POLY shapes (5 : 0) & save layout (gds format)**
   - **Check for extra POLY shapes (5 : 0) that are drawn on the NMOS device**
   - **Remove the extra shape**



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PV(Physical Verification) : Use ICV (IC Validator tool)

7. **Remove A1 text & save layout (gds format)**
   - **Save layout after remove text**



   - **Check gds file after save layout**

```
  < user@linux Input >> pwd
/slowfs/dgscratch2/asic/ASIC_Project/2_BACKEND/1_pnr/rundir/08_pv/LVS/Input
  < user@linux Input >> ls
01_text_short_Error_fix.gds   03_device_extraction_Error_fix.gds  LVS_lab.rs
01_text_short_Error.gds       03_device_extraction_Error.gds      LVS_lab.sp
02_unused_text_Error_fix.gds  04_unmatched_device_Error_fix.gds
02_unused_text_Error.gds      04_unmatched_device_Error.gds
```


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8. **Execute the following ICV command**
   - **Use 01_text_short_Error_fix.gds**

```
  < user@linux LVS >> icv -host_init 2 -vue -vueshort -create_lvs_short_output -c add4 -i ./Input/
04_unmatched_device_Error_fix.gds -s ./Input/LVS_lab.sp -sf SPICE ./Input/LVS_lab.rs

                                IC Validator

          Version U-2022.12-SP5 for linux64 - Oct 26, 2023 cl#9338971

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Time Started: 2023-12-20 11:48:44
Running on: us01odcvde31334.internal.synopsys.com 3.10.0-1160.31.1.el7.x86_64

-------------

Called as: icv -host_init 2 -vue -vueshort -create_lvs_short_output -c add4 -i ./Input/04_unmatche
d_device_Error_fix.gds -s ./Input/LVS_lab.sp -sf SPICE ./Input/LVS_lab.rs
```



```
IC Validator Machine Memory Report
us01odcvde31334.internal.synopsys.com : Average = 0.166 GB, Peak = 0.166 GB

Overall Disk Usage Disk=0.009 GB
Network Disk Usage Peak=0.009 GB
Group File Disk Usage Peak=0.000 GB
Overall engine Time=0:00:07 Highest command Mem=0.031 GB

Overall Master Mem=0.438 GB

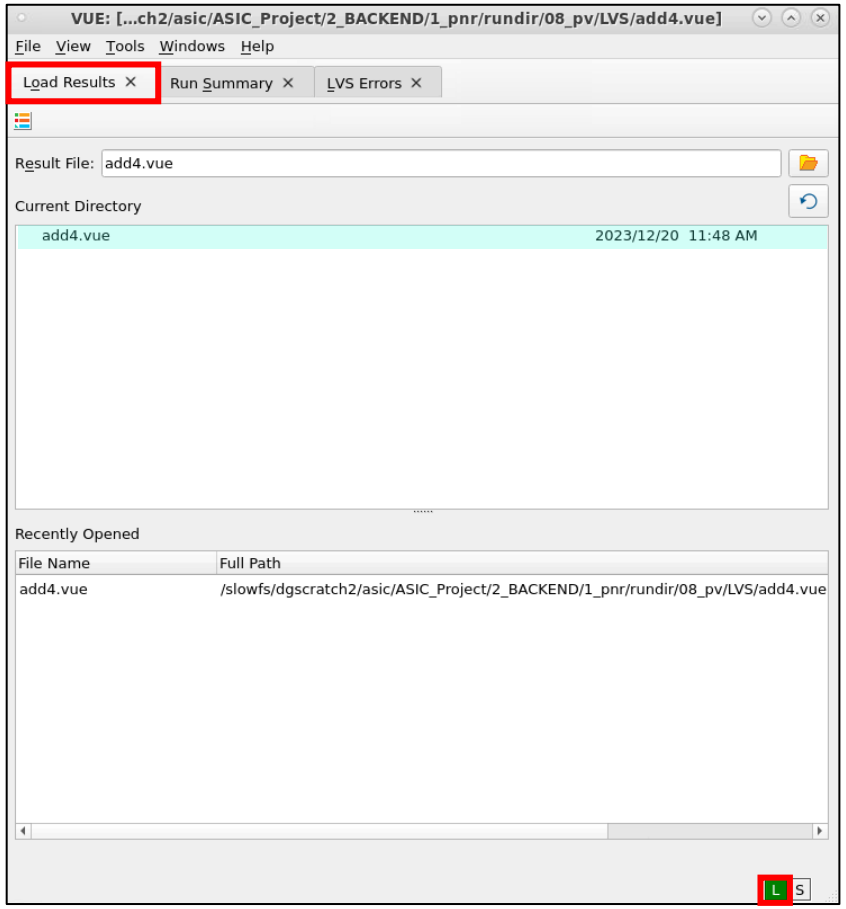
Time Ended: 2023-12-19 13:52:52

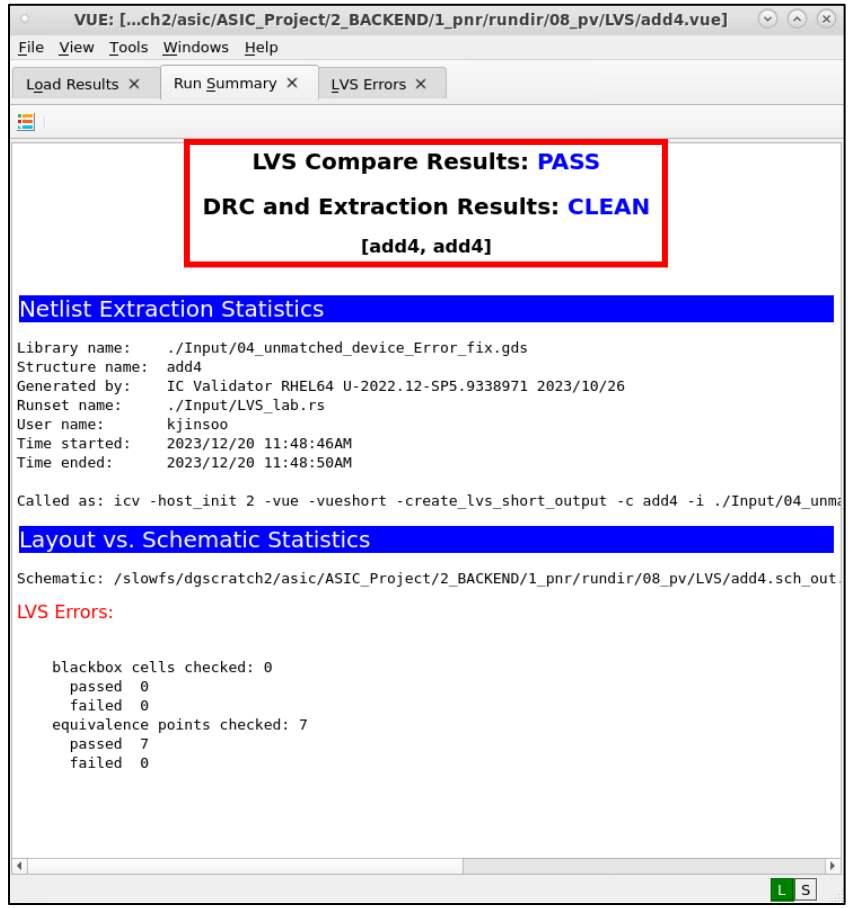
IC Validator is done.
```


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9. Inside of the "Load Results" tab, click on the add4.vue file that was generated in your LVS run directory





Check LVS Compare Results & DRC and Extraction Results

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Thank You

